

VERSION 1 CAPS

GRADE 11 PHYSICAL SCIENCES

WRITTEN BY VOLUNTEERS

EVERYTHING SCIENCE

BY



SIYAVULA
TECHNOLOGY-POWERED LEARNING



basic education

Department:
Basic Education
REPUBLIC OF SOUTH AFRICA

METROPOLITAN



1 IA																										18 0																					
2,1 H 1,01		Periodic Table of the Elements																2		He		4,00																									
		2 IIA														13 IIIA		14 IVA		15 VA		16 VIA		17 VIIA																							
1,0 4 1,5 Li 6,94		Be 9,01		No EN Element AMU												5 2,0 B 10,8		6 2,5 C 12,0		7 3,0 N 14,0		8 3,5 O 16,0		9 4,0 F 19,0		10 Ne 20,2																					
1 0,9 12 1,2 Na 23,0		Mg 24,3														13 1,5 Al 27,0		14 1,8 Si 28,1		15 2,1 P 31,0		16 2,5 S 32,1		17 3,0 Cl 35,45		18 Ar 39,9																					
19 0,8 20 1,0 K 39,1		Ca 40,1														21 1,3 Sc 45,0		22 1,5 Ti 47,9		23 1,6 V 50,9		24 1,6 Cr 52,0		25 1,5 Mn 54,9		26 1,8 Fe 55,8		27 1,8 Co 58,9		28 1,8 Ni 58,7		29 1,9 Cu 63,5		30 1,6 Zn 65,4		31 1,6 Ga 69,7		32 1,8 Ge 72,6		33 2,0 As 74,9		34 2,4 Se 79,0		35 2,8 Br 79,9		36 Kr 83,8	
37 0,8 38 1,0 Rb 85,5		Sr 87,6														39 1,2 Y 88,9		40 1,4 Zr 91,2		41 1,6 Nb 92,9		42 1,8 Mo 95,9		43 1,9 Tc (98)		44 2,2 Ru 101,1		45 2,2 Rh 102,9		46 2,2 Pd 106,4		47 1,9 Ag 107,9		48 1,7 Cd 112,4		49 1,7 In 114,8		50 1,8 Sn 118,7		51 1,9 Sb 121,8		52 2,1 Te 127,6		53 2,5 I 126,9		54 Xe 131,3	
55 0,7 56 0,9 Cs 132,9		Ba 137,3														57-71 La-Lu Lanthanides		72 1,6 Hf 178,5		73 1,5 Ta 180,9		74 2,4 W 183,8		75 1,9 Re 186,2		76 2,2 Os 190,2		77 2,2 Ir 192,2		78 2,2 Pt 195,1		79 2,5 Au 197,0		80 2,0 Hg 200,6		81 1,6 Tl 204,4		82 1,8 Pb 207,2		83 1,9 Bi 209,0		84 2,0 Po (209)		85 2,5 At (210)		86 Rn (222)	
87 0,7 88 0,9 Fr (223)		Ra 226,0		89-103 Ac-Lr Actinides		104 (261) Rf (261)		105 (262) Db (262)		106 (263) Sg (263)		107 (262) Bh (262)		108 (265) Hs (265)		109 (266) Mt (266)		110 (269) Ds (269)		111 (272) Rg (272)		112 (277) Cn (277)		113 (284) Uut (284)		114 (289) Uuq (289)		115 (288) Uup (288)		116 (293) Uuh (293)		117 (282) Uus (282)		118 (282) Uuo (282)													

- Transition Metal
- Metal
- Metalloid
- Non-metal
- Noble Gas
- Lanthanide
- Actinide

57	1,1	58	1,1	59	1,1	60	1,1	61		62	1,1	63		64	1,2	65		66	1,2	67	1,2	68	1,2	69	1,3	70		71	1,3
La		Ce		Pr		Nd		Pm		Sm		Eu		Gd		Tb		Dy		Ho		Er		Tm		Yb		Lu	
138,9		140,1		140,9		144,2		(145)		150,4		152,0		157,3		158,9		162,5		164,9		167,3		168,9		173,0		175,0	
89	1,1	90	1,3	91	1,5	92	1,4	93	1,3	94	1,3	95	1,3	96	1,3	97	1,3	98	1,3	99	1,3	100	1,3	101	1,3	102	1,3	103	
Ac		Th		Pa		U		Np		Pu		Am		Cm		Bk		Cf		Es		Fm		Md		No		Lr	
227,0		232,0		231,0		238,0		237,0		(244)		(243)		(247)		(247)		(251)		(252)		(257)		(258)		(258)		(260)	

EVERYTHING SCIENCE

GRADE 11 PHYSICAL SCIENCES

VERSION 1 CAPS

WRITTEN BY VOLUNTEERS

COPYRIGHT NOTICE

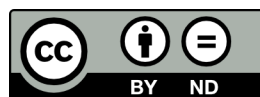
Your freedom to legally copy this book

You are allowed and encouraged to copy any of the Everything Maths and Everything Science textbooks. You can legally photocopy any page or even the entire book. You can download it from www.everythingmaths.co.za and www.everythingscience.co.za, read it on your phone, tablet, iPad, or computer. You can burn it to CD, put on your flash drive, e-mail it around or upload it to your website.

The only restriction is that you have to keep this book, its cover, title, contents and short-codes unchanged.

This book was derived from the original Free High School Science Texts written by volunteer academics, educators and industry professionals. Everything Maths and Everything Science are trademarks of Siyavula Education.

For more information about the Creative Commons Attribution-NoDerivs 3.0 Unported (CC BY-ND 3.0) license see <http://creativecommons.org/licenses/by-nd/3.0/>



AUTHORS AND CONTRIBUTORS

Siyavula Education

Siyavula Education is a social enterprise launched in 2012 with capital and support from the PSG Group Limited and the Shuttleworth Foundation. The Everything Maths and Science series is one of the titles developed and openly released by Siyavula. For more information about the writing and distribution of these or other openly licensed titles:

www.siyavula.com

info@siyavula.com

021 469 4771

Siyavula Authors

Dr. Mark Horner; Heather Williams

Siyavula and DBE team

Jayanthi SK Maharaj (Veena); Marongwa Masemula; Ewald Zietsman; Bridget Nash; Prof. Gilberto Isquierdo; Karen Kornet; Dr. Kevin Reddy; Enoch Ndwamato Makhado; Clive Mhaka

Siyavula and Free High School Science Text contributors

Dr. Mark Horner; Dr. Samuel Halliday; Dr. Sarah Blyth; Dr. Rory Adams; Dr. Spencer Wheaton

Iesrafeel Abbas; Sarah Abel; Dr. Rory Adams; Andrea Africa; Wiehan Agenbag; Matthew Amundsen; Ben Anhalt; Prashant Arora; Amos Baloyi; Bongani Baloyi; Raymond Barbour; Caro-Joy Barendse; Richard Baxter; Tara Beckerling; Tim van Beek; Mariaan Bester; Jennifer de Beyer; Dr. Sarah Blyth; Sebastian Bodenstein; Martin Bongers; Thinus Booysen; Gareth Boxall; Stephan Brandt; Hannes Breytenbach; Alexander Briell; Wilbur Britz; Graeme Broster; Craig Brown; Michail Brynard; Deanne de Bude; Richard Burge; Bianca B hmer; Jan Buys; George Calder-Potts; Eleanor Cameron; Mark Carolissen; Shane Carolisson; Richard Case; Sithembile Cele; Alice Chang; Richard Cheng; Fanny Cherblanc; Dr. Christine Chung; Brett Cocks; Roch  Compaan; Willem Conradie; Stefaan Conradie; Rocco Coppejans; Tim Craib; Andrew Craig; Tim Crombie; Dan Crytser; Jock Currie; Dr. Anne Dabrowski; Laura Daniels; Gareth Davies; Sandra Dickson; Sean Dobbs; Buhle Donga; William Donkin; Esmi Dreyer; Matthew Duddy; Christel Durie; Fernando Durrell; Dr. Dan Dwyer; Frans van Eeden; Alexander Ellis; Tom Ellis; Andrew Fisher; Giovanni Franzoni; Olivia Gillett; Ingrid von Glehn; Tamara von Glehn; Lindsay Glesener; Kevin Godby; Dr. Vanessa Godfrey; Terence Goldberg; Dr. Johan Gonzalez; Saaligha Gool; Hemant Gopal; Dr. Stephanie Gould; Umeshree Govender; Heather Gray; Lynn Greeff; Jaco Greyling; Martli Greyvenstein; Carine Grobbelaar; Suzanne Grov ; Dr. Tom Gutierrez; Brooke Haag; Kate Hadley; Alex Hall; Dr. Sam Halliday; Asheena Hanuman; Dr. Melanie Dymond Harper; Ebrahim Harris; Dr. Nicholas Harrison; Neil Hart; Nicholas Hatcher; Jason Hayden; Laura Hayward; Dr. William P. Heal; Pierre van Heerden; Dr. Fritha Hennessy; Dr. Colleen Henning; Shaun Hewitson; Millie Hilgart; Grant Hillebrand; Nick Hobbs; Chris Holdsworth; Dr. Benne Holwerda; Dr. Mark Horner; Robert Hovden; Mfandaidza Hove; Jennifer Hsieh; George Hugo; Laura Huss; Prof. Ed Jacobs

Hester Jacobs; Stefan Jacobs; Rowan Jelley; Grant Jelley; Clare Johnson; Francois Jooste; Luke Jordan; Tana Joseph; Corli Joubert; Dr. Fabian Jutz; Brian Kamanzi; Herman Kamper; Dr. Lutz Kampmann; Simon Katende; Natalia Kavalenia; Rabia Khan; Nothando Khumalo; Paul Kim; Lizl King; Melissa Kistner; Dr. Jennifer Klay; Andrea Koch; Grove Koch; Bishop Komolafe; Dr. Timo Kriel; Lara Kruger; Sihle Kubheka; Andrew Kubik; Luca Lategan; Dr. Jannie Leach; Nkoana Lebaka; Dr. Marco van Leeuwen; Dr. Tom Leinster; Ingrid Lezar; Henry Liu; Christopher Loetscher; Linda Loots; Michael Loseby; Bets Lourens; Chris Louw; Amandla Mabona; Malothe Mabutho; Stuart Macdonald; Dr. Anton Machacek; Tshepo Madisha; Batsirai Magunje; Dr. Komal Maheshwari; Michael Malahe; Masoabi Malunga; Kosma von Maltitz; Masilo Mapaila; Bryony Martin; Nicole Masureik; Jacques Masuret ; John Mathew; Dr. Will Matthews; Chiedza Matuso; JoEllen McBride; Nikolai Meures; Margaretha Meyer; Riana Meyer; Filippo Miatto; Jenny Miller; Rossouw Minnaar; Abdul Mirza; Colin Mkhize; Mapholo Modise; Carla Morderdyk; Tshwarelo Mohlala; Relebohile Molaoa; Marasi Monyau; Asogan Moodaly; Jothi Moodley; Robert Moon; Calvin Moore; Bhavani Morarjee; Kholofelo Moyaba; Nina Gitau Muchunu; Christopher Muller; Helgard Muller; Johan Muller; Caroline Munyonga; Alban Murewi; Kate Murphy; Emmanuel Musonza; Tom Mutabazi; David Myburgh; Johann Myburgh; Kamie Naidu; Nolene Naidu; Gokul Nair; Vafa Naraghi; Bridget Nash; Eduan Naudé; Tyrone Negus; Theresa Nel; Huw Newton-Hill; Buntu Ngcebetsha; Towan Nothling; Dr. Markus Oldenburg; Adekunle Oyewo; Thomas O'Donnell; Dr. Jaynie Padayachee; Poveshen Padayachee; Masimba Paradza; Quinton Paulse; Dave Pawson; Justin Pead; Carli Pengilly; Nicolette Pekeur; Joan Pienaar; Petrus Pieter; Sirika Pillay; Jacques Plaut; Jaco du Plessis; Barry Povey; Barry Povey; Andrea Prinsloo; David Prinsloo; Joseph Raimondo; Sanya Rajani; Alastair Ramlakan; Thinus Ras; Dr. Matina J. Rassias; Ona Rautenbach; Dr. Jocelyn Read; Jonathan Reader; Jane Reddick; Robert Reddick; Dr. Matthew Reece; Chris Reeders; Razvan Remsing; Laura Richter; Max Richter; Sean Riddle; Dr. David Roberts; Christopher Roberts; Helen Robertson; Evan Robinson; Christian Roelofse; Raoul Rontsch; Dr. Andrew Rose; Katie Ross; Jeanne-Mari Roux; Karen Roux; Mark Roux; Bianca Ruddy; Heinrich Rudman; Nitin Rughoonauth; Katie Russell; Steven Sam; Jason Avron Samuels; Dr. Carl Scheffler; Nathaniel Schwartz; Duncan Scott; Christovan van Schalkwyk; Rhoda van Schalkwyk; Helen Seals; Relebohile Sefako; Prof. Sergey Rakityansky; Sandra Serumaga-Zake; Paul Shangase; Cameron Sharp; Ian Sherratt; Dr. James Short; Cho Hee Shrader; Roger Sieloff; Brandon Sim; Bonga Skozana; Clare Slotow; Bradley Smith; Greg Solomon; Nicholas Spaul; Hester Spies; Dr. Andrew Stacey; Dr. Jim Stasheff; Mike Stay; Nicol Steenkamp; Dr. Fred Strassberger; Mike Stringer; Stephanie Strydom; Abdulhuck Suliman; Masixole Swartbooi; Tshenolo Tau; Tim Teatro; Ben Thompson; Shen Tian; Xolani Timbile; Liezel du Toit; Nicola du Toit; Dr. Francois Toerien; Renè Toerien; Dr. Johan du Toit; Robert Torregrosa; Jimmy Tseng; Pieter Vergeer; Rizmari Versfeld; Nina Verwey; Mfundo Vezi; Mpilonhle Vilakazi; Wetsie Visser; Alexander Volkwyn; Mia de Vos; Dr. Karen Wallace; John Walmsley; Helen Waugh; Leandra Webb; Dr. Dawn Webber; Michelle Wen; Dr. Rufus Wesi; Francois Wessels; Wessel Wessels; Neels van der Westhuizen; Sabet van der Westhuizen; Dr. Alexander Wetzler; Dr. Spencer Wheaton; Vivian White; Dr. Gerald Wigger; Harry Wiggins; Heather Williams; Wendy Williams; Julie Wilson; Timothy Wilson; Andrew Wood; Emma Wormauld; Dr. Sahal Yacoob; Jean Youssef; Ewald Zietsman; Johan Zietsman; Marina van Zyl

SPONSOR

This text book was developed with corporate social investment funding from MMI Holdings.



Well structured, impactful Corporate Social Investment (CSI) has the ability to contribute positively to nation building and drive positive change in the communities. MMI's commitment to social investment means that we are constantly looking for ways in which we can assist some of South Africa's most vulnerable citizens to expand their horizons and gain greater access to life's opportunities. This means that we do not view social investment as a nice to have or as an exercise in marketing or sponsorship but rather as a critical part of our contribution to society.

The merger between Metropolitan and Momentum was lauded for the complementary fit between two companies. This complementary fit is also evident in the focus areas of CSI programmes where Metropolitan and Momentum together cover and support the most important sectors and where the greatest need is in terms of social participation.

HIV/AIDS is becoming a manageable disease in many developed countries but in a country such as ours, it remains a disease where people are still dying of this scourge unnecessarily. Metropolitan continues to make a difference in making sure that HIV/AIDS moves away from being a death sentence to a manageable disease. Metropolitan's other focus area is education which remains the key to economic prosperity for our country.

Momentum's focus on persons with disabilities ensures that this community is included and allowed to make their contribution to society. Orphaned and vulnerable children are another focus area for Momentum and projects supported ensure that children are allowed to grow up safely, to assume their role along with other children in inheriting a prosperous future.

EVERYTHING MATHS AND SCIENCE

The Everything Mathematics and Science series covers Mathematics, Physical Sciences, Life Sciences and Mathematical Literacy.

The Siyavula Everything
Science textbooks



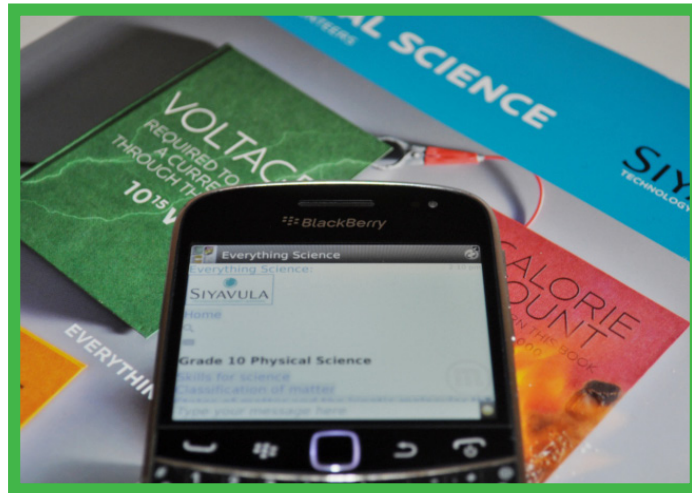
The Siyavula Everything
Maths textbooks



READ ON MOBILE

- **MOBI-WEB**

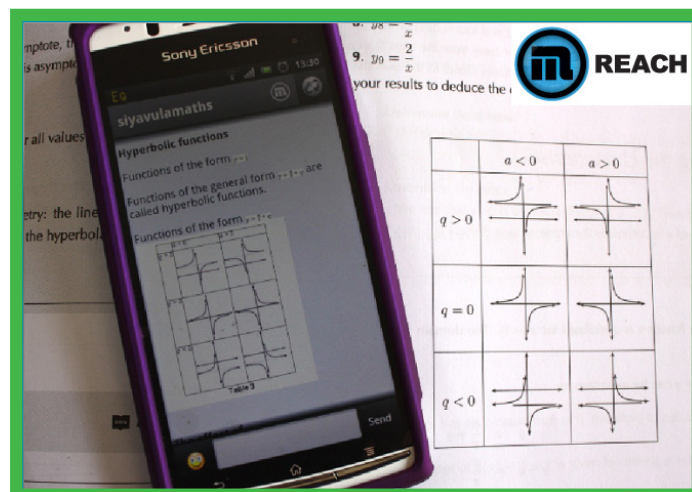
You can read all of the Everything Series textbooks on your mobile phone. Visit the Everything Maths and Everything Science mobi sites at:



m.everythingmaths.co.za and m.everythingscience.co.za

- **MXIT**

All Mxit users can read their Everything Series textbooks on Mxit Reach. Add Everything Maths and Everything Science as Mxit contacts or browse to the books on Mxit Reach.

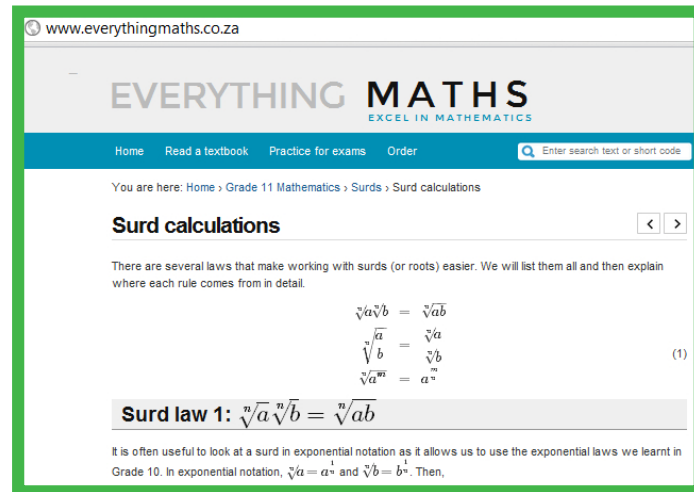


[mxit>tradepost>reach>education>everything maths or everything science](#)

DIGITAL TEXTBOOKS

- **READ ONLINE**

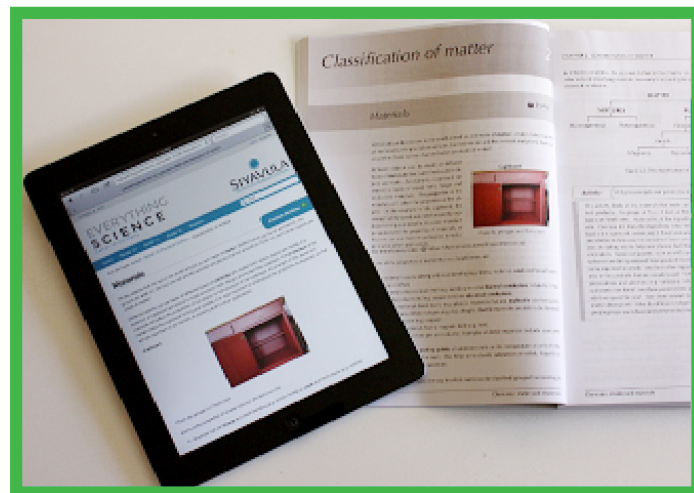
The on-line books feature videos, presentations, simulations and fully worked solutions to the questions and exercises found in the book.



www.everythingmaths.co.za and www.everythingscience.co.za

- **DOWNLOAD FOR TABLETS**

For off-line reading on your PC, tablet, iPad and Kindle you can download a digital copy of the Everything Series textbooks. Visit the Everything Maths and Everything Science websites and download the books.

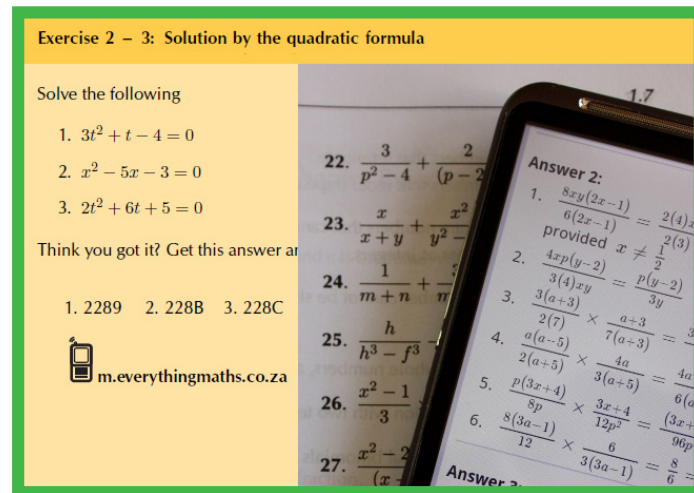


www.everythingmaths.co.za and www.everythingscience.co.za

PRACTISE INTELLIGENTLY

- **CHECK YOUR ANSWERS ON YOUR PHONE**

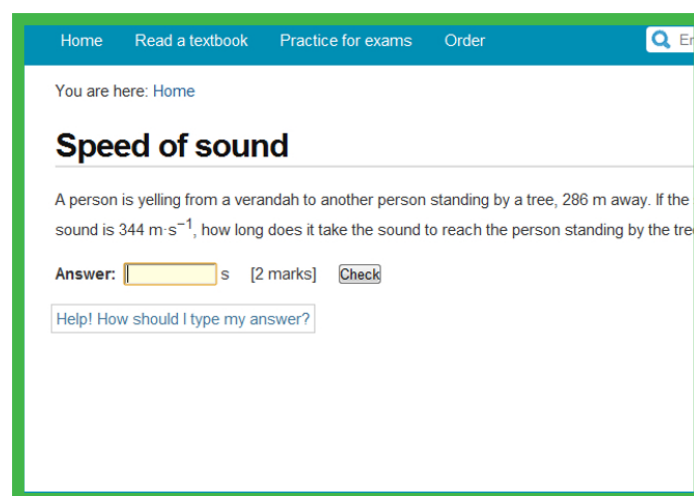
You can check your answer to any question in this textbook on your mobile phone by entering the shortcode found in the textbook into the search box on the mobi-site.



m.everythingmaths.co.za and m.everythingscience.co.za

- **PRACTISE FOR TESTS AND EXAMS ON YOUR PHONE**

To do well in tests and exams you need practice. Practise the exercises from this textbook, additional exercises and questions from past exam papers on m.everythingmaths.co.za and m.everythingscience.co.za and Mxit Reach.

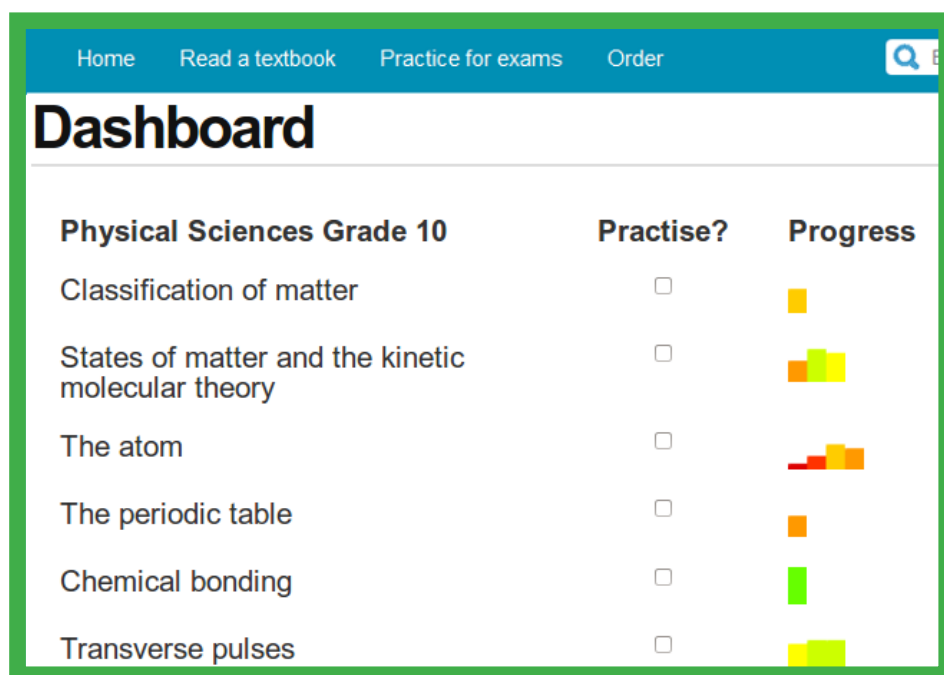


m.everythingmaths.co.za and m.everythingscience.co.za

MANAGE YOUR STUDIES

- **YOUR DASHBOARD**

If you complete your practice homework and test questions at m.everythingmaths.co.za or m.everythingscience.co.za, you can track your work. Your dashboard will show you your progress and mastery for every topic in the book and help you to manage your studies. You can use your dashboard to show your teachers, parents, universities or bursary institutions what you have done during the year.



EVERYTHING SCIENCE

When we look outside at everything in nature, look around us at everything manufactured or look up at everything in space we cannot but be struck by the incredible diversity and complexity of life; so many things, that look so different, operating in such unique ways. The physical universe really contains incredible complexity.

Yet, what is even more remarkable than this seeming complexity is the fact that things in the physical universe are knowable. We can investigate them, analyse them and understand them. It is this ability to understand the physical universe that allows us to transform elements and make technological progress possible.

If we look back at some of the things that developed over the last century – space travel, advances in medicine, wireless communication (from television to mobile phones) and materials a thousand times stronger than steel we see they are not the consequence of magic or some inexplicable phenomena. They were all developed through the study and systematic application of the physical sciences. So as we look forward at the 21st century and some of the problems of poverty, disease and pollution that face us, it is partly to the physical sciences we need to turn.

For however great these challenges seem, we know that the physical universe is knowable and that the dedicated study thereof can lead to the most remarkable advances. There can hardly be a more exciting challenge than laying bare the seeming complexity of the physical universe and working with the incredible diversity therein to develop products and services that add real quality to people's lives.

Physical sciences is far more wonderful, exciting and beautiful than magic! It is everywhere.

Contents

1	Vectors in two dimensions	4
1.1	Introduction	4
1.2	Resultant of perpendicular vectors	4
1.3	Components of vectors	36
1.4	Chapter summary	50
2	Newton's laws	56
2.1	Introduction	56
2.2	Force	56
2.3	Newton's laws	74
2.4	Forces between masses	109
2.5	Chapter summary	119
3	Atomic combinations	136
3.1	Chemical bonds	136
3.2	Molecular shape	148
3.3	Electronegativity	153
3.4	Energy and bonding	159
3.5	Chapter summary	160
4	Intermolecular forces	166
4.1	Intermolecular and interatomic forces	166
4.2	The chemistry of water	184
4.3	Chapter summary	189
5	Geometrical optics	194
5.1	Summary of properties of light	194
5.2	Light rays	194
5.3	Properties of light: revision	196
5.4	The speed of light	202
5.5	Refraction	202
5.6	Snell's Law	211
5.7	Critical angles and total internal reflection	221
5.8	Summary	229
6	2D and 3D wavefronts	234
6.1	Introduction	234
6.2	Wavefronts	234
6.3	Huygens principle	235
6.4	Diffraction	237
6.5	Diffraction through a single slit	238
6.6	Chapter summary	248
7	Ideal gases	252
7.1	Motion of particles	252
7.2	Ideal gas laws	255

7.3 Chapter summary	284
8 Quantitative aspects of chemical change	290
8.1 Gases and solutions	290
8.2 Stoichiometric calculations	298
8.3 Volume relationships in gaseous reactions	310
8.4 Chapter summary	312
9 Electrostatics	318
9.1 Introduction	318
9.2 Coulomb's law	318
9.3 Electric field	328
9.4 Chapter summary	341
10 Electromagnetism	346
10.1 Introduction	346
10.2 Magnetic field associated with a current	346
10.3 Faraday's law of electromagnetic induction	357
10.4 Chapter summary	369
11 Electric circuits	372
11.1 Introduction	372
11.2 Ohm's Law	372
11.3 Power and energy	399
11.4 Chapter summary	413
12 Energy and chemical change	418
12.1 Energy changes in chemical reactions	418
12.2 Exothermic and endothermic reactions	425
12.3 Activation energy and the activated complex	429
12.4 Chapter summary	432
13 Types of reactions	438
13.1 Acids and bases	438
13.2 Acid-base reactions	443
13.3 Redox reactions	452
13.4 Chapter summary	465
14 The lithosphere	470
14.1 Introduction	470
14.2 The lithosphere	470
14.3 Mining and mineral processing	477
14.4 Energy resources	485
14.5 Summary	486
15 Quantities used in the book	490
Solutions to exercises	491
List of Definitions	508
Image Attribution	510

Vectors in two dimensions

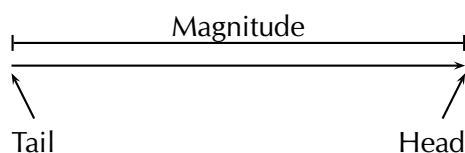
1.1	<i>Introduction</i>	4
1.2	<i>Resultant of perpendicular vectors</i>	4
1.3	<i>Components of vectors</i>	36
1.4	<i>Chapter summary</i>	50

1.1 Introduction

ESBK2

In grade 10 you learnt about vectors in one dimension. Now we will take these concepts further and learn about vectors in two dimensions as well as components of vectors.

As a very short recap, a vector has both a magnitude and a direction. There are many physical quantities, like forces, that are well described by vectors (called or known as vector quantities). We often use arrows to represent vectors visually because the length of the arrow can be related to the magnitude and the arrowhead can indicate the direction. We will talk about the head, tail and magnitude of a vector when using arrows to represent them. Below is a diagram showing a vector (the arrow). The magnitude is indicated by the length and the labels show the the tail and the head of the vector. The direction of the vector is indicated by the direction in which the arrow is pointing.



When we write the symbol for a physical quantity represented by a vector we draw an arrow over it to signify that it is a vector. If the arrow is left out then we are referring only to the magnitude of the vector quantity.

Key Mathematics Concepts

- Theorem of Pythagoras — Mathematics, Grade 10, Analytical geometry
- Units and unit conversions — Physical Sciences, Grade 10, Science skills
- Equations — Mathematics, Grade 10, Equations and inequalities
- Trigonometry — Mathematics, Grade 10, Trigonometry
- Graphs — Mathematics, Grade 10, Functions and graphs

1.2 Resultant of perpendicular vectors

ESBK3

In grade 10 you learnt about the resultant vector in one dimension, we are going to extend this to two dimensions. As a reminder, if you have a number of vectors (think forces for now) acting at the same time you can represent the result of all of them together with a single vector known as the resultant. The resultant vector will have the **same** effect as all the vectors adding together.

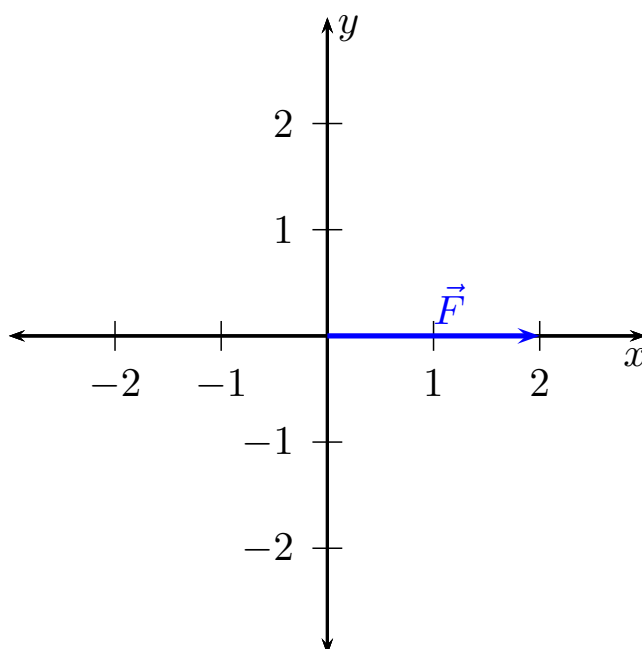
We will focus on examples involving forces but it is very **important** to remember that this applies to all physical quantities that can be described by vectors, forces, displacements, accelerations, velocities and more.

Vectors on the Cartesian plane

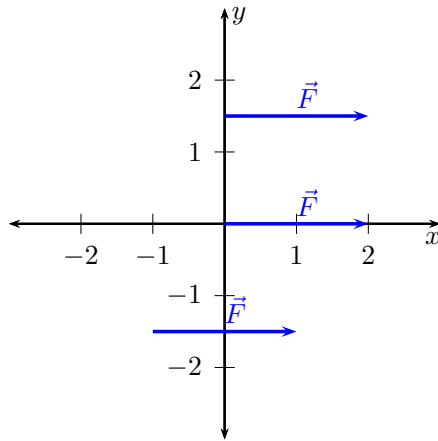
ESBK4

The first thing to make a note of is that in Grade 10 we worked with vectors all acting in a line, on a single axis. We are now going to go further and start to deal with two dimensions. We can represent this by using the Cartesian plane which consists of two perpendicular (at a right angle) axes. The axes are a x -axis and a y -axis. We normally draw the x -axis from left to right (horizontally) and the y -axis up and down (vertically).

We can draw vectors on the Cartesian plane. For example, if we have a force, $\vec{F}$, of magnitude 2 N acting in the positive x -direction we can draw it as a vector on the Cartesian plane.



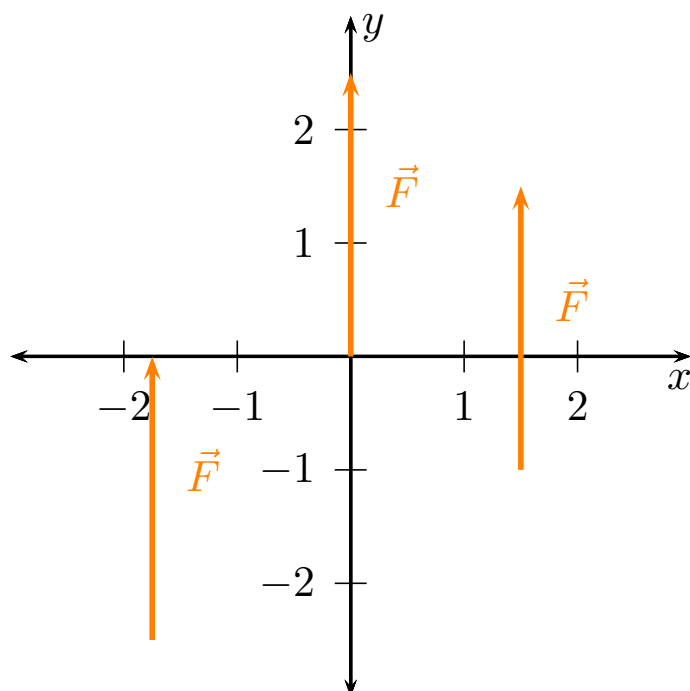
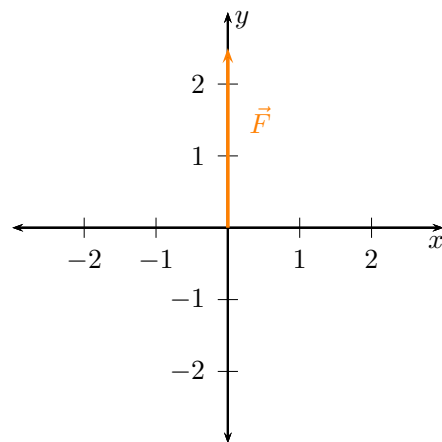
Notice that the length of the vector as measured using the axes is 2, the magnitude specified. A vector doesn't have to start at the origin but can be placed anywhere on the Cartesian plane. Where a vector starts on the plane doesn't affect the physical quantity as long as the magnitude and direction remain the same. That means that all of the vectors in the diagram below can represent the same force. This property is known as *equality of vectors*.



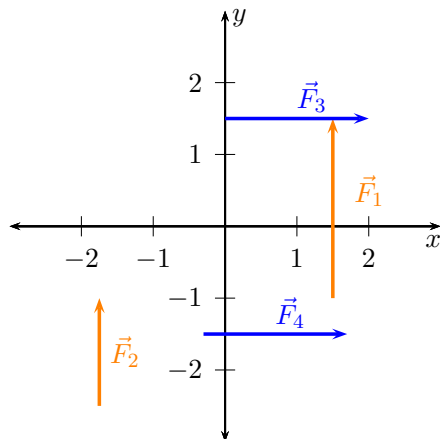
In the diagram the vectors have the same magnitude because the arrows are the same **length** and they have the same **direction**. They are all parallel to the x -direction and parallel to each other.

This applies equally in the y -direction. For example, if we have a force, $\vec{F}$, of magnitude 2,5 N acting in the positive y -direction we can draw it as a vector on the Cartesian plane.

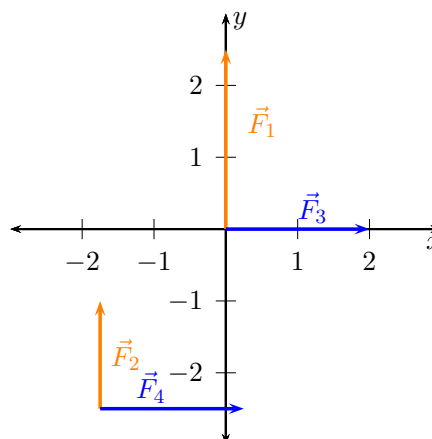
Just as in the case of the x -direction, a vector doesn't have to start at the origin but can be placed anywhere on the Cartesian plane. All of the vectors in the diagram below can represent the same force.



The following diagram shows an example of four force vectors, two vectors that are parallel to each other and the y -axis as well as two that are parallel to each other and the x -axis.



To emphasise that the vectors are perpendicular you can see in the figure below that when originating from the same point the vector are at right angles.



Exercise 1 – 1:

1. Draw the following forces as vectors on the Cartesian plane originating at the origin:

- $\vec{F}_1 = 1,5 \text{ N}$ in the positive x -direction
- $\vec{F}_2 = 2 \text{ N}$ in the positive y -direction

2. Draw the following forces as vectors on the Cartesian plane:

- $\vec{F}_1 = 3 \text{ N}$ in the positive x -direction
- $\vec{F}_2 = 1 \text{ N}$ in the negative x -direction
- $\vec{F}_3 = 3 \text{ N}$ in the positive y -direction

3. Draw the following forces as vectors on the Cartesian plane:

- $\vec{F}_1 = 3 \text{ N}$ in the positive x -direction
- $\vec{F}_2 = 1 \text{ N}$ in the positive x -direction
- $\vec{F}_3 = 2 \text{ N}$ in the negative x -direction
- $\vec{F}_4 = 3 \text{ N}$ in the positive y -direction

4. Draw the following forces as vectors on the Cartesian plane:

- $\vec{F}_1 = 2 \text{ N}$ in the positive y -direction
- $\vec{F}_2 = 1,5 \text{ N}$ in the negative y -direction
- $\vec{F}_3 = 2,5 \text{ N}$ in the negative x -direction
- $\vec{F}_4 = 3 \text{ N}$ in the positive y -direction

Think you got it? Get this answer and more practice on our Intelligent Practice Service

1. 23F3 2. 23F4 3. 23F5 4. 23F6

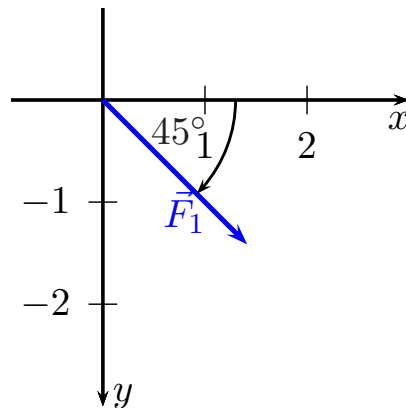
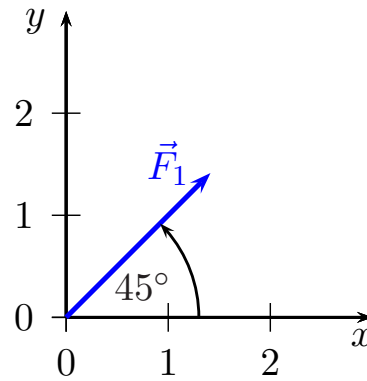


www.everythingscience.co.za



m.everythingscience.co.za

Vectors in two dimensions are not always parallel to an axis. We might know that a force acts at an angle to an axis so we still know the direction of the force and if we know the magnitude we can draw the force vector. For example, we can draw $\vec{F}_1 = 2 \text{ N}$ acting at 45° to the positive x -direction:



We always specify the angle as being anti-clockwise from the positive x -axis. So if we specified a negative angle we would measure it clockwise from the x -axis. For example, $\vec{F}_1 = 2 \text{ N}$ acting at -45° to the positive x -direction:

We can use many other ways of specifying the direction of a vector. The direction just needs to be unambiguous. We have used the Cartesian coordinate system and an angle with the x -axis so far but there are other common ways of specifying direction that you need to be aware of and comfortable to handle.

Compass directions

ESBK5

We can use compass directions when appropriate to specify the direction of a vector. For example, if we were describing the forces of tectonic plates (the sections of the earth's crust that move) to talk about the forces involved in earthquakes we could talk the force that the moving plates exert on each other.

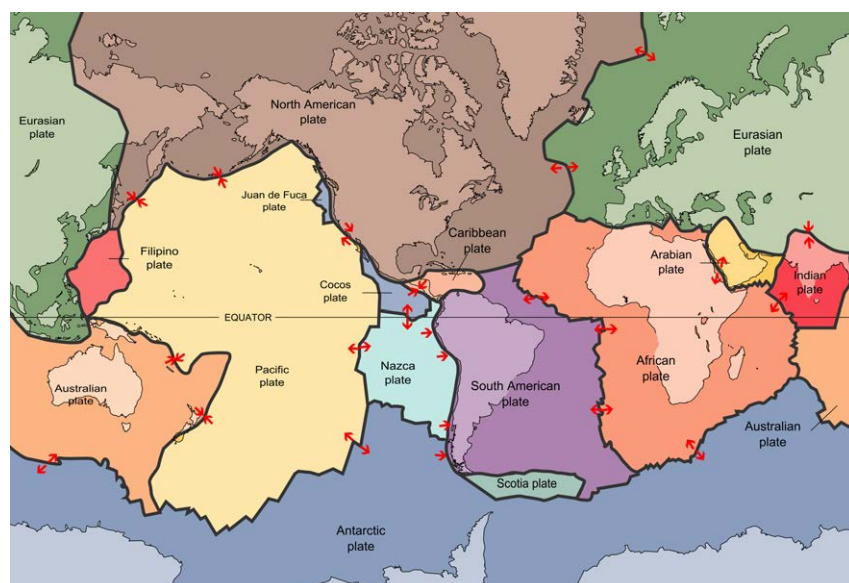


Figure 1.1: A map of the 15 major tectonic plates that make up the Earth's crust.

The four cardinal directions are North, South, East and West when using a compass. They are shown in this figure:

When specifying a direction of a vector using a compass directions are given by name, North or South. If the direction is directly between two directions we can combine the names, for example North-East is half-way between North and East. This can only happen for directions at right angles to each other, you cannot say North-South as it is ambiguous.



Figure 1.2: A sketch of the compass directions.

Bearings

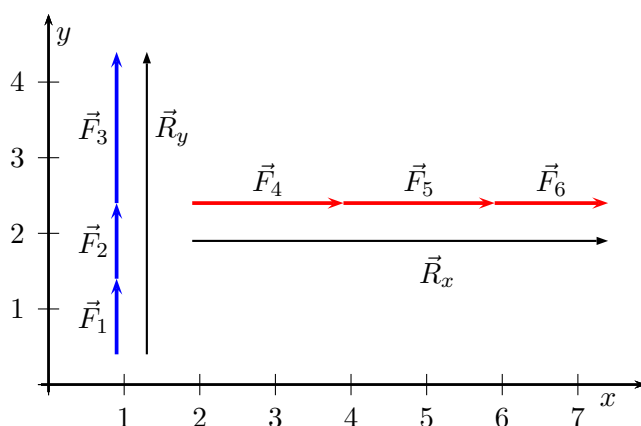
ESBK6

Another way of using the compass to specify direction in a numerical way is to use bearings. A bearing is an angle, usually measured clockwise from North. **Note** that this is different to the Cartesian plane where angles are anti- or counter-clockwise from the positive x -direction.

The resultant vector

ESBK7

In grade 10 you learnt about adding vectors together in one dimension. The same principle can be applied for vectors in two dimensions. The following examples show addition of vectors. Vectors that are parallel can be shifted to fall on a line. Vectors falling on the same line are called *co-linear* vectors. To add co-linear vectors we use the tail-to-head method you learnt in Grade 10. In the figure below we remind you of the approach of adding co-linear vectors to get a resultant vector.



In the above figure the blue vectors are in the y -direction and the red vectors are in the x -direction. The two black vectors represent the resultants of the co-linear vectors graphically.

What we have done is implement the tail-to-head method of vector addition for the vertical set of vectors and the horizontal set of vectors.

Worked example 1: Revision: head-to-tail addition in one dimension

QUESTION

Use the graphical head-to-tail method to determine the resultant force on a rugby player if two players on his team are pushing him forwards with forces of $\vec{F}_1 = 600 \text{ N}$ and $\vec{F}_2 = 900 \text{ N}$ respectively and two players from the opposing team are pushing him backwards with forces of $\vec{F}_3 = 1000 \text{ N}$ and $\vec{F}_4 = 650 \text{ N}$ respectively.

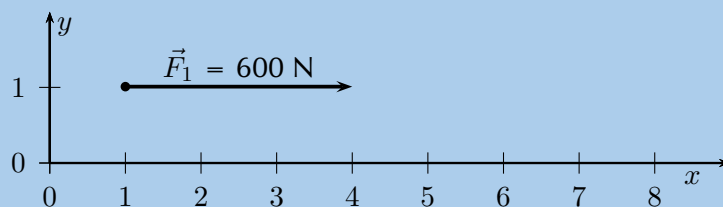
SOLUTION

Step 1: Choose a scale and a reference direction

Let's choose a scale of 100 N: 0,5 cm and for our diagram we will define the positive direction as *to the right*.

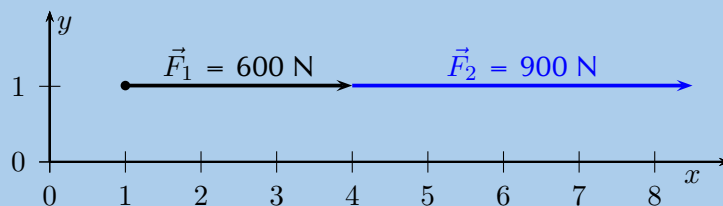
Step 2: Choose one of the vectors and draw it as an arrow of the correct length in the correct direction

We will start with drawing the vector $\vec{F}_1 = 600 \text{ N}$ pointing in the positive direction. Using our scale of 0,5 cm : 100 N, the length of the arrow must be 3 cm pointing to the right.



Step 3: Take the next vector and draw it starting at the arrowhead of the previous vector

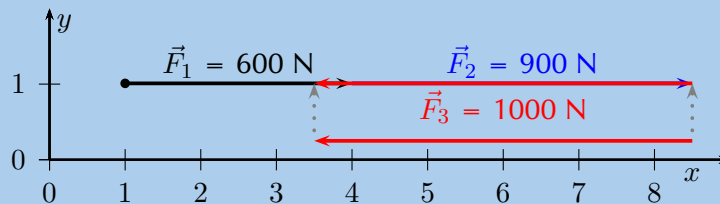
The next vector is $\vec{F}_2 = 900 \text{ N}$ in the same direction as $\vec{F}_1$. Using the scale, the arrow should be 4,5 cm long and pointing to the right.



Step 4: Take the next vector and draw it starting at the arrowhead of the previous vector

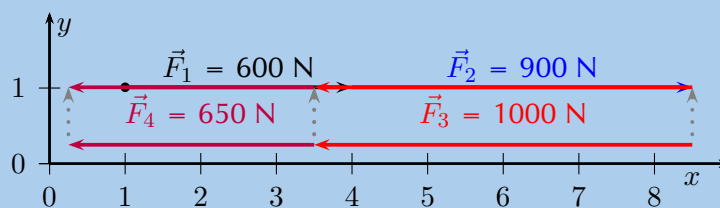
The next vector is $\vec{F}_3 = 1000 \text{ N}$ in the *opposite* direction. Using the scale, this arrow should be 5 cm long and point to the *left*.

Note: We are working in one dimension so this arrow would be drawn on top of the first vectors to the left. This will get confusing so we'll draw it next to the actual line as well to show you what it looks like.



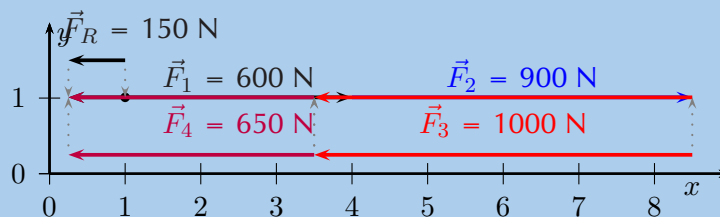
Step 5: Take the next vector and draw it starting at the arrowhead of the previous vector

The fourth vector is $\vec{F}_4 = 650 \text{ N}$ in the opposite direction. Using the scale, this arrow must be 3,25 cm long and point to the left.



Step 6: Draw the resultant, measure its length and find its direction

We have now drawn all the force vectors that are being applied to the player. The resultant vector is the arrow which starts at the tail of the first vector and ends at the head of the last drawn vector.



The resultant vector measures 0,75 cm which, using our scale is equivalent to 150 N and points to the left (*or the negative direction or the direction the opposing team members are pushing in*).

Exercise 1 – 2:

- Find the resultant in the x -direction, R_x , and y -direction, R_y for the following forces:
 - $\vec{F}_1 = 1,5 \text{ N}$ in the positive x -direction
 - $\vec{F}_2 = 1,5 \text{ N}$ in the positive x -direction
 - $\vec{F}_3 = 2 \text{ N}$ in the negative x -direction
- Find the resultant in the x -direction, $\vec{R}_x$, and y -direction, $\vec{R}_y$ for the following forces:
 - $\vec{F}_1 = 2,3 \text{ N}$ in the positive x -direction
 - $\vec{F}_2 = 1 \text{ N}$ in the negative x -direction
 - $\vec{F}_3 = 2 \text{ N}$ in the positive y -direction
 - $\vec{F}_4 = 3 \text{ N}$ in the negative y -direction
- Find the resultant in the x -direction, $\vec{R}_x$, and y -direction, $\vec{R}_y$ for the following forces:
 - $\vec{F}_1 = 3 \text{ N}$ in the positive x -direction
 - $\vec{F}_2 = 1 \text{ N}$ in the positive x -direction
 - $\vec{F}_3 = 2 \text{ N}$ in the negative x -direction
 - $\vec{F}_4 = 3 \text{ N}$ in the positive y -direction
- Find the resultant in the x -direction, $\vec{R}_x$, and y -direction, $\vec{R}_y$ for the following forces:
 - $\vec{F}_1 = 2 \text{ N}$ in the positive y -direction
 - $\vec{F}_2 = 1,5 \text{ N}$ in the negative y -direction
 - $\vec{F}_3 = 2,5 \text{ N}$ in the negative x -direction
 - $\vec{F}_4 = 3 \text{ N}$ in the positive y -direction
- Find a force in the x -direction, F_x , and y -direction, F_y , that you can add to the following forces to make the resultant in the x -direction, R_x , and y -direction, R_y zero:
 - $\vec{F}_1 = 2,4 \text{ N}$ in the positive y -direction
 - $\vec{F}_2 = 0,7 \text{ N}$ in the negative y -direction
 - $\vec{F}_3 = 2,8 \text{ N}$ in the negative x -direction
 - $\vec{F}_4 = 3,3 \text{ N}$ in the positive y -direction

Think you got it? Get this answer and more practice on our Intelligent Practice Service

1. 23F7 2. 23F8 3. 23F9 4. 23FB 5. 23FC



www.everythingscience.co.za



m.everythingscience.co.za

We apply the same principle to vectors that are at right angles or perpendicular to each other.

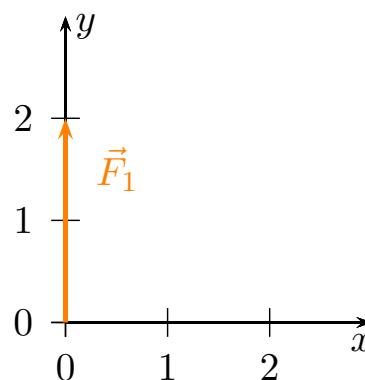
Sketching tail-to-head method

The tail of the one vector is placed at the head of the other but in two dimensions the vectors may not be co-linear. The approach is to draw all the vectors, one at a time. For the first vector begin at the origin of the Cartesian plane, for the second vector draw it from the head of the first vector. The third vector should be drawn from the head of the second and so on. Each vector is drawn from the head of the vector that preceded it. The order doesn't matter as the resultant will be the same if the order is different.

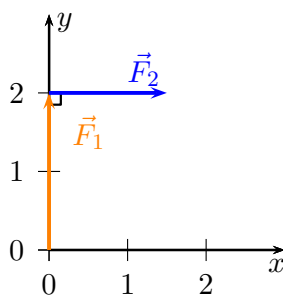
Let us apply this procedure to two vectors:

- $\vec{F}_1 = 2 \text{ N}$ in the positive y -direction
- $\vec{F}_2 = 1,5 \text{ N}$ in the positive x -direction

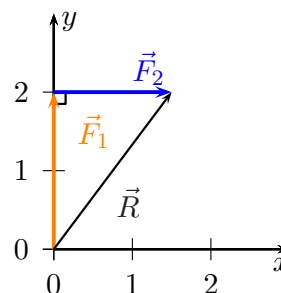
First, we first draw a Cartesian plane with the first vector originating at the origin as shown on the right.



The second step is to take the second vector and draw it from the head of the first vector:

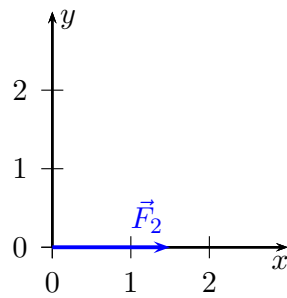


The resultant, $\vec{R}$, is the vector connecting the tail of the first vector drawn to the head of the last vector drawn:

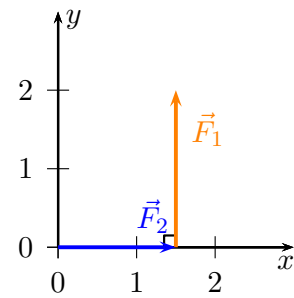


It is important to remember that the order in which we draw the vectors doesn't matter. If we had drawn them in the opposite order we would have the same resultant, $\vec{R}$. We can repeat the process to demonstrate this:

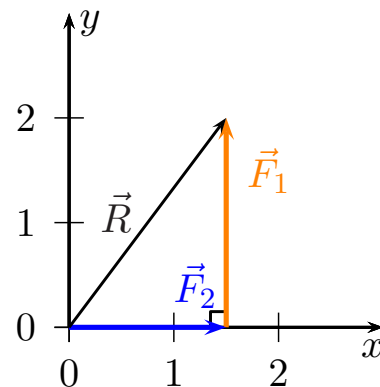
We first draw a Cartesian plane with the second vector originating at the origin:



The next step is to take the other vector and draw it from the head of the vector we have already drawn:



The resultant, $\vec{R}$, is the vector connecting the tail of the first vector drawn to the head of the last vector drawn (the vector from the start point to the end point):



Worked example 2: Sketching vectors using tail-to-head

QUESTION

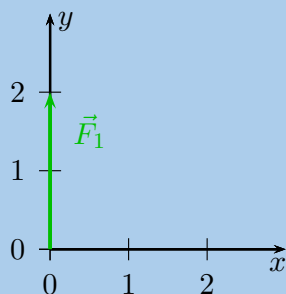
Sketch the resultant of the following force vectors using the tail-to-head method:

- $\vec{F}_1 = 2 \text{ N}$ in the positive y -direction
- $\vec{F}_2 = 1,5 \text{ N}$ in the positive x -direction
- $\vec{F}_3 = 1,3 \text{ N}$ in the negative y -direction
- $\vec{F}_4 = 1 \text{ N}$ in the negative x -direction

SOLUTION

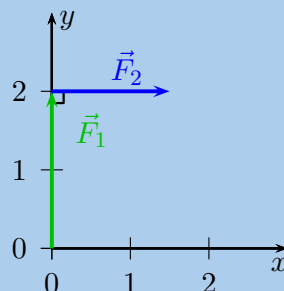
Step 1: Draw the Cartesian plane and the first vector

First draw the Cartesian plane and force, $\vec{F}_1$ starting at the origin:



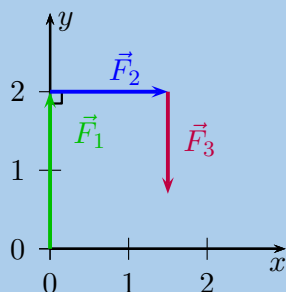
Step 2: Draw the second vector

Starting at the head of the first vector we draw the tail of the second vector:



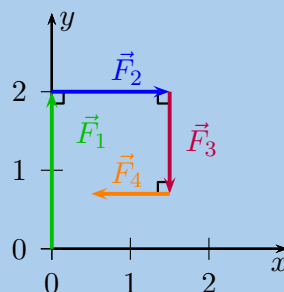
Step 3: Draw the third vector

Starting at the head of the second vector we draw the tail of the third vector:



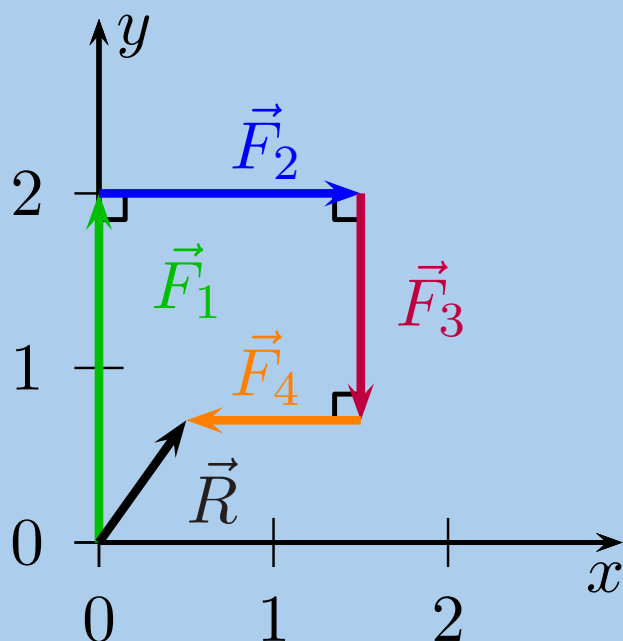
Step 4: Draw the fourth vector

Starting at the head of the third vector we draw the tail of the fourth vector:



Step 5: Draw the resultant vector

Starting at the origin draw the resultant vector to the head of the fourth vector:



Worked example 3: Sketching vectors using tail-to-head

QUESTION

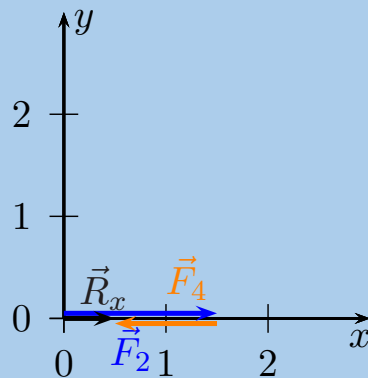
Sketch the resultant of the following force vectors using the tail-to-head method by first determining the resultant in the x - and y -directions:

- $\vec{F}_1 = 2 \text{ N}$ in the positive y -direction
- $\vec{F}_2 = 1,5 \text{ N}$ in the positive x -direction
- $\vec{F}_3 = 1,3 \text{ N}$ in the negative y -direction
- $\vec{F}_4 = 1 \text{ N}$ in the negative x -direction

SOLUTION

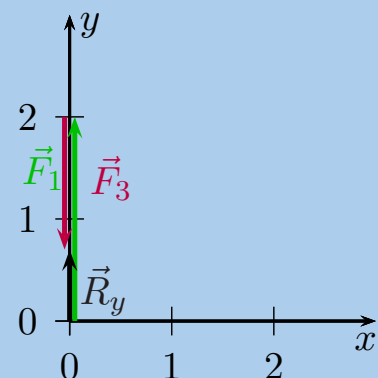
Step 1: First determine $\vec{R}_x$

First draw the Cartesian plane with the vectors in the x -direction:

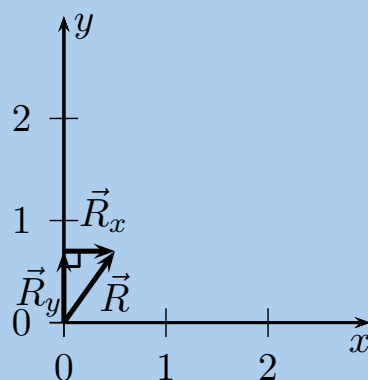


Step 2: Secondly determine $\vec{R}_y$

Next we draw the Cartesian plane with the vectors in the y -direction:

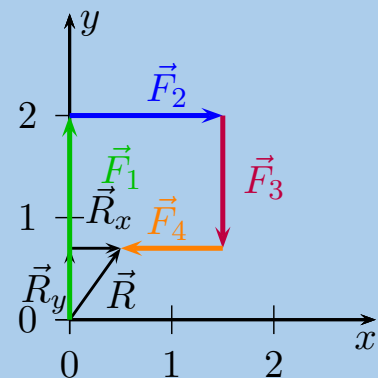


Step 3: Draw the resultant vectors, $\vec{R}_y$ and $\vec{R}_x$ head-to-tail



Step 4: Comparison of results

To double check, we can replot all the vectors again as we did in the previous worked example to see that the outcome is the same:



Exercise 1 – 3:

1. Sketch the resultant of the following force vectors using the tail-to-head method:

- $\vec{F}_1 = 2,1 \text{ N}$ in the positive y -direction
- $\vec{F}_2 = 1,5 \text{ N}$ in the negative x -direction

2. Sketch the resultant of the following force vectors using the tail-to-head method:

- $\vec{F}_1 = 12 \text{ N}$ in the positive y -direction
- $\vec{F}_2 = 10 \text{ N}$ in the positive x -direction
- $\vec{F}_3 = 5 \text{ N}$ in the negative y -direction
- $\vec{F}_4 = 5 \text{ N}$ in the negative x -direction

3. Sketch the resultant of the following force vectors using the tail-to-head method by first determining the resultant in the x - and y -directions:

- $\vec{F}_1 = 2 \text{ N}$ in the positive y -direction
- $\vec{F}_2 = 1,5 \text{ N}$ in the negative y -direction
- $\vec{F}_3 = 1,3 \text{ N}$ in the negative y -direction
- $\vec{F}_4 = 1 \text{ N}$ in the negative x -direction

4. Sketch the resultant of the following force vectors using the tail-to-head method by first determining the resultant in the x - and y -directions:

- $\vec{F}_1 = 6 \text{ N}$ in the positive y -direction
- $\vec{F}_2 = 3,5 \text{ N}$ in the negative x -direction
- $\vec{F}_3 = 8,7 \text{ N}$ in the negative y -direction
- $\vec{F}_4 = 3 \text{ N}$ in the negative y -direction

Think you got it? Get this answer and more practice on our Intelligent Practice Service

1. 23FD 2. 23FF 3. 23FG 4. 23FH



www.everythingscience.co.za



m.everythingscience.co.za

FACT

When dealing with more than two vectors the procedure is repetitive. First find the resultant of any two of the vectors to be added. Then use the same method to add the resultant from the first two vectors with a third vector. This new resultant is then added to the fourth vector and so on, until there are no more vectors to be added.

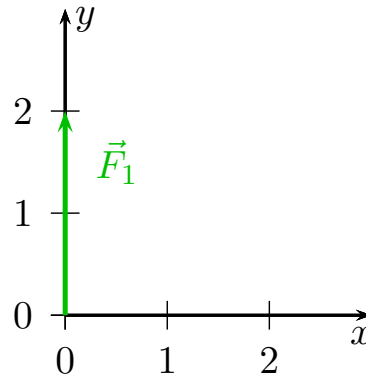
Sketching tail-to-tail method

In this method we draw the two vectors with their tails on the origin. Then we draw a line parallel to the first vector from the head of the second vector and vice versa. Where the parallel lines intersect is the head of the resultant vector that will also start at the origin. We will only deal with perpendicular vectors but this procedure works for any vectors.

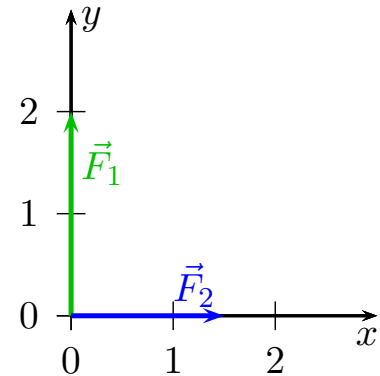
Let us apply this procedure to the same two vectors we used to illustrate the head-to-tail method:

- $\vec{F}_1 = 2 \text{ N}$ in the positive y -direction
- $\vec{F}_2 = 1,5 \text{ N}$ in the positive x -direction

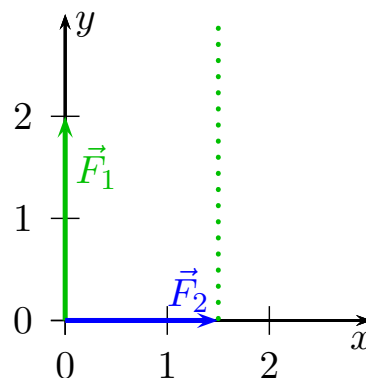
1. We first draw a Cartesian plane with the first vector originating at the origin:



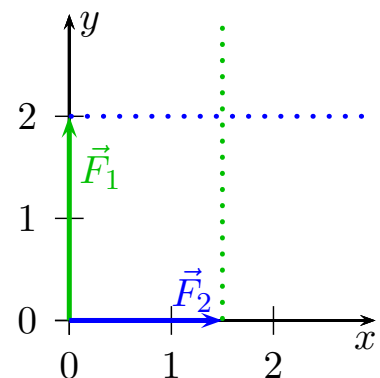
2. Then we add the second vector but also originating from the origin so that the vectors are drawn tail-to-tail:



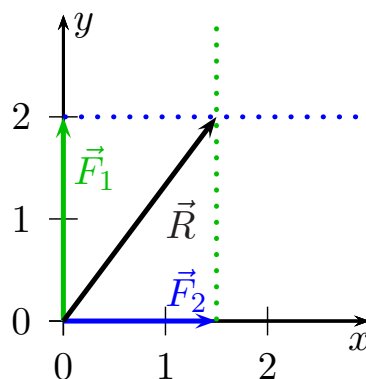
3. Now we draw a line parallel to $\vec{F}_1$ from the head of $\vec{F}_2$:



4. Next we draw a line parallel to $\vec{F}_2$ from the head of $\vec{F}_1$:



5. Where the two lines intersect is the head of the resultant vector which will originate at the origin so:



You might be asking what you would do if you had more than 2 vectors to add together. In this case all you need to do is first determine $\vec{R}_x$ by adding all the vectors that are parallel to the x -direction and $\vec{R}_y$ by adding all the vectors that are parallel to the y -direction. Then you use the tail-to-tail method to find the resultant of $\vec{R}_x$ and $\vec{R}_y$.

Exercise 1 – 4:

1. Sketch the resultant of the following force vectors using the tail-to-tail method:

- $\vec{F}_1 = 2,1$ N in the positive y -direction
- $\vec{F}_2 = 1,5$ N in the negative x -direction

2. Sketch the resultant of the following force vectors using the tail-to-tail method by first determining the resultant in the x - and y -directions:

- $\vec{F}_1 = 2$ N in the positive y -direction
- $\vec{F}_2 = 1,5$ N in the negative y -direction
- $\vec{F}_3 = 1,3$ N in the negative y -direction
- $\vec{F}_4 = 1$ N in the negative x -direction

3. Sketch the resultant of the following force vectors using the tail-to-tail method by first determining the resultant in the x - and y -directions:

- $\vec{F}_1 = 6$ N in the positive y -direction
- $\vec{F}_2 = 3,5$ N in the negative x -direction
- $\vec{F}_3 = 8,7$ N in the negative y -direction
- $\vec{F}_4 = 3$ N in the negative y -direction

Think you got it? Get this answer and more practice on our Intelligent Practice Service

1. [23FJ](#) 2. [23FK](#) 3. [23FM](#)



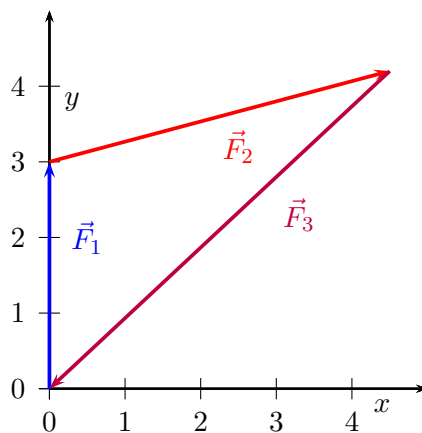
www.everythingscience.co.za



m.everythingscience.co.za

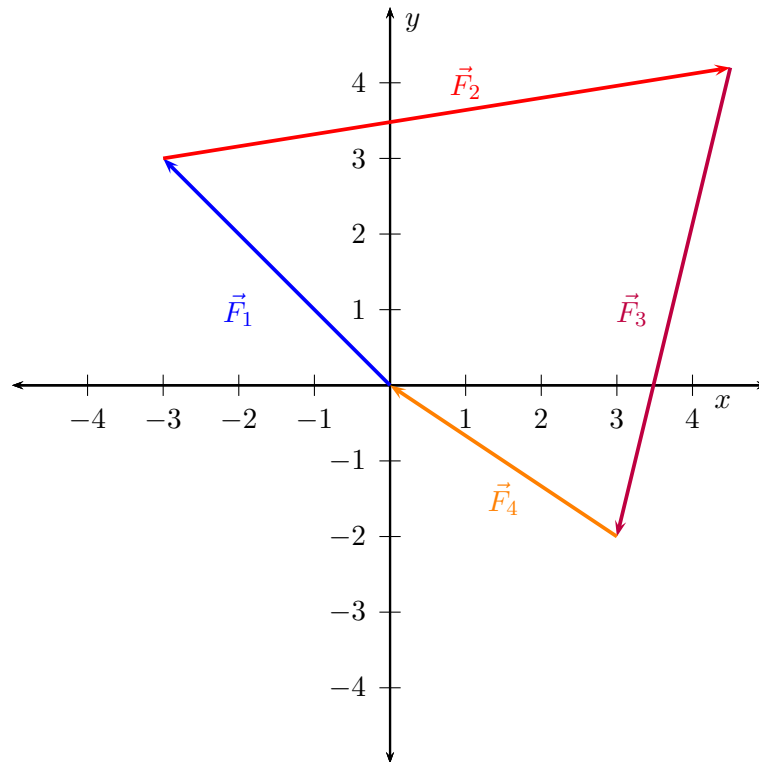
Closed vector diagrams

A closed vector diagram is a set of vectors drawn on the Cartesian using the tail-to-head method and that has a resultant with a magnitude of zero. This means that if the first vector starts at the origin the last vector drawn must end at the origin. The vectors form a closed polygon, no matter how many of them are drawn. Here are a few examples of closed vector diagrams:



In this case there were 3 force vectors. When drawn tail-to-head with the first force starting at the origin the last force drawn ends at the origin. The resultant would have a magnitude of zero. The resultant is drawn from the tail of the first vector to the head of the final vector.

In the diagram below there are 4 vectors that also form a closed vector diagram.

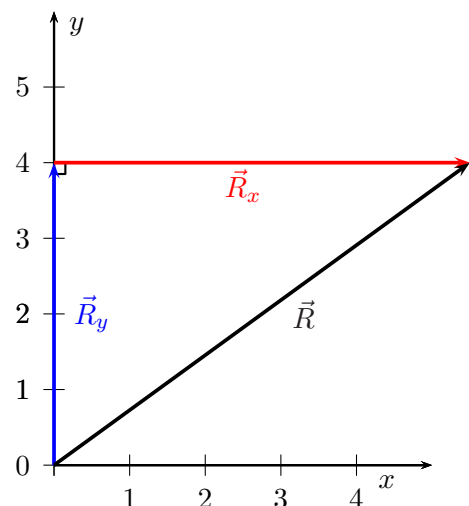


In this case with 4 vectors, the shape is a 4-sided polygon. Any polygon made up of vectors drawn tail-to-head will be a closed vector diagram because a polygon has no gaps.

Using Pythagoras' theorem to find magnitude

If we wanted to know the resultant of the three blue vectors and the three red vectors in Figure 1.2 we can use the resultant vectors in the x - and y -directions to determine this.

The black arrow represents the resultant of the vectors $\vec{R}_x$ and $\vec{R}_y$. We can find the magnitude of this vector using the theorem of Pythagoras because the three vectors form a right angle triangle. If we had drawn the vectors to scale we would be able to measure the magnitude of the resultant as well.



What we've actually sketched out already is our approach to finding the resultant of many vectors using components so remember this example when we get there a little later.

Worked example 4: Finding the magnitude of the resultant

QUESTION

The force vectors in Figure 1.2 have the following magnitudes: 1 N, 1 N, 2 N for the blue ones and 2 N, 2 N and 1,5 N for the red ones. Determine the magnitude of the resultant.

SOLUTION

Step 1: Determine the resultant of the vectors parallel to the y -axis

The resultant of the vectors parallel to the y -axis is found by adding the magnitudes (lengths) of three vectors because they all point in the same direction. The answer is $\vec{R}_y = 1\text{ N} + 1\text{ N} + 2\text{ N} = 4\text{ N}$ in the positive y -direction.

Step 2: Determine the resultant of the vectors parallel to the x -axis

The resultant of the vectors parallel to the x -axis is found by adding the magnitudes (lengths) of three vectors because they all point in the same direction. The answer is $\vec{R}_x = 2\text{ N} + 2\text{ N} + 1,5\text{ N} = 5,5\text{ N}$ in the positive x -direction.

Step 3: Determine the magnitude of the resultant

We have a right angled triangle. We also know the length of two of the sides. Using Pythagoras we can find the length of the third side. From what we know about resultant vectors this length will be the magnitude of the resultant vector.

The resultant is:

$$\begin{aligned}R_x^2 + R_y^2 &= R^2 \text{ (Pythagoras' theorem)} \\(5,5)^2 + (4)^2 &= R^2 \\R &= 6,8\end{aligned}$$

Step 4: Quote the final answer

Magnitude of the resultant: 6,8 N

Note: we did not determine the resultant vector in the worked example above because we only determined the magnitude. A vector needs a **magnitude** and a **direction**. We did not determine the direction of the resultant vector.

Graphical techniques

In grade 10 you learnt how to add vectors in one dimension graphically.

We can expand these ideas to include vectors in two-dimensions. The following worked example shows this.

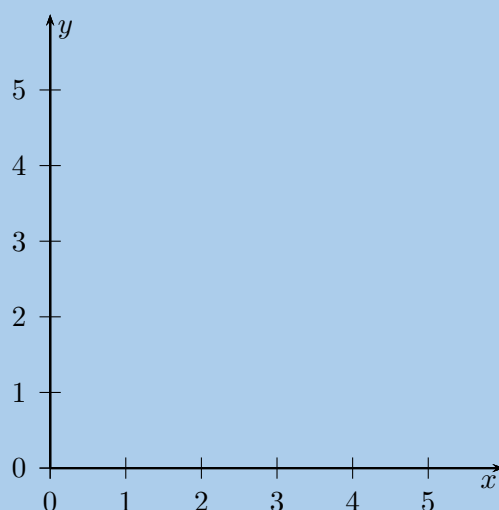
Worked example 5: Finding the magnitude of the resultant in two dimensions graphically
QUESTION

Given two vectors, $\vec{R}_y = 4 \text{ N}$ in the positive y -direction and $\vec{R}_x = 3 \text{ N}$ in the positive x -direction, use the tail-to-head method to find the resultant of these vectors graphically.

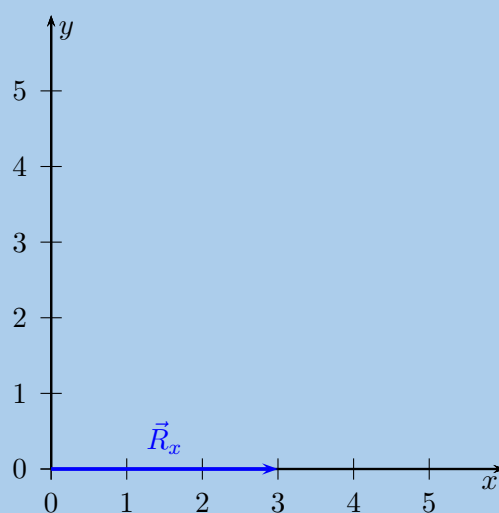
SOLUTION
Step 1: Choose a scale and draw axes

The vectors we have do not have very big magnitudes so we can choose simple scale, we can use $1 \text{ N} : 1 \text{ cm}$ as our scale for the drawing.

Then we draw axes that the vector diagram should fit in. The largest vector has length 4 N and both vectors are in the positive direction so we can draw axes from the origin to 5 and expect the vectors to fit.

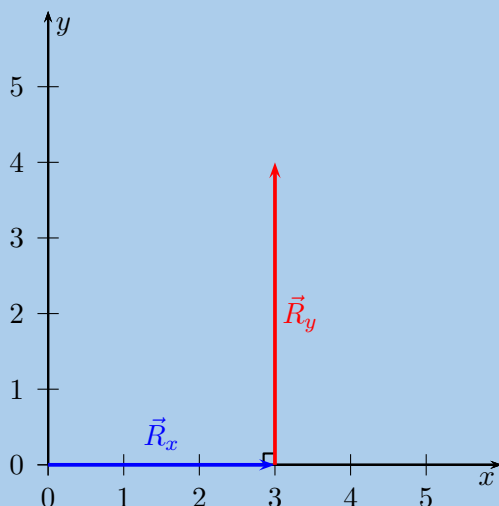

Step 2: Draw $\vec{R}_x$

The magnitude of $\vec{R}_x$ is 3 N so the arrow we need to draw must be 3 cm long. The arrow must point in the positive x -direction.

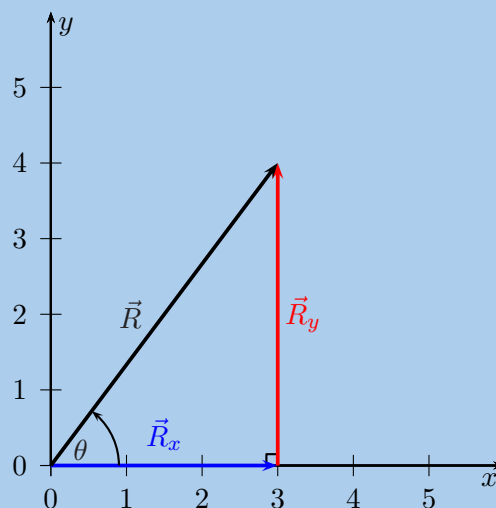


Step 3: Draw $\vec{R}_y$

The length of $\vec{R}_y$ is 4 so the arrow we need to draw must be 4 cm long. The arrow must point in the positive y -direction. The important fact to note is that we are implementing the head-to-tail method so the vector must start at the end (head) of $\vec{R}_x$.

**Step 4: Draw the resultant vector, $\vec{R}$**

The resultant vector is the vector from the tail of the first vector we drew directly to the head of the last vector we drew. This means we need to draw a vector from the tail of $\vec{R}_x$ to the head of $\vec{R}_y$.

**Step 5: Measure the resultant, $\vec{R}$**

We are solving the problem graphically so we now need to measure the magnitude of the vector and use the scale we chose to convert our answer from the diagram to the actual result. In the last diagram the resultant, $\vec{R}$ is 5 cm long therefore the magnitude of the vector is 5 N.

The direction of the resultant, θ , we need to measure from the diagram using a protractor. The angle that the vector makes with the x -axis is 53° .

Step 6: Quote the final answer

$\vec{R}$ is 5 N at 53° from the positive x -direction.

In the case where you have to find the resultant of more than two vectors first apply the tail-to-head method to all the vectors parallel to the one axis and then all the vectors parallel to the other axis. For example, you would first calculate $\vec{R}_y$ from all the vectors parallel to the y -axis and then $\vec{R}_x$ from all the vectors parallel to the x -axis. After that you apply the same procedure as in the previous worked example to get the final resultant.

Worked example 6: Finding the magnitude of the resultant in two dimensions graphically

QUESTION

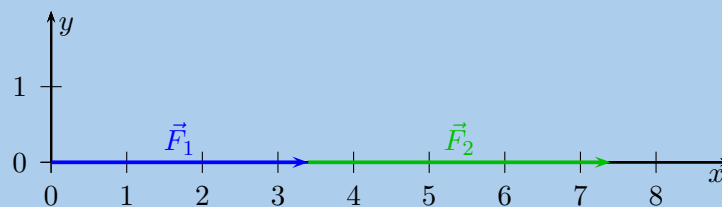
Given the following three force vectors, determine the resultant force:

- $\vec{F}_1 = 3,4 \text{ N}$ in the positive x -direction
- $\vec{F}_2 = 4 \text{ N}$ in the positive x -direction
- $\vec{F}_3 = 3 \text{ N}$ in the negative y -direction

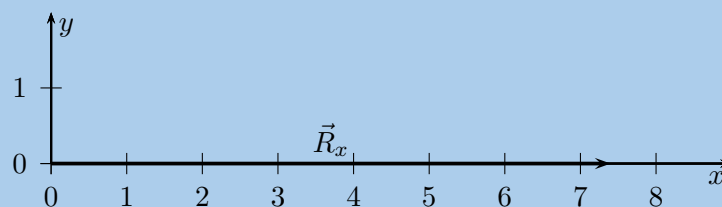
SOLUTION

Step 1: Determine $\vec{R}_x$

First we determine the resultant of all the vectors that are parallel to the x -axis. There are two vectors $\vec{F}_1$ and $\vec{F}_2$ that we need to add. We do this using the tail-to-head method for co-linear vectors.



The single vector, $\vec{R}_x$, that would give us the same outcome is:



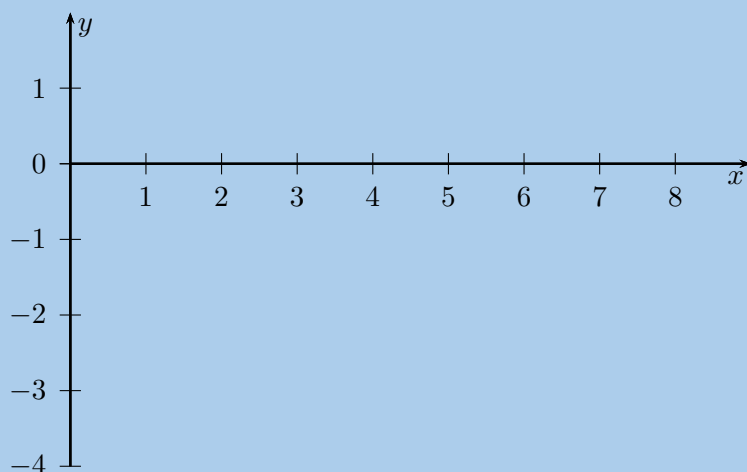
Step 2: Determine $\vec{R}_y$

There is only one vector in the y -direction, $\vec{F}_3$, therefore $\vec{R}_y = \vec{F}_3$.

Step 3: Choose a scale and draw axes

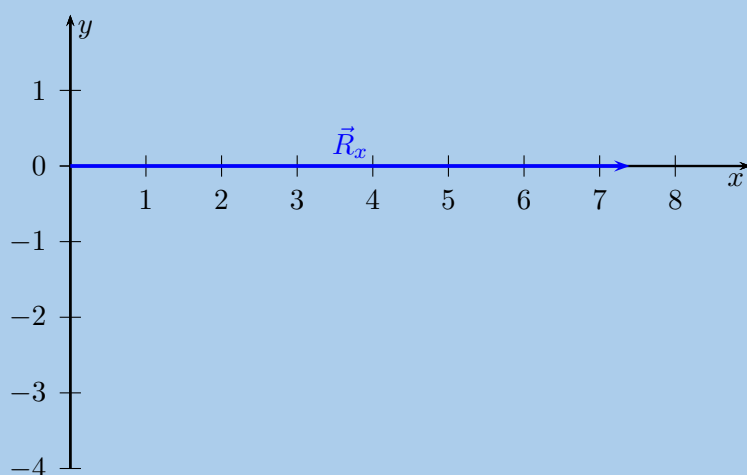
The vectors we have do not have very big magnitudes so we can choose simple scale, we can use $1 \text{ N} : 1 \text{ cm}$ as our scale for the drawing.

Then we draw axes that the diagram should fit on. The longest vector has length $7,4 \text{ N}$. We need our axes to extend just further than the vectors aligned with each axis. Our axes need to start at the origin and go beyond $7,4 \text{ N}$ in the positive x -direction and further than 3 N in the negative y -direction. Our scale choice of $1 \text{ N} : 1 \text{ cm}$ means that our axes actually need to extend $7,4 \text{ cm}$ in the positive x -direction and further than 3 cm in the negative y -direction



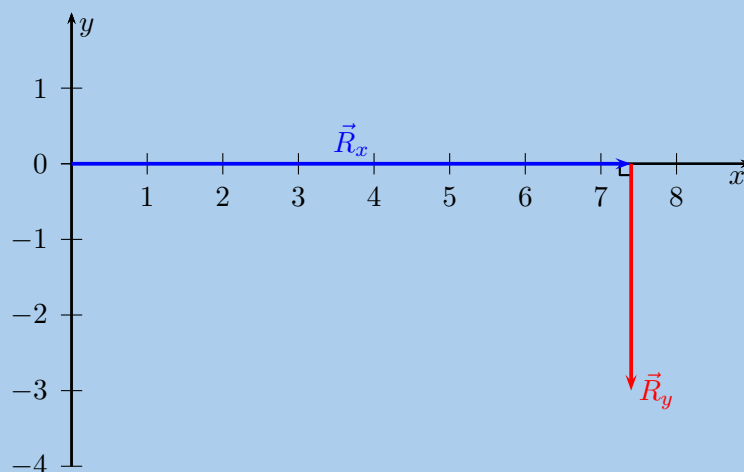
Step 4: Draw $\vec{R}_x$

The magnitude of $\vec{R}_x$ is 7,4 N so the arrow we need to draw must be 7,4 cm long. The arrow must point in the positive x -direction.



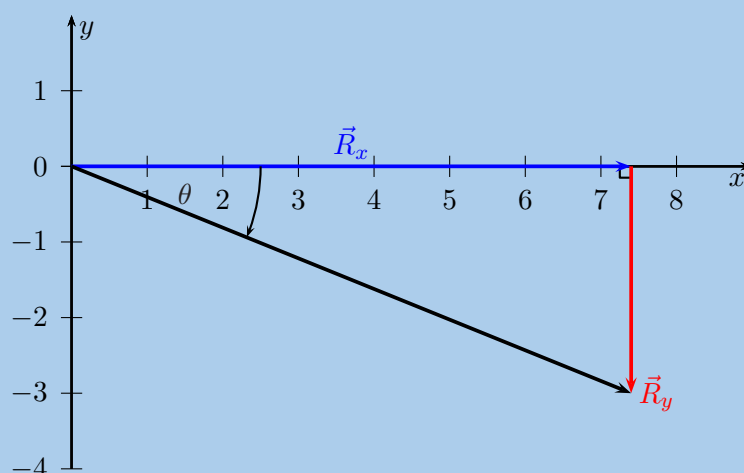
Step 5: Draw $\vec{R}_y$

The magnitude of $\vec{R}_y$ is 3 N so the arrow we need to draw must be 3 cm long. The arrow must point in the negative y -direction. The important fact to note is that we are implementing the head-to-tail method so the vector must start at the end (head) of $\vec{R}_x$.



Step 6: Draw the resultant vector, $\vec{R}$

The resultant vector is the vector from the tail of the first vector we drew directly to the head of the last vector we drew. This means we need to draw a vector from the tail of $\vec{R}_x$ to the head of $\vec{R}_y$.



Step 7: Measure the resultant, $\vec{R}$

We are solving the problem graphically so we now need to measure the magnitude of the vector and use the scale we chose to convert our answer from the diagram to the actual result. In the last diagram the resultant, $\vec{R}$ is 8,0 cm long therefore the magnitude of the vector is 8,0 N.

The direction of the resultant we need to measure from the diagram using a protractor. The angle that the vector makes with the x -axis is 22° .

Step 8: Quote the final answer

$\vec{R}$ is 8,0 N at -22° from the positive x -direction.

QUESTION

Given the following three force vectors, determine the resultant force:

- $\vec{F}_1 = 2,3 \text{ N}$ in the positive x -direction
- $\vec{F}_2 = 4 \text{ N}$ in the positive y -direction
- $\vec{F}_3 = 3,3 \text{ N}$ in the negative y -direction
- $\vec{F}_4 = 2,1 \text{ N}$ in the negative y -direction

SOLUTION

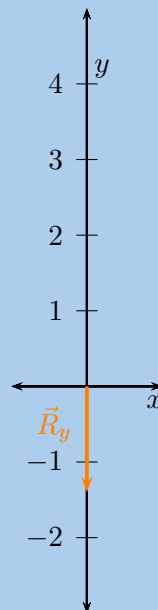
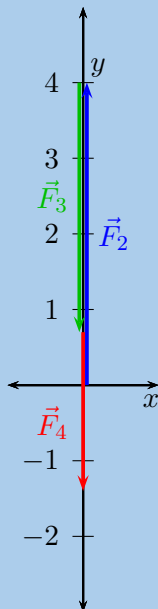
Step 1: Determine $\vec{R}_x$

There is only one vector in the x -direction, $\vec{F}_1$, therefore $\vec{R}_x = \vec{F}_1$.

Step 2: Determine $\vec{R}_y$

Then we determine the resultant of all the vectors that are parallel to the y -axis. There are three vectors $\vec{F}_2$, $\vec{F}_3$ and $\vec{F}_4$ that we need to add. We do this using the tail-to-head method for co-linear vectors.

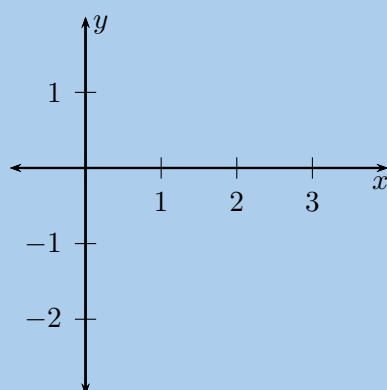
The single vector, $\vec{R}_y$, that would give us the same effect is:



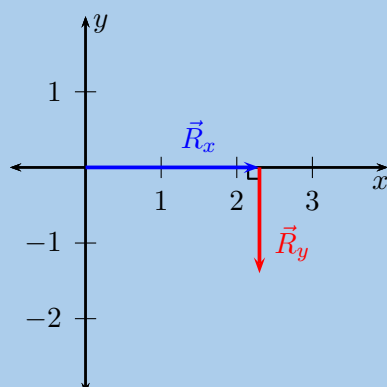
Step 3: Choose a scale and draw axes

We choose a scale 1 N : 1 cm for the drawing.

Then we draw axes that the diagram should fit in. We need our axes to extend just further than the vectors aligned with each axis. Our axes need to start at the origin and go beyond 2,3 N in the positive x -direction and further than 1,4 N in the negative y -direction. Our scale choice of 1 N = 1 cm means that our axes actually need to extend 2,3 cm in the positive x -direction and further than 1,4 cm in the negative y -direction

**Step 5: Draw $\vec{R}_y$**

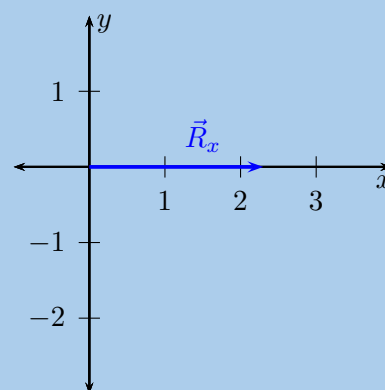
The magnitude of $\vec{R}_y$ is 1,4 N so the arrow we need to draw must be 1,4 cm long. The arrow must point in the negative y -direction. The important fact to note is that we are implementing the head-to-tail method so the vector must start at the end (head) of $\vec{R}_x$.

**Step 7: Measure the resultant, $\vec{R}$**

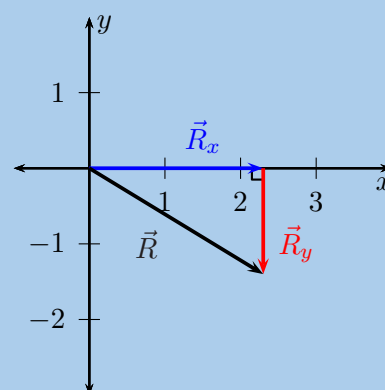
We are solving the problem graphically so we now need to measure the magnitude of the vector and use the scale we chose to convert our answer from the diagram to the magnitude of the vector. In the last diagram the resultant, $\vec{R}$ is 2,7 cm long therefore the magnitude of the vector is 2,7 N.

Step 4: Draw $\vec{R}_x$

The magnitude of $\vec{R}_x$ is 2,3 N so the arrow we need to draw must be 2,3 cm long. The arrow must point in the positive x -direction.

**Step 6: Draw the resultant vector, $\vec{R}$**

The resultant vector is the vector from the tail of the first vector we drew directly to the head of the last vector we drew. This means we need to draw a vector from the tail of $\vec{R}_x$ to the head of $\vec{R}_y$.



The direction of the resultant we need to measure from the diagram using a protractor. The angle that the vector makes with the x -axis is 31 degrees.

Step 8: Quote the final answer

$\vec{R}$ is 2,7 N at -31° from the positive x -direction.

Worked example 8: Finding the resultant in two dimensions graphically

QUESTION

A number of tugboats are trying to manoeuvre a submarine in the harbour but they are not working as a team. Each tugboat is exerting a different force on the submarine.



Given the following force vectors, determine the resultant force:

- $\vec{F}_1 = 3,4 \text{ kN}$ in the positive x -direction
- $\vec{F}_2 = 4000 \text{ N}$ in the positive y -direction
- $\vec{F}_3 = 300 \text{ N}$ in the negative y -direction
- $\vec{F}_4 = 7 \text{ kN}$ in the negative y -direction

SOLUTION

Step 1: Convert to consistent S.I. units

To use the graphical method of finding the resultant we need to work in the same units. Strictly speaking in this problem all the vectors are in newtons but they have different factors which will affect the choice of scale. These need to be taken into account and the simplest approach is to convert them all to a consistent unit and factor. We could use kN or N, the choice does not matter. We will choose kN. Remember that k represents a factor of $\times 10^3$.

$\vec{F}_1$ and $\vec{F}_4$ do not require any adjustment because they are both in kN. To convert N to kN we use:

$$\begin{aligned} \text{kN} &= \times 10^3 \\ \frac{\text{N}}{\text{kN}} &= \frac{1}{\times 10^3} \\ \text{N} &= \times 10^{-3} \text{ kN} \end{aligned}$$

To convert the magnitude of $\vec{F}_2$ to kN:

$$\begin{aligned} F_2 &= 4000 \text{ N} \\ F_2 &= 4000 \times 10^{-3} \text{ kN} \\ F_2 &= 4 \text{ kN} \end{aligned}$$

Therefore $\vec{F}_2 = 4 \text{ kN}$ in the positive y -direction.

To convert the magnitude of $\vec{F}_3$ to kN:

$$\begin{aligned} F_3 &= 300 \text{ N} \\ F_3 &= 300 \times 10^{-3} \text{ kN} \\ F_3 &= 0,3 \text{ kN} \end{aligned}$$

Therefore $\vec{F}_3 = 0,3 \text{ kN}$ in the negative y -direction. So:

- $\vec{F}_1 = 3,4 \text{ kN}$ in the positive x -direction
- $\vec{F}_2 = 4 \text{ kN}$ in the positive y -direction
- $\vec{F}_3 = 0,3 \text{ kN}$ in the negative y -direction
- $\vec{F}_4 = 7 \text{ kN}$ in the negative y -direction

Step 2: Choose a scale and draw axes

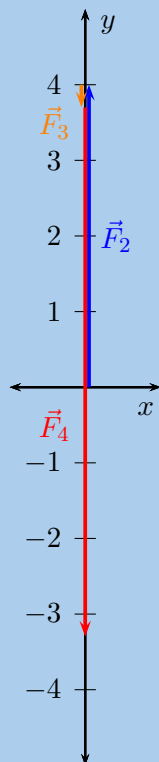
The vectors we have do have very big magnitudes so we need to choose a scale that will allow us to draw them in a reasonable space, we can use 1 kN : 1 cm as our scale for the drawings.

Step 3: Determine $\vec{R}_x$

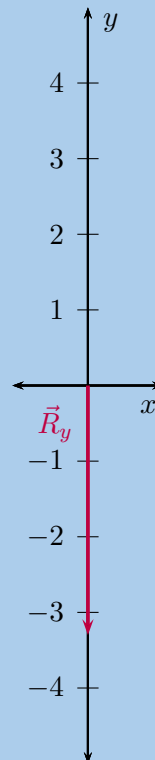
There is only one vector in the x -direction, $\vec{F}_1$, therefore $\vec{R}_x = \vec{F}_1$.

Step 4: Determine $\vec{R}_y$

Then we determine the resultant of all the vectors that are parallel to the y -axis. There are three vectors $\vec{F}_2$, $\vec{F}_3$ and $\vec{F}_4$ that we need to add. We do this using the tail-to-head method for co-linear vectors.

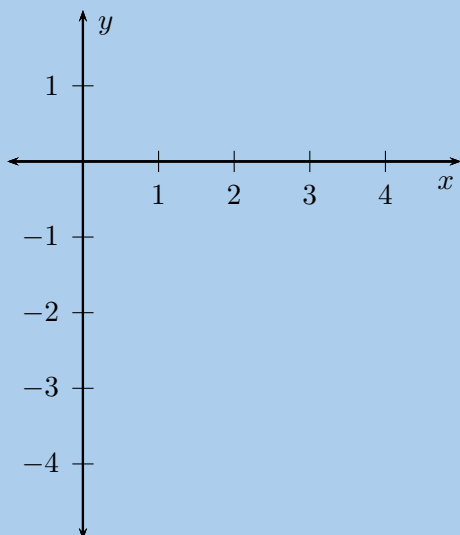


The single vector, $\vec{R}_y$, that would give us the same outcome is:



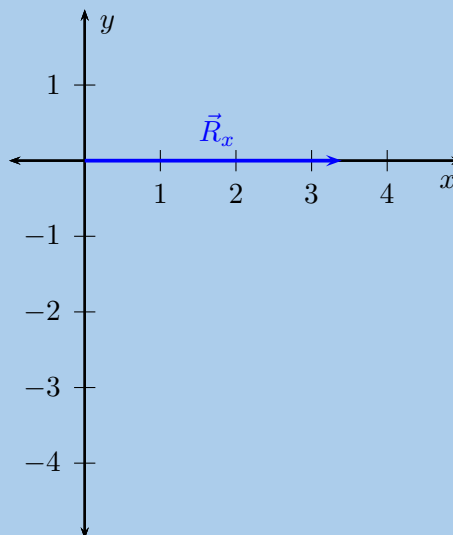
Step 5: Draw axes

Then we draw axes that the diagram should fit on. We need our axes to extend just further than the vectors aligned with each axis. Our axes need to start at the origin and go beyond 3,4 kN in the positive x -direction and further than 3,3 kN in the negative y -direction. Our scale choice of 1 kN : 1 cm means that our axes actually need to extend 3,4 cm in the positive x -direction and further than 3,3 cm in the negative y -direction



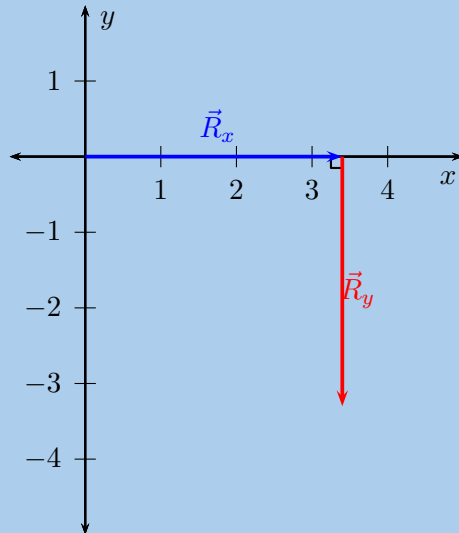
Step 6: Draw $\vec{R}_x$

The length of $\vec{R}_x$ is 3,4 kN so the arrow we need to draw must be 3,4 cm long. The arrow must point in the positive x -direction.

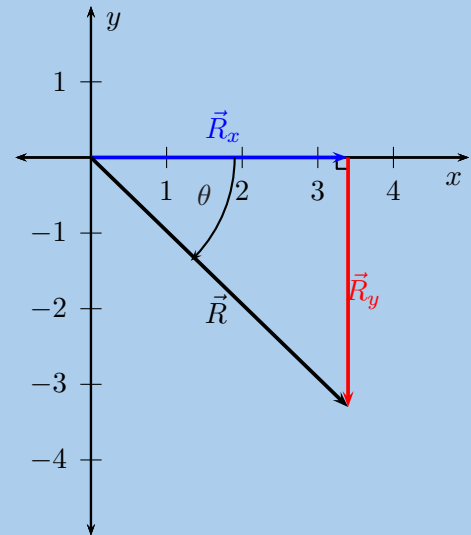


Step 7: Draw $\vec{R}_y$

The length of $\vec{R}_y$ is 3,3 kN so the arrow we need to draw must be 3,3 cm long. The arrow must point in the negative y -direction. The important fact to note is that we are implementing the head-to-tail method so the vector must start at the end (head) of $\vec{R}_x$.

**Step 8: Draw the resultant vector, $\vec{R}$**

The resultant vector is the vector from the tail of the first vector we drew directly to the head of the last vector we drew. This means we need to draw a vector from the tail of $\vec{R}_x$ to the head of $\vec{R}_y$.

**Step 9: Measure the resultant, $\vec{R}$**

We are solving the problem graphically so we now need to measure the magnitude of the vector and use the scale we chose to convert our answer from the diagram to the actual result. In the last diagram the resultant, $\vec{R}$ is 4,7 cm long therefore the magnitude of the vector is 4,7 kN.

The direction of the resultant we need to measure from the diagram using a protractor. The angle that the vector makes with the x -axis is 44° .

Step 10: Quote the final answer

$\vec{R}$ is 4,7 kN at -44° from the positive x -direction.

Algebraic methods

ESBKB

Algebraic addition and subtraction of vectors

In grade 10 you learnt about addition and subtraction of vectors in one dimension. The following worked example provides a refresher of the concepts.

Worked example 9: Adding vectors algebraically

QUESTION

A force of 5 N to the right is applied to a crate. A second force of 2 N to the left is also applied to the crate. Calculate algebraically the resultant of the forces applied to the crate.

SOLUTION

Step 1: Draw a sketch

A simple sketch will help us understand the problem.



Step 2: Decide which method to use to calculate the resultant

Remember that force is a vector. Since the forces act along a straight line (i.e. the x -direction), we can use the algebraic technique of vector addition.

Step 3: Choose a positive direction

Choose the **positive** direction to be to the right. This means that the **negative** direction is to the left.

Rewriting the problem using the choice of a positive direction gives us a force of 5 N in the positive x -direction and force of 2 N in the negative x -direction being applied to the crate.

Step 4: Now define our vectors algebraically

$$\begin{aligned}\vec{F}_1 &= 5 \text{ N} \\ \vec{F}_2 &= -2 \text{ N}\end{aligned}$$

Step 5: Add the vectors

Thus, the resultant force is:

$$\begin{aligned}\vec{F}_1 + \vec{F}_2 &= (5) + (-2) \\ &= 3 \text{ N}\end{aligned}$$

Step 6: Quote the resultant

Remember that in this case a positive force means to the right: 3 N to the right.

We can now expand on this work to include vectors in two dimensions.

Worked example 10: Algebraic solution in two dimensions

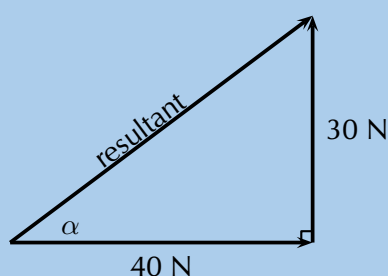
QUESTION

A force of 40 N in the positive x -direction acts simultaneously (at the same time) to a force of 30 N in the positive y -direction. Calculate the magnitude of the resultant force.

SOLUTION

Step 1: Draw a rough sketch

As before, the rough sketch looks as follows:



Step 2: Determine the length of the resultant

Note that the triangle formed by the two force vectors and the resultant vector is a right-angle triangle. We can thus use the Theorem of Pythagoras to determine the length of the resultant. Let R represent the length of the resultant vector. Then:

$$\begin{aligned}F_x^2 + F_y^2 &= R^2 \text{ Pythagoras' theorem} \\(40)^2 + (30)^2 &= R^2 \\R &= 50 \text{ N}\end{aligned}$$

Step 3: Quote the resultant

The magnitude of the resultant force is then 50 N.

Direction

For two dimensional vectors we have only covered finding the magnitude of vectors algebraically. We also need to know the direction. For vectors in one dimension this was simple. We chose a positive direction and then the resultant was either in the positive or in the negative direction. In grade 10 you learnt about the different ways to specify direction. We will now look at using trigonometry to determine the direction of the resultant vector.

We can use simple trigonometric identities to calculate the direction. We can calculate the direction of the resultant in the previous worked example.

Worked example 11: Direction of the resultant

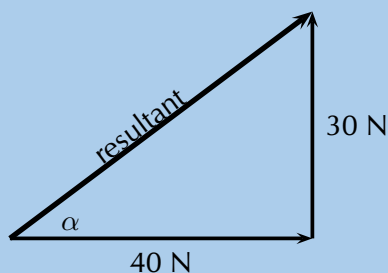
QUESTION

A force of 40 N in the positive x -direction acts simultaneously (at the same time) to a force of 30 N in the positive y -direction. Calculate the magnitude of the resultant force.

SOLUTION

Step 1: Magnitude

We determined the magnitude of the resultant vector in the previous worked example to be 50 N. The sketch of the situation is:



Step 2: Determine the direction of the resultant

To determine the direction of the resultant force, we calculate the angle α between the resultant force vector and the positive x -axis, by using simple trigonometry:

$$\begin{aligned}\tan \alpha &= \frac{\text{opposite side}}{\text{adjacent side}} \\ \tan \alpha &= \frac{30}{40} \\ \alpha &= \tan^{-1}(0,75) \\ \alpha &= 36,87^\circ\end{aligned}$$

Step 3: Quote the resultant

The resultant force is then 50 N at $36,9^\circ$ to the positive x -axis.

Exercise 1 – 5: Algebraic addition of vectors

1. A force of 17 N in the positive x -direction acts simultaneously (at the same time) to a force of 23 N in the positive y -direction. Calculate the resultant force.
2. A force of 23,7 N in the negative x -direction acts simultaneously to a force of 9 N in the positive y -direction. Calculate the resultant force.
3. Four forces act simultaneously at a point, find the resultant if the forces are:

- $\vec{F}_1 = 2,3 \text{ N}$ in the positive x -direction
- $\vec{F}_2 = 4 \text{ N}$ in the positive y -direction
- $\vec{F}_3 = 3,3 \text{ N}$ in the negative y -direction
- $\vec{F}_4 = 2,1 \text{ N}$ in the negative y -direction

4. The following forces act simultaneously on a pole, if the pole suddenly snaps in which direction will it be pushed:

- $\vec{F}_1 = 2,3 \text{ N}$ in the negative x -direction
- $\vec{F}_2 = 11,7 \text{ N}$ in the negative y -direction
- $\vec{F}_3 = 6,9 \text{ N}$ in the negative y -direction
- $\vec{F}_4 = 1,9 \text{ N}$ in the negative y -direction

Think you got it? Get this answer and more practice on our Intelligent Practice Service

1. 23FN 2. 23FP 3. 23FQ 4. 23FR



www.everythingscience.co.za



m.everythingscience.co.za

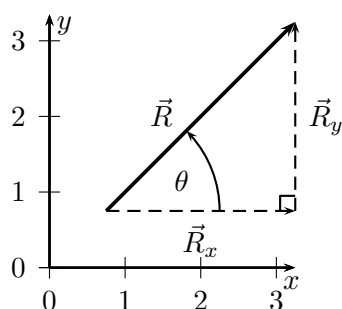
1.3 Components of vectors

ESBKC

In the discussion of vector addition we saw that a number of vectors acting together can be combined to give a single vector (the resultant). In much the same way a single vector can be broken down into a number of vectors which when added give that original vector. These vectors which sum to the original are called **components** of the original vector. The process of breaking a vector into its components is called **resolving into components**.

In practise it is most useful to resolve a vector into components which are at right angles to one another, usually horizontal and vertical. Think about all the problems we've solved so far. If we have vectors parallel to the x - and y -axes problems are straightforward to solve.

Any vector can be resolved into a horizontal and a vertical component. If $\vec{R}$ is a vector, then the horizontal component of $\vec{R}$ is $\vec{R}_x$ and the vertical component is $\vec{R}_y$.



When resolving into components that are parallel to the x - and y -axes we are always dealing with a right-angled triangle. This means that we can use trigonometric identities to determine the magnitudes of the components (we know the directions because they are aligned with the axes).

From the triangle in the diagram above we know that

$$\cos(\theta) = \frac{R_x}{R}$$

$$\frac{R_x}{R} = \cos(\theta)$$

$$R_x = R \cos(\theta)$$

and

$$\sin \theta = \frac{R_y}{R}$$

$$\frac{R_y}{R} = \sin(\theta)$$

$$R_y = R \sin(\theta)$$

$$R_x = R \cos(\theta)$$

$$R_y = R \sin(\theta)$$

Note that the angle is measured counter-clockwise from the positive x -axis.

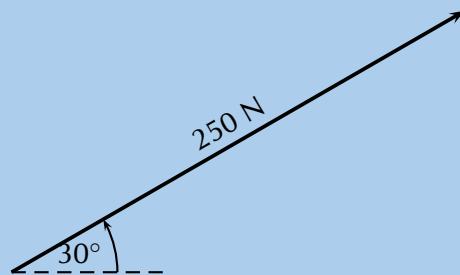
Worked example 12: Resolving a vector into components

QUESTION

A force of 250 N acts at an angle of 30° to the positive x -axis. Resolve this force into components parallel to the x - and y -axes.

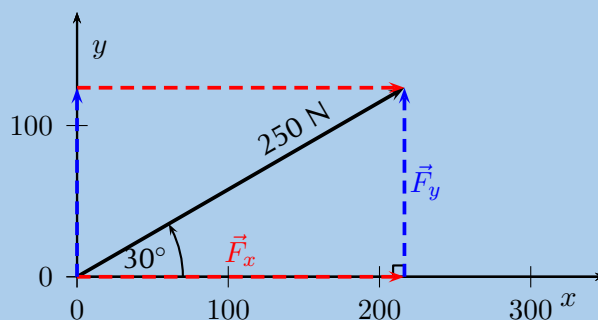
SOLUTION

Step 1: Draw a rough sketch of the original vector



Step 2: Determine the vector components

Next we resolve the force into components parallel to the axes. Since these directions are perpendicular to one another, the components form a right-angled triangle with the original force as its hypotenuse.



Notice how the two components acting together give the original vector as their resultant.

Step 3: Determine the magnitudes of the component vectors

Now we can use trigonometry to calculate the magnitudes of the components of the original displacement:

$$\begin{aligned}F_y &= 250 \sin(30^\circ) \\ &= 125 \text{ N}\end{aligned}$$

and

$$\begin{aligned}F_x &= 250 \cos(30^\circ) \\ &= 216,5 \text{ N}\end{aligned}$$

Remember F_x and F_y are the magnitudes of the components. $\vec{F}_x$ is in the positive x -direction and $\vec{F}_y$ is in the positive y -direction.

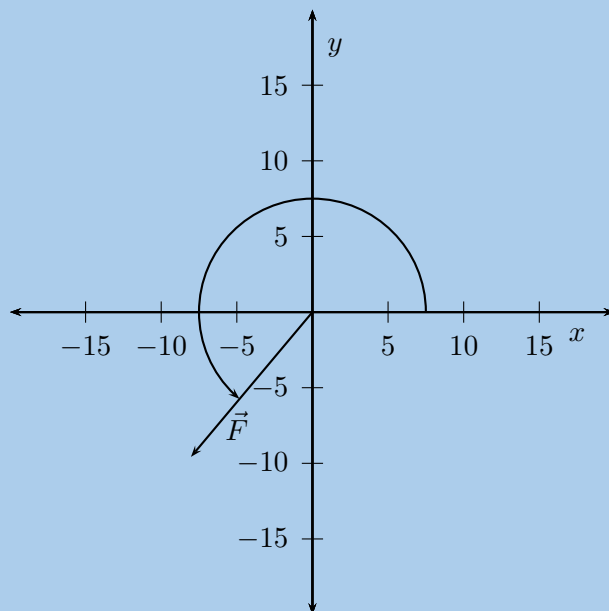
Worked example 13: Resolving a vector into components

QUESTION

A force of 12,5 N acts at an angle of 230° to the positive x -axis. Resolve this force into components parallel to the x - and y -axes.

SOLUTION

Step 1: Draw a rough sketch of the original vector



Step 2: Determine the vector components

Next we resolve the force into components parallel to the axes. Since these directions are perpendicular to one another, the components form a right-angled triangle with the original force as its hypotenuse.

Now we can use trigonometry to calculate the magnitudes of the components of the original force:

$$\begin{aligned}F_y &= 12,5 \sin(230^\circ) \\ &= -9,58 \text{ N}\end{aligned}$$

and

$$\begin{aligned}F_x &= 12,5 \cos(230^\circ) \\ &= -8,03 \text{ N}\end{aligned}$$

Notice that by using the full angle we actually get the correct signs for the components if we use the standard Cartesian coordinates. $\vec{F}_x$ is in the negative x -direction and $\vec{F}_y$ is in the negative y -direction.

Exercise 1 – 6:

1. Resolve each of the following vectors into components:

- $\vec{F}_1 = 5 \text{ N}$ at 45° to the positive x -axis.
- $\vec{F}_2 = 15 \text{ N}$ at 63° to the positive x -axis.
- $\vec{F}_3 = 11,3 \text{ N}$ at 127° to the positive x -axis.
- $\vec{F}_4 = 125 \text{ N}$ at 245° to the positive x -axis.

2. Resolve each of the following vectors into components:

- $\vec{F}_1 = 11 \times 10^4 \text{ N}$ at 33° to the positive x -axis.
- $\vec{F}_2 = 15 \text{ GN}$ at 28° to the positive x -axis.
- $\vec{F}_3 = 11,3 \text{ kN}$ at 193° to the positive x -axis.
- $\vec{F}_4 = 125 \times 10^5 \text{ N}$ at 317° to the positive x -axis.

Think you got it? Get this answer and more practice on our Intelligent Practice Service

1. [23FS](#) 2. [23FT](#)



www.everythingscience.co.za



m.everythingscience.co.za

Vector addition using components

ESBKD

Components can also be used to find the resultant of vectors. This technique can be applied to both graphical and algebraic methods of finding the resultant. The method is straightforward:

1. make a rough sketch of the problem;
2. find the horizontal and vertical components of each vector;
3. find the sum of all horizontal components, $\vec{R}_x$;

4. find the sum of all the vertical components, $\vec{R}_y$;
5. then use them to find the resultant, $\vec{R}$.

Consider the two vectors, $\vec{F}_1$ and $\vec{F}_2$, in Figure 1.3, together with their resultant, $\vec{R}$.

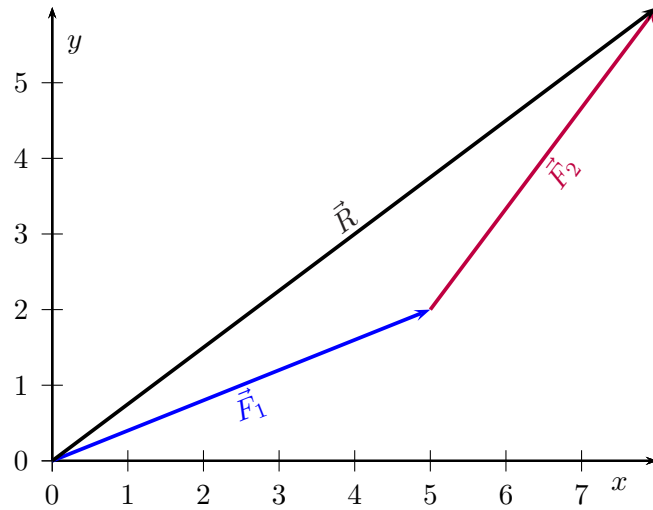


Figure 1.3: An example of two vectors being added to give a resultant.

Each vector in Figure 1.3 can be broken down into one component in the x -direction (horizontal) and one in the y -direction (vertical). These components are two vectors which when added give you the original vector as the resultant. This is shown in Figure 1.4 below:

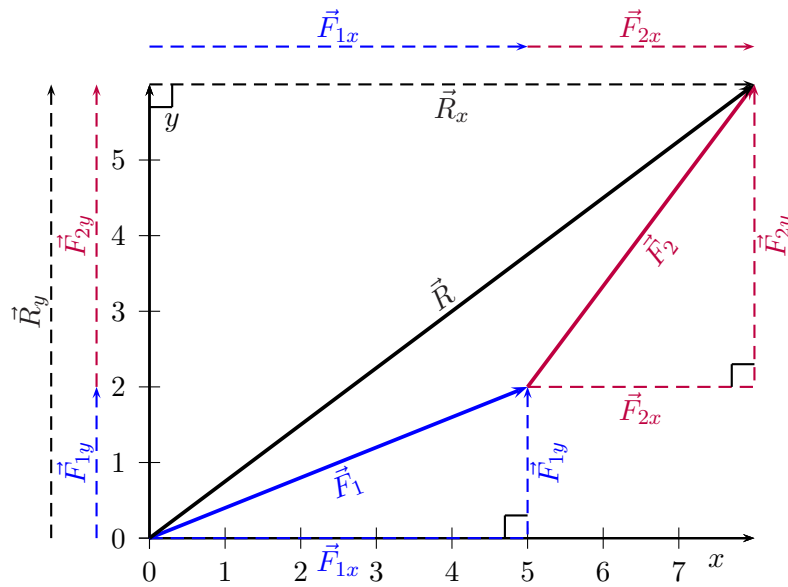


Figure 1.4: Adding vectors using components.

We can see that:

$$\begin{aligned}\vec{F}_1 &= \vec{F}_{1x} + \vec{F}_{1y} \\ \vec{F}_2 &= \vec{F}_{2x} + \vec{F}_{2y} \\ \vec{R} &= \vec{R}_x + \vec{R}_y\end{aligned}$$

$$\text{But, } \vec{R}_x = \vec{F}_{1x} + \vec{F}_{2x}$$

$$\text{and } \vec{R}_y = \vec{F}_{1y} + \vec{F}_{2y}$$

In summary, addition of the x -components of the two original vectors gives the x -component of the resultant. The same applies to the y -components. So if we just added all the components together we would get the **same answer!** This is another important property of vectors.

Worked example 14: Adding vectors using components

QUESTION

If in Figure 1.4, $\vec{F}_1 = 5,385 \text{ N}$ at an angle of $21,8^\circ$ to the horizontal and $\vec{F}_2 = 5 \text{ N}$ at an angle of $53,13^\circ$ to the horizontal, find the resultant force, $\vec{R}$.

SOLUTION

Step 1: Decide how to tackle the problem

The first thing we must realise is that the order that we add the vectors does not matter. Therefore, we can work through the vectors to be added in any order. We also draw up the following table to help us work through the problem:

Vector	x -component	y -component	Total
$\vec{F}_1$			
$\vec{F}_2$			
Resultant			

Step 2: Resolve $\vec{F}_1$ into components

We find the components of $\vec{F}_1$ by using known trigonometric ratios.

First we find the magnitude of the vertical component, F_{1y} :

$$\sin(\theta) = \frac{F_{1y}}{F_1}$$

$$\sin(21,8^\circ) = \frac{F_{1y}}{5,385}$$

$$F_{1y} = (\sin(21,8^\circ)) (5,385)$$

$$= 2,00 \text{ N}$$

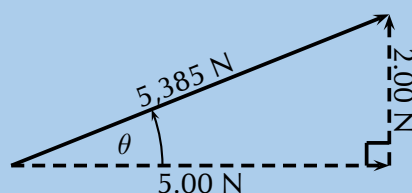
Secondly we find the magnitude of the horizontal component, F_{1x} :

$$\cos(\theta) = \frac{F_{1x}}{F_1}$$

$$\cos(21,8^\circ) = \frac{F_{1x}}{5,385}$$

$$F_{1x} = (\cos(21,8^\circ)) (5,385)$$

$$= 5,00 \text{ N}$$



The components give the sides of the right angle triangle, for which the original vector, $\vec{F}_1$, is the hypotenuse.

Vector	x-component	y-component	Resultant
$\vec{F}_1$	5,00 N	2,00 N	5,385 N
$\vec{F}_2$			
Resultant			

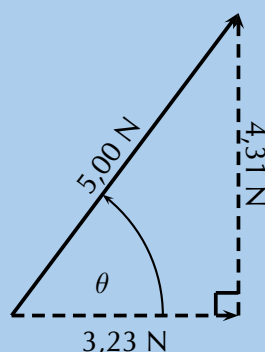
Step 3: Resolve $\vec{F}_2$ into components

We find the components of $\vec{F}_2$ by using known trigonometric ratios. First we find the magnitude of the vertical component, F_{2y} :

$$\begin{aligned}\sin(\theta) &= \frac{F_{2y}}{F_2} \\ \sin(53,13^\circ) &= \frac{F_{2y}}{5} \\ F_{2y} &= (\sin(53,13^\circ)) (5) \\ &= 4,00 \text{ N}\end{aligned}$$

Secondly we find the magnitude of the horizontal component, F_{2x} :

$$\begin{aligned}\cos(\theta) &= \frac{F_{2x}}{F_2} \\ \cos(53,13^\circ) &= \frac{F_{2x}}{5} \\ F_{2x} &= (\cos(53,13^\circ)) (5) \\ &= 3,00 \text{ N}\end{aligned}$$



Vector	x-component	y-component	Total
$\vec{F}_1$	5,00 N	2,00 N	5,385 N
$\vec{F}_2$	3,00 N	4,00 N	5 N
Resultant			

Step 4: Determine the components of the resultant vector

Now we have all the components. If we add all the horizontal components then we will have the x-component of the resultant vector, $\vec{R}_x$. Similarly, we add all the vertical components then we will have the y-component of the resultant vector, $\vec{R}_y$.

$$\begin{aligned}R_x &= F_{1x} + F_{2x} \\ &= 5,00 \text{ N} + 3,00 \text{ N} \\ &= 8,00 \text{ N}\end{aligned}$$

$$\begin{aligned}R_y &= F_{1y} + F_{2y} \\ &= 2,00 \text{ N} + 4,00 \text{ N} \\ &= 6,00 \text{ N}\end{aligned}$$

Therefore, $\vec{R}_x$ is 8 N to the right.

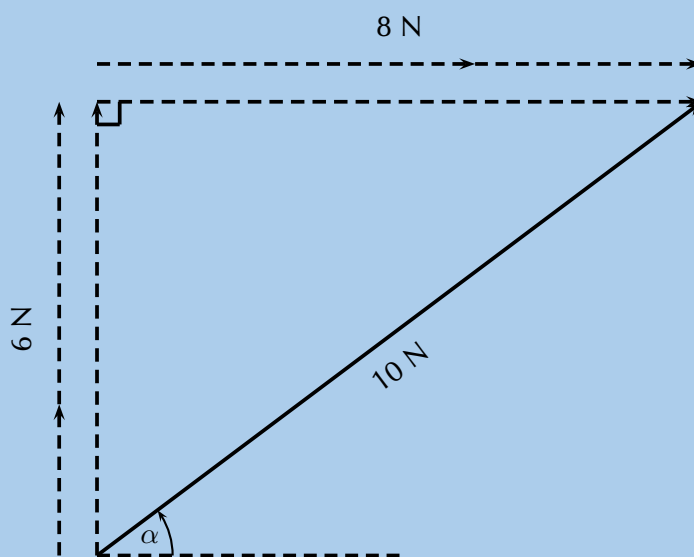
Therefore, $\vec{R}_y$ is 6 N up.

Vector	x -component	y -component	Total
$\vec{F}_1$	5,00 N	2,00 N	5,385 N
$\vec{F}_2$	3,00 N	4,00 N	5 N
Resultant	8,00 N	6,00 N	

Step 5: Determine the magnitude and direction of the resultant vector

Now that we have the components of the resultant, we can use the Theorem of Pythagoras to determine the magnitude of the resultant, R .

$$\begin{aligned}
 R^2 &= (R_y)^2 + (R_x)^2 \\
 &= (6,00)^2 + (8,00)^2 \\
 &= 100,00 \\
 R &= 10,00 \text{ N}
 \end{aligned}$$



The magnitude of the resultant, R is 10,00 N. So all we have to do is calculate its direction. We can specify the direction as the angle the vectors makes with a known direction. To do this you only need to visualise the vector as starting at the origin of a coordinate system. We have drawn this explicitly below and the angle we will calculate is labelled α .

Using our known trigonometric ratios we can calculate the value of α :

$$\begin{aligned}
 \tan \alpha &= \frac{6,00}{8,00} \\
 \alpha &= \tan^{-1} \frac{6,00}{8,00} \\
 \alpha &= 36,9^\circ
 \end{aligned}$$

Step 6: Quote the final answer

$\vec{R}$ is 10 m at an angle of $36,9^\circ$ to the positive x -axis.

QUESTION

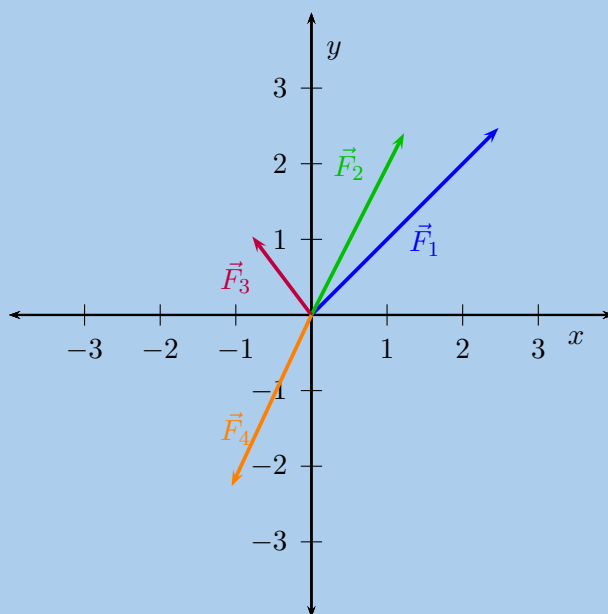
Determine, by resolving into components, the resultant of the following four forces acting at a point:

- $\vec{F}_1 = 3,5 \text{ N}$ at 45° to the positive x -axis.
- $\vec{F}_2 = 2,7 \text{ N}$ at 63° to the positive x -axis.
- $\vec{F}_3 = 1,3 \text{ N}$ at 127° to the positive x -axis.
- $\vec{F}_4 = 2,5 \text{ N}$ at 245° to the positive x -axis.

SOLUTION

Step 1: Sketch the problem

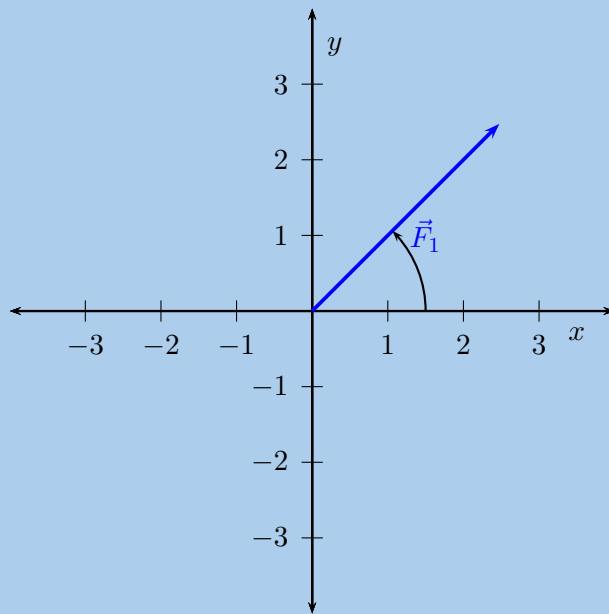
Draw all of the vectors on the Cartesian plane. This does not have to be precisely accurate because we are solving algebraically but vectors need to be drawn in the correct quadrant and with the correct relative positioning other.



We are going to record the various components in a table to help us manage keep track of the calculation. For each vector we need to determine the components in the x - and y -directions.

Vector	x -component	y -component	Total
$\vec{F}_1$			3,5 N
$\vec{F}_2$			2,7 N
$\vec{F}_3$			1,3 N
$\vec{F}_4$			2,5 N
$\vec{R}$			

Step 2: Determine components of $\vec{F}_1$



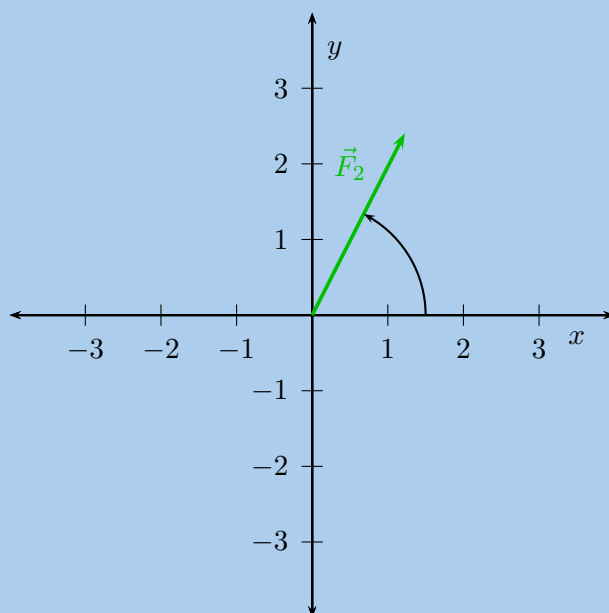
Firstly we find the magnitude of the vertical component, F_{1y} :

$$\begin{aligned}\sin(\theta) &= \frac{F_{1y}}{F_1} \\ \sin(45^\circ) &= \frac{F_{1y}}{3,5} \\ F_{1y} &= (\sin(45^\circ)) (3,5) \\ &= 2,47 \text{ N}\end{aligned}$$

Secondly we find the magnitude of the horizontal component, F_{1x} :

$$\begin{aligned}\cos(\theta) &= \frac{F_{1x}}{F_1} \\ \cos(45^\circ) &= \frac{F_{1x}}{3,5} \\ F_{1x} &= (\cos(45^\circ)) (3,5) \\ &= 2,47 \text{ N}\end{aligned}$$

Step 3: Determine components of $\vec{F}_2$



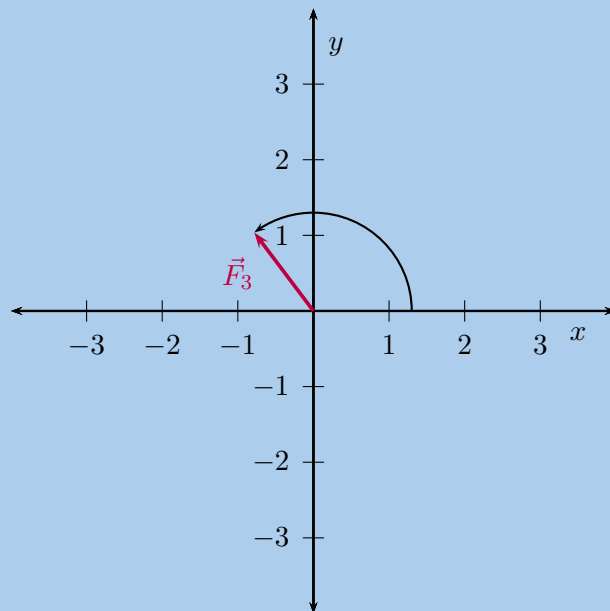
Firstly we find the magnitude of the vertical component, F_{2y} :

$$\begin{aligned}\sin(\theta) &= \frac{F_{2y}}{F_2} \\ \sin(63^\circ) &= \frac{F_{2y}}{2,7} \\ F_{2y} &= (\sin(63^\circ)) (2,7) \\ &= 2,41 \text{ N}\end{aligned}$$

Secondly we find the magnitude of the horizontal component, F_{2x} :

$$\begin{aligned}\cos \theta &= \frac{F_{2x}}{F_2} \\ \cos(63^\circ) &= \frac{F_{2x}}{2,7} \\ F_{2x} &= (\cos(63^\circ)) (2,7) \\ &= 1,23 \text{ N}\end{aligned}$$

Step 4: Determine components of $\vec{F}_3$



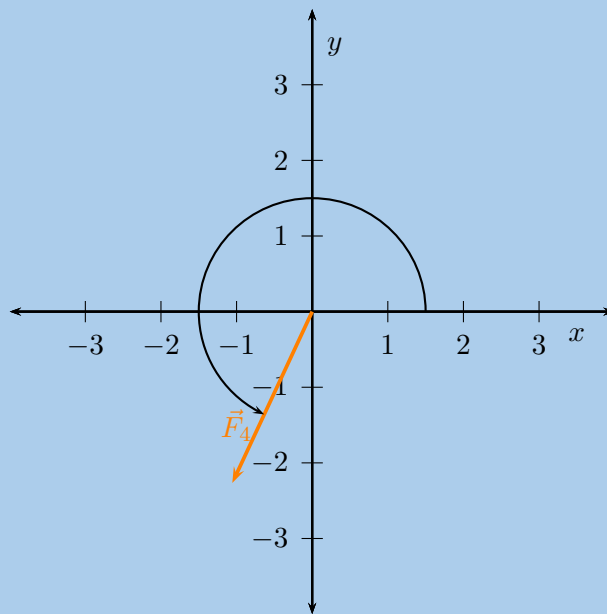
Firstly we find the magnitude of the vertical component, F_{3y} :

$$\begin{aligned}\sin(\theta) &= \frac{F_{3y}}{F_3} \\ \sin(127^\circ) &= \frac{F_{3y}}{1,3} \\ F_{3y} &= (\sin 127^\circ) (1,3) \\ &= 1,04 \text{ N}\end{aligned}$$

Secondly we find the magnitude of the horizontal component, F_{3x} :

$$\begin{aligned}\cos(\theta) &= \frac{F_{3x}}{F_3} \\ \cos(127^\circ) &= \frac{F_{3x}}{1,3} \\ F_{3x} &= (\cos 127^\circ) (1,3) \\ &= -0,78 \text{ N}\end{aligned}$$

Step 5: Determine components of $\vec{F}_4$



Firstly we find the magnitude of the vertical component, F_{4y} :

$$\begin{aligned}\sin(\theta) &= \frac{F_{4y}}{F_4} \\ \sin(245^\circ) &= \frac{F_{4y}}{2,5} \\ F_{4y} &= (\sin(245^\circ)) (2,5) \\ &= -2,27 \text{ N}\end{aligned}$$

Secondly we find the magnitude of the horizontal component, F_{4x} :

$$\begin{aligned}\cos(\theta) &= \frac{F_{4x}}{F_4} \\ \cos(245^\circ) &= \frac{F_{4x}}{2,5} \\ F_{4x} &= (\cos(245^\circ)) (2,5) \\ &= -1,06 \text{ N}\end{aligned}$$

Step 6: Determine components of resultant

Sum the various component columns to determine the components of the resultant. Remember that if the component was negative don't leave out the negative sign in the summation.

Vector	x -component	y -component	Total
$\vec{F}_1$	2,47 N	2,47 N	3,5 N
$\vec{F}_2$	1,23 N	2,41 N	2,7 N
$\vec{F}_3$	-0,78 N	1,04 N	1,3 N
$\vec{F}_4$	-1,06 N	-2,27 N	2,5 N
$\vec{R}$	1,86 N	3,65 N	

Now that we have the components of the resultant, we can use the Theorem of Pythagoras to determine the magnitude of the resultant, R .

$$\begin{aligned}R^2 &= (R_y)^2 + (R_x)^2 \\ &= (1,86)^2 + (3,65)^2 \\ &= 16,78 \\ R &= 4,10 \text{ N}\end{aligned}$$

We can also determine the angle with the positive x -axis.

$$\begin{aligned}\tan(\alpha) &= \frac{1,86}{3,65} \\ \alpha &= \tan^{-1}\left(\frac{3,65}{1,86}\right) \\ \alpha &= 27,00^\circ\end{aligned}$$

Step 7: Quote the final answer

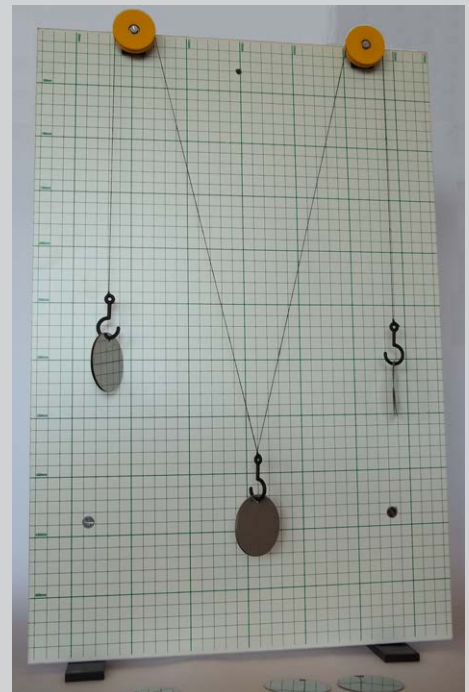
The resultant has a magnitude of 4,10 N at an angle of $27,00^\circ$ to the positive x -direction.

Informal experiment: Force board

Aim:

Determine the resultant of three non-linear forces using a force board.

Apparatus and materials:



You will need:

- blank paper
- force board
- 4 spring balances
- assortment of weights

Method:

Before beginning the detailed method think about the strategy. By connecting a cord to the ring, running it over a pulley and hanging weights off it you can get a force exerted on the ring. The more weights or heavier the weight you hang the greater the force. The force is in the direction of the cord. If you run a number of cords over pulleys you are exerting more forces, all in different directions, on the ring. So we have a system where we can change the magnitude and direction of the forces acting on the ring. By putting a spring balance between the cord and the ring, we can measure the force. Putting a piece of paper under the ring allows us to draw the directions and the readings on the spring balances allows us to measure the magnitude.

We are going to use this information to measure the forces acting on the rings and then we can determine the resultants graphically.

1. Set up the forceboard and place a piece of paper under the ring.
2. Set up four different forces by connecting a spring balance to the ring on the one side and some cord on the other side. Run the cord over a pulley and attach some weights to it. Work in a group to do this effectively.
3. Draw a line along each cord being careful not to move any of them.
4. Make a note of the reading on each spring balance.
5. Now remove the paper.
6. Working on the paper, draw each line back towards the where the centre of the ring had been. The lines should all intersect at a point. Make this point the centre of your Cartesian coordinate system.
7. Now choose an appropriate scale to relate the length of arrows to the readings on the spring balances. Using the appropriate spring balance and correct line on the paper, draw an arrow to represent each of the forces.

Results:

For two different choices of 3 of the force vectors we will determine the resultant. To determine the resultant we need to add the vectors together. The easiest way to do this is to replicate the vectors using a ruler and protractor and draw them tail-to-head.

Conclusions and questions:

Note the direction and the magnitudes of the resultants of the various combinations.

1. How does the calculated resultant compare to the vector that wasn't used to calculate the resultant in each case?
2. What general relationship should exist between the resultant and the fourth vector and why do you think this is the case?
3. Would this be the same if we had more or less forces in the problem? Justify your answer.

► See simulation: 23FV at www.everythingscience.co.za

► See video: 23FW at www.everythingscience.co.za

1.4 Chapter summary

ESBKF

► See presentation: 23FX at www.everythingscience.co.za

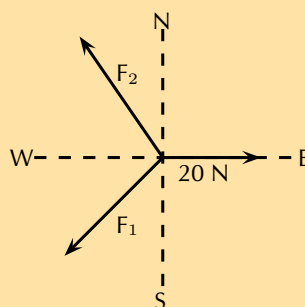
- A vector has a magnitude and direction.
- Vectors can be used to represent many physical quantities that have a magnitude and direction, like forces.
- Vectors may be represented as arrows where the length of the arrow indicates the magnitude and the arrowhead indicates the direction of the vector.
- Vectors in two dimensions can be drawn on the Cartesian plane.
- Vectors can be added graphically using the head-to-tail method or the tail-to-tail method.
- A closed vector diagram is a set of vectors drawn on the Cartesian using the tail-to-head method and that has a resultant with a magnitude of zero.
- Vectors can be added algebraically using Pythagoras' theorem or using components.
- The direction of a vector can be found using simple trigonometric calculations.
- The components of a vector are a series of vectors that, when combined, give the original vector as their resultant.
- Components are usually created that align with the Cartesian coordinate axes. For a vector $\vec{F}$ that makes an angle of θ with the positive x -axis the x -component is $\vec{R}_x = R \cos(\theta)$ and the y -component is $\vec{R}_y = R \sin(\theta)$.

Exercise 1 – 7:

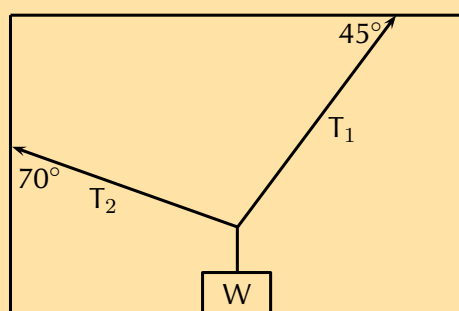
1. Draw the following forces as vectors on the Cartesian plane originating at the origin:
 - $\vec{F}_1 = 3,7 \text{ N}$ in the positive x -direction
 - $\vec{F}_2 = 4,9 \text{ N}$ in the positive y -direction
2. Draw the following forces as vectors on the Cartesian plane:
 - $\vec{F}_1 = 4,3 \text{ N}$ in the positive x -direction
 - $\vec{F}_2 = 1,7 \text{ N}$ in the negative x -direction
 - $\vec{F}_3 = 8,3 \text{ N}$ in the positive y -direction
3. Find the resultant in the x -direction, R_x , and y -direction, R_y for the following forces:

- $\vec{F}_1 = 1,5 \text{ N}$ in the positive x -direction
 - $\vec{F}_2 = 1,5 \text{ N}$ in the positive x -direction
 - $\vec{F}_3 = 2 \text{ N}$ in the negative x -direction
4. Find the resultant in the x -direction, R_x , and y -direction, R_y for the following forces:
- $\vec{F}_1 = 4,8 \text{ N}$ in the positive x -direction
 - $\vec{F}_2 = 3,2 \text{ N}$ in the negative x -direction
 - $\vec{F}_3 = 1,9 \text{ N}$ in the positive y -direction
 - $\vec{F}_4 = 2,1 \text{ N}$ in the negative y -direction
5. Find the resultant in the x -direction, R_x , and y -direction, R_y for the following forces:
- $\vec{F}_1 = 2,7 \text{ N}$ in the positive x -direction
 - $\vec{F}_2 = 1,4 \text{ N}$ in the positive x -direction
 - $\vec{F}_3 = 2,7 \text{ N}$ in the negative x -direction
 - $\vec{F}_4 = 1,7 \text{ N}$ in the negative y -direction
6. Sketch the resultant of the following force vectors using the tail-to-head method:
- $\vec{F}_1 = 4,8 \text{ N}$ in the positive y -direction
 - $\vec{F}_2 = 3,3 \text{ N}$ in the negative x -direction
7. Sketch the resultant of the following force vectors using the tail-to-head method:
- $\vec{F}_1 = 0,7 \text{ N}$ in the positive y -direction
 - $\vec{F}_2 = 6 \text{ N}$ in the positive x -direction
 - $\vec{F}_3 = 3,8 \text{ N}$ in the negative y -direction
 - $\vec{F}_4 = 11,9 \text{ N}$ in the negative x -direction
8. Sketch the resultant of the following force vectors using the tail-to-head method by first determining the resultant in the x - and y -directions:
- $\vec{F}_1 = 5,2 \text{ N}$ in the positive y -direction
 - $\vec{F}_2 = 7,5 \text{ N}$ in the negative y -direction
 - $\vec{F}_3 = 4,8 \text{ N}$ in the positive y -direction
 - $\vec{F}_4 = 6,3 \text{ N}$ in the negative x -direction
9. Sketch the resultant of the following force vectors using the tail-to-head method by first determining the resultant in the x - and y -directions:
- $\vec{F}_1 = 6,7 \text{ N}$ in the positive y -direction
 - $\vec{F}_2 = 4,2 \text{ N}$ in the negative x -direction
 - $\vec{F}_3 = 9,9 \text{ N}$ in the negative y -direction
 - $\vec{F}_4 = 3,4 \text{ N}$ in the negative y -direction
10. Sketch the resultant of the following force vectors using the tail-to-tail method:

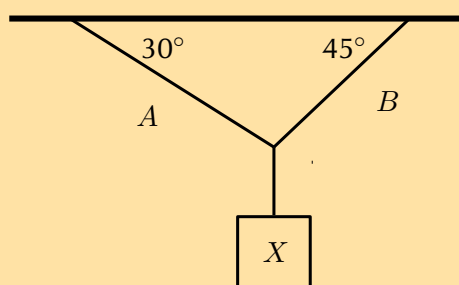
- $\vec{F}_1 = 6,1 \text{ N}$ in the positive y -direction
 - $\vec{F}_2 = 4,5 \text{ N}$ in the negative x -direction
11. Sketch the resultant of the following force vectors using the tail-to-tail method by first determining the resultant in the x - and y -directions:
- $\vec{F}_1 = 2,3 \text{ N}$ in the positive y -direction
 - $\vec{F}_2 = 11,8 \text{ N}$ in the negative y -direction
 - $\vec{F}_3 = 7,9 \text{ N}$ in the negative y -direction
 - $\vec{F}_4 = 3,2 \text{ N}$ in the negative x -direction
12. Four forces act simultaneously at a point, find the resultant if the forces are:
- $\vec{F}_1 = 2,3 \text{ N}$ in the positive x -direction
 - $\vec{F}_2 = 4,9 \text{ N}$ in the positive y -direction
 - $\vec{F}_3 = 4,3 \text{ N}$ in the negative y -direction
 - $\vec{F}_4 = 3,1 \text{ N}$ in the negative y -direction
13. Resolve each of the following vectors into components:
- $\vec{F}_1 = 105 \text{ N}$ at $23,5^\circ$ to the positive x -axis.
 - $\vec{F}_2 = 27 \text{ N}$ at $58,9^\circ$ to the positive x -axis.
 - $\vec{F}_3 = 11,3 \text{ N}$ at 323° to the positive x -axis.
 - $\vec{F}_4 = 149 \text{ N}$ at 245° to the positive x -axis.
 - $\vec{F}_5 = 15 \text{ N}$ at 375° to the positive x -axis.
 - $\vec{F}_6 = 14,9 \text{ N}$ at $75,6^\circ$ to the positive x -axis.
 - $\vec{F}_7 = 11,3 \text{ N}$ at $123,4^\circ$ to the positive x -axis.
 - $\vec{F}_8 = 169 \text{ N}$ at 144° to the positive x -axis.
14. A point is acted on by two forces and the resultant is zero. The forces
- have equal magnitudes and directions.
 - have equal magnitudes but opposite directions.
 - act perpendicular to each other.
 - act in the same direction.
15. A point in equilibrium is acted on by three forces. Force F_1 has components 15 N due south and 13 N due west. What are the components of force F_2 ?
- 13 N due north and 20 N due west
 - 13 N due north and 13 N due west
 - 15 N due north and 7 N due west
 - 15 N due north and 13 N due east



16. Two vectors act on the same point. What should the angle between them be so that a maximum resultant is obtained?
- a) 0° b) 90° c) 180° d) cannot tell
17. Two forces, 4 N and 11 N, act on a point. Which one of the following cannot be the magnitude of a resultant?
- a) 4 N b) 7 N c) 11 N d) 15 N
18. An object of weight W is supported by two cables attached to the ceiling and wall as shown. The tensions in the two cables are T_1 and T_2 respectively. Tension $T_1 = 1200$ N. Determine the tension T_2 by accurate construction and measurement or by calculation.



19. An object X is supported by two strings, A and B , attached to the ceiling as shown in the sketch. Each of these strings can withstand a maximum force of 700 N. The weight of X is increased gradually.



- a) Draw a rough sketch of the triangle of forces, and use it to explain which string will break first.
- b) Determine the maximum weight of X which can be supported.

Think you got it? Get this answer and more practice on our Intelligent Practice Service

- | | | | | | |
|---------------------------|---------------------------|---------------------------|---------------------------|---------------------------|---------------------------|
| 1. 23FY | 2. 23FZ | 3. 23G2 | 4. 23G3 | 5. 23G4 | 6. 23G5 |
| 7. 23G6 | 8. 23G7 | 9. 23G8 | 10. 23G9 | 11. 23GB | 12. 23GC |
| 13a. 23GD | 13b. 23GF | 13c. 23GG | 13d. 23GH | 13e. 23GJ | 13f. 23GK |
| 13g. 23GM | 13h. 23GN | 14. 23GP | 15. 23GQ | 16. 23GR | 17. 23GS |
| 18. 23GT | 19. 23GV | | | | |



www.everythingscience.co.za



m.everythingscience.co.za

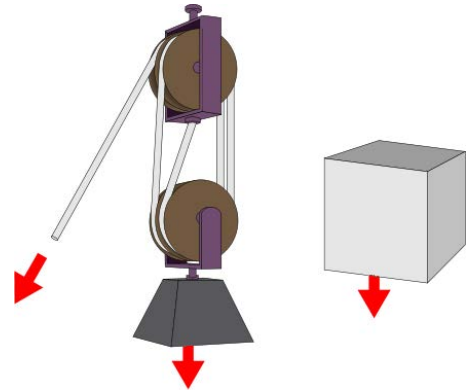
Newton's laws

2.1	<i>Introduction</i>	56
2.2	<i>Force</i>	56
2.3	<i>Newton's laws</i>	74
2.4	<i>Forces between masses</i>	109
2.5	<i>Chapter summary</i>	119

2.1 Introduction

ESBKG

In this chapter we will learn how a net force is needed to modify the motion of an object. We will recall what a force is and learn about how force and motion are related. We are also introduced to Newton's three laws and we will learn more about the force of gravity.



Key Mathematics Concepts

- Ratio and proportion — Physical Sciences, Grade 10, Science skills
- Equations — Mathematics, Grade 10, Equations and inequalities
- Units and unit conversions — Physical Sciences, Grade 10, Science skills

2.2 Force

ESBKH

What is a *force*?

ESBKJ

A force is anything that can cause a change to objects. Forces can do things like:

- change the shape of an object,
- accelerate or stop an object, and
- change the direction of a moving object.

A force can be classified as either a *contact force* or a *non-contact force*.



Figure 2.1: Contact forces



Figure 2.2: Contact forces

A contact force must touch or *be in contact* with an object to cause a change. Examples of contact forces are:

- the force that is used to push or pull things, like on a door to open or close it
- the force that a sculptor uses to turn clay into a pot
- the force of the wind to turn a windmill

A non-contact force does not have to touch an object to cause a change. Examples of non-contact forces are the forces due to:

- gravity, like the Earth pulling the Moon towards itself;
- electricity, like a proton and an electron attracting each other; and
- magnetism, like a magnet pulling a paper clip towards itself.

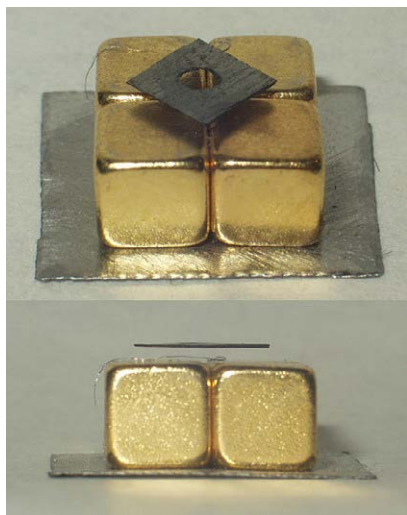


Figure 2.3: Non-contact forces

The unit of force in the international system of units (S.I. units) is the *newton* (symbol N). This unit is named after Sir Isaac Newton who first defined force. Force is a vector quantity and so it has a magnitude and a direction. We use the symbol $\vec{F}$ for force.

This chapter will often refer to the *resultant force* acting on an object. The resultant force is simply the vector sum of all the forces acting on the object. It is very important to remember that all the forces must be acting on the *same* object. The resultant force is the force that has the same effect as all the other forces added together.

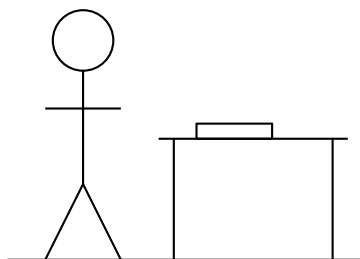
A lot of the physics topics you will study revolve around the impact or effect of forces. Although there are many different forces, will learn some fundamental principles for approaching the problems and applications in this book no matter which force applies

Physics is the study of the natural world and you probably know a lot more physics than you think. You see things happening everyday that are governed by the laws of physics but you probably aren't thinking about physics at the time. If you throw a stone up in the air it eventually falls to the ground. A lot of physics can be learnt by analysing an everyday situation.

We are going to learn about some forces in the next few sections but before we start lets describe an everyday situation in which they all play a role so you can visualise what is happening. You need a table and a three books that all have different masses. Take any book and put it on the table. Nothing happens, the book just rests on the table if the table is flat. If you slowly lift one side of the table so that the top of the table is tilted the book doesn't move immediately. As you lift the table more and more the book suddenly starts to slide off the table. You can repeat this with all three books and see how much you have to tilt the table before the books start to slide.

This real world situation illustrates a lot of the physics we want to learn about in this chapter.

The normal force



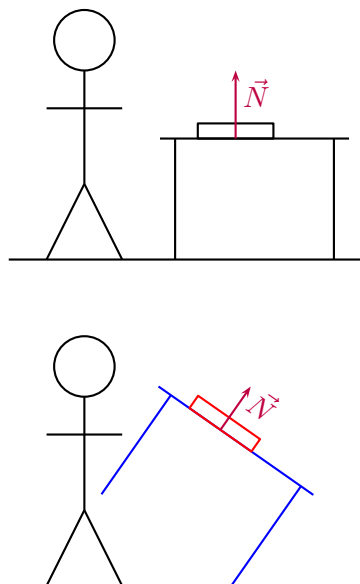
When an object is placed on a surface, for example think of the case of putting a book on a table, there are a number of forces acting. Firstly, if the table were not there the book would fall to the floor. The force that causes this is gravity. The table stops the book falling to the floor. The only way this can happen is for the table to exert a force on the book. The force that the table exerts on the book must balance out the force of gravity. This tells us a few things immediately! Gravity is a force pulling the book down, it is a vector. The force that the table exerts must balance this out and it can only do this if it has the same magnitude and acts in the opposite direction.

This occurs often, gravity pulls a person towards the earth but when you are standing on the ground something must be balancing it, if you put a heavy box on the ground the gravitational force is balanced. If you put a brick on water it will sink because nothing balances the gravitational force. We give the force that a surface (any surface) exerts to balance the forces on an object in contact with that surface the **normal** force.

The normal force is a force that acts on the object as a result of the interaction with the surface and is perpendicular to the surface. This last part might seem unexpected (counter-intuitive) because if we tilt the table slightly the direction of the gravitational force hasn't changed but the direction of the normal force has a little (the normal is not always directly opposite gravity). Don't panic, this will all make sense before the end of this chapter. Remember: the normal force is **always perpendicular (at a right angle)** to the surface.

DEFINITION: *Normal force*

The normal force, $\vec{N}$, is the force exerted by a surface on an object in contact with it.



Friction forces

Why does a box sliding on a surface eventually come to a stop? The answer is **friction**. Friction arises where two surfaces are in contact and moving relative to each other.

For an everyday example, press your hands together and move one backwards and forwards, we have two surfaces in contact with one moving relative to the other. Your

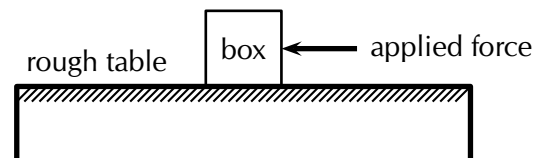
hands get warm, you will have experienced this before and probably rub your hands together in winter to warm them up. The heat is generated through friction.

Friction arises because the surfaces interact with each other. Think about sandpaper with lots of bumps on the surface. If you rub sandpaper the bumps will slot into any groove .

When the surface of one object slides over the surface of another, each body exerts a frictional force on the other. For example if a book slides across a table, the table exerts a frictional force onto the book and the book exerts a frictional force onto the table. Frictional forces act **parallel to surfaces**.

DEFINITION: *Frictional force*

Frictional force is the force that opposes the motion of an object in contact with a surface and it acts parallel to the surface the object is in contact with.



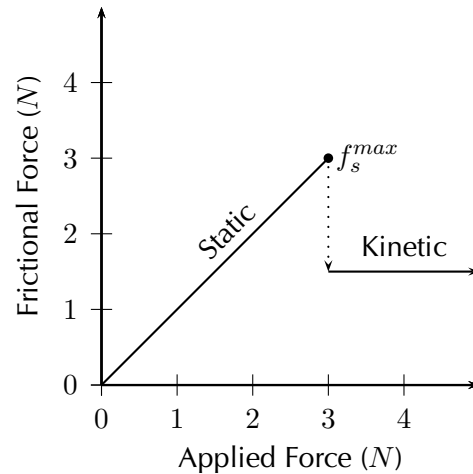
The magnitude of the frictional force depends on the surface and the magnitude of the normal force. Different surfaces will give rise to different frictional forces, even if the normal force is the same. Frictional forces are proportional to the magnitude of the normal force.

$$F_{\text{friction}} \propto N$$

For every surface we can determine a constant factor, the coefficient of friction, that allows us to calculate what the frictional force would be if we know the magnitude of the normal force. We know that static friction and kinetic friction have different magnitudes so we have different coefficients for the two types of friction:

- μ_s is the coefficient of **static** friction
- μ_k is the coefficient of **kinetic** friction

A force is not always large enough to make an object move, for example a small applied force might not be able to move a heavy crate. The frictional force opposing the motion of the crate is equal to the applied force but acting in the opposite direction. This frictional force is called *static friction*. When we increase the applied force (push harder), the frictional force will also increase until it reaches a maximum value. When the applied force is larger than the maximum force of static friction the object will move. The static frictional force can vary from zero (when no other forces are present and the object is stationary) to a maximum that depends on the surfaces.



For static friction the force can vary up to some maximum value after which friction has been overcome and the object starts to move. So we define a maximum value for the static friction: $f_s^{max} = \mu_s N$.

When the applied force is greater than the maximum, static frictional force, the object moves but still experiences friction. This is called *kinetic friction*. For kinetic friction the value remains the same regardless of the magnitude of the applied force. The magnitude of the kinetic friction is: $f_k = \mu_k N$.



Remember that static friction is present when the object is not moving and kinetic friction while the object is moving. For example when you drive at constant velocity in a car on a tar road you have to keep the accelerator pushed in slightly to overcome the friction between the tar road and the wheels of the car. However, while moving at a constant velocity the wheels of the car are rolling, so this is not a case of two surfaces “rubbing” against each other and we are in fact looking at static friction. If you should break hard, causing the car to skid to a halt, we would be dealing with two surfaces rubbing against each other and hence kinetic friction. The higher the value for the coefficient of friction, the more ‘sticky’ the surface is and the lower the value, the more ‘slippery’ the surface is.

Friction is very useful. If there was no friction and you tried to prop a ladder up against a wall, it would simply slide to the ground. Rock climbers use friction to maintain their grip on cliffs. The brakes of cars would be useless if it wasn’t for friction!



Early humans made use of friction to create fire. Friction can create a lot of heat and

the early humans used this fact when they rubbed two sticks together to start a fire. When you rub your hands together fast and pressing hard you will feel that they get warm. This is heat created by the friction. You can use this to start a fire.

To start a fire you need two pieces of wood, one long straight, round piece approximately the same thickness as your finger and about 40 cm long as well as a thicker flat piece of wood. The flat thick piece of wood needs a hole that the long straight one can fit into. Then you put the flat piece on the ground, the long straight one in the hole and rub it between your hands applying downwards pressure to increase the normal force and the amount of friction. Where the two pieces of wood rub against each other the friction results in many tiny pieces of wood being rubbed off and getting hot.



▶ See video: [23GW](https://www.everythingscience.co.za/23GW) at www.everythingscience.co.za

After a while the hole will start to smoke. At this point the smoking wood bits, called the ember, need to be gently tipped out of the hole into a small bed of dry grass. You cover the ember completely and blow gently. The grass should start to burn. Then you use the burning grass to light some dry twigs, and keep working your way up to bigger pieces of wood.

To make it even easier, a bow from wood with a string can be used to cause the wood to turn. By twisting the string from the bow around the long piece of wood it can be driven without requiring a person use their hands.

Worked example 1: Static friction

QUESTION

A box resting on a surface experiences a normal force of magnitude 30 N and the coefficient of static friction between the surface and the box, μ_s , is 0,34. What is the maximum static frictional force?

SOLUTION

Step 1: Maximum static friction

We know that the relationship between the maximum static friction, f_s^{max} , the coefficient of static friction, μ_s and the normal, N , to be:

$$f_s^{max} = \mu_s N$$

We have been given that $\mu_s = 0,34$ and $N = 30$ N. This is all of the information required to do the calculation.

Step 2: Calculate the result

$$\begin{aligned}
 f_s^{max} &= \mu_s N \\
 &= (0,34)(30) \\
 &= 10,2
 \end{aligned}$$

The maximum magnitude of static friction is 10,2 N.

Worked example 2: Static friction

QUESTION

The forwards of your school's rugby team are trying to push their scrum machine. The normal force exerted on the scrum machine is 10 000 N. The machine isn't moving at all. If the coefficient of static friction is 0,78 what is the minimum force they need to exert to get the scrum machine to start moving?

SOLUTION

Step 1: Minimum or maximum

The question asks what the minimum force required to get the scrum machine moving will be. We don't know a relationship for this but we do know how to calculate the maximum force of static friction. The forwards need to exert a force greater than this so the minimum amount they can exert is in fact equal to the maximum force of static friction.

Step 2: Maximum static friction

We know that the relationship between the maximum static friction, f_s^{max} , the coefficient of static friction, μ_s and the normal, N , to be:

$$f_s^{max} = \mu_s N$$

We have been given that $\mu_s = 0,78$ and $N = 10\,000$ N. This is all of the information required to do the calculation.

Step 3: Calculate the result

$$\begin{aligned}
 f_s^{max} &= \mu_s N \\
 &= (0,78)(10\,000) \\
 &= 7800 \text{ N}
 \end{aligned}$$

The maximum magnitude of static friction is 7800 N.

QUESTION

The normal force exerted on a pram is 100 N. The pram's brakes are locked so that the wheels cannot turn. The owner tries to push the pram but it doesn't move. The owner pushes harder and harder until it suddenly starts to move when the applied force is three quarters of the normal force. After that the owner is able to keep it moving with a force that is half of the force at which it started moving. What is the magnitude of the applied force at which it starts moving and what are the coefficients of static and kinetic friction?

SOLUTION**Step 1: Maximum static friction**

The owner of the pram increases the force he is applying until suddenly the pram starts to move. This will be equal to the maximum static friction which we know is given by:

$$f_s^{max} = \mu_s N$$

We are given that the magnitude of the applied force is three quarters of the normal force magnitude, so:

$$\begin{aligned} f_s^{max} &= \frac{3}{4}N \\ &= \frac{3}{4}(100) \\ &= 75 \text{ N} \end{aligned}$$

Step 2: Coefficient of static friction

We now know both the maximum magnitude of static friction and the magnitude of the normal force so we can find the coefficient of static friction:

$$\begin{aligned} f_s^{max} &= \mu_s N \\ 75 &= \mu_s (100) \\ \mu_s &= 0,75 \end{aligned}$$

Step 3: Coefficient of kinetic friction

The magnitude of the force required to keep the pram moving is half of the magnitude of the force required to get it to start moving so we can determine it from:

$$\begin{aligned} f_k &= \frac{1}{2}f_s^{max} \\ &= \frac{1}{2}(75) \\ &= 37,5 \text{ N} \end{aligned}$$

We know the relationship between the magnitude of the kinetic friction, magnitude of the normal force and coefficient of kinetic friction. We can use it to solve for the

coefficient of kinetic friction:

$$\begin{aligned}f_k &= \mu_k N \\37,5 &= \mu_k (100) \\ \mu_k &= 0,375\end{aligned}$$

Worked example 4: Coefficient of static friction

QUESTION

A block of wood experiences a normal force of 32 N from a rough, flat surface. There is a rope tied to the block. The rope is pulled parallel to the surface and the tension (force) in the rope can be increased to 8 N before the block starts to slide. Determine the coefficient of static friction.

SOLUTION

Step 1: Analyse the question and determine what is asked

The normal force is given (32 N) and we know that the block does not move until the applied force is 8 N.

We are asked to find the coefficient for static friction μ_s .

Step 2: Find the coefficient of static friction

$$\begin{aligned}F_f &= \mu_s N \\8 &= \mu_s (32) \\ \mu_s &= 0,25\end{aligned}$$

Note that the coefficient of friction does not have a unit as it shows a ratio. The value for the coefficient of friction can have any value up to a maximum of 0,25. When a force less than 8 N is applied, the coefficient of friction will be less than 0,25.

Worked example 5: Static friction

QUESTION

A box resting on an inclined plane experiences a normal force of magnitude 130 N and the coefficient of static friction, μ_s , between the box and the surface is 0,47. What

is the maximum static frictional force?

SOLUTION

Step 1: Maximum static friction

We know that the relationship between the maximum static friction, f_s^{max} , the coefficient of static friction, μ_s and the normal, N , to be:

$$f_s^{max} = \mu_s N.$$

This does not depend on whether the surface is inclined or not. Changing the inclination of the surface will affect the magnitude of the normal force but the method of determining the frictional force remains the same.

We have been given that $\mu_s = 0,47$ and $N = 130 \text{ N}$. This is all of the information required to do the calculation.

Step 2: Calculate the result

$$\begin{aligned} f_s^{max} &= \mu_s N \\ &= (0,47)(130) \\ &= 61,1 \text{ N} \end{aligned}$$

The maximum magnitude of static friction is 61,1 N.

Informal experiment: Normal forces and friction

Aim:

To investigate the relationship between normal force and friction.

Apparatus:

Spring balance, several blocks, of the same material, with hooks attached to one end, several rough and smooth surfaces, bricks or blocks to incline the surfaces

Method:

- Attach each of the blocks to the spring balance in turn and note the reading.
- Take one block and attach it to the spring balance. Now slide the block along each of the surfaces in turn. Note the readings. Repeat for the other blocks.
- Repeat the above step but incline the surfaces at different angles.

Results:

Tension

Tension is the magnitude of the force that exists in objects like ropes, chains and struts that are providing support. For example, there are tension forces in the ropes supporting a child's swing hanging from a tree.

Exercise 2 – 1:

1. A box is placed on a rough surface. It has a normal force of magnitude 120 N. A force of 20 N applied to the right cannot move the box. Calculate the magnitude and direction of the friction forces.
2. A block rests on a horizontal surface. The normal force is 20 N. The coefficient of static friction between the block and the surface is 0,40 and the coefficient of dynamic friction is 0,20.
 - a) What is the magnitude of the frictional force exerted on the block while the block is at rest?
 - b) What will the magnitude of the frictional force be if a horizontal force of magnitude 5 N is exerted on the block?
 - c) What is the minimum force required to start the block moving?
 - d) What is the minimum force required to keep the block in motion once it has been started?
 - e) If the horizontal force is 10 N, determine the frictional force.
3. A lady injured her back when she slipped and fell in a supermarket. She holds the owner of the supermarket accountable for her medical expenses. The owner claims that the floor covering was not wet and meets the accepted standards. He therefore cannot accept responsibility. The matter eventually ends up in court. Before passing judgement, the judge approaches you, a science student, to determine whether the coefficient of static friction of the floor is a minimum of 0,5 as required. He provides you with a tile from the floor, as well as one of the shoes the lady was wearing on the day of the incident.
 - a) Write down an expression for the coefficient of static friction.
 - b) Plan an investigation that you will perform to assist the judge in his judgement. Follow the steps outlined below to ensure that your plan meets the requirements.
 - i. Formulate an investigation question.
 - ii. Apparatus: List *all* the other apparatus, except the tile and the shoe, that you will need.
 - iii. A stepwise method: How will you perform the investigation? Include a relevant, labelled diagram.
 - iv. Results: What will you record?
 - v. Conclusion: How will you interpret the results to draw a conclusion?

Think you got it? Get this answer and more practice on our Intelligent Practice Service

1. 23GX 2. 23GY 3. 23GZ



www.everythingscience.co.za

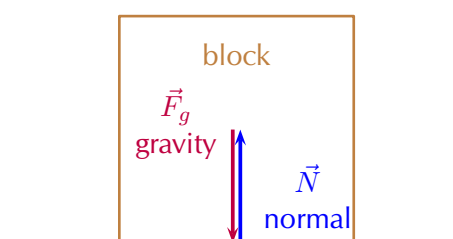


m.everythingscience.co.za

Force diagrams

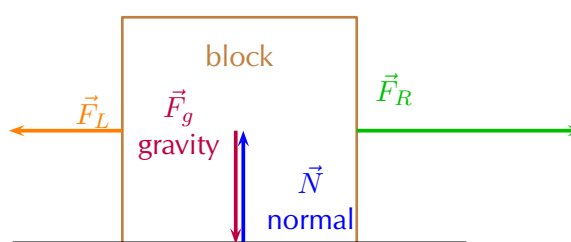
ESBKM

Force diagrams are sketches of the physical situation you are dealing with, with arrows for all the forces acting drawn on the system. For example, if a block rests on a surface then there is a force from gravity pulling the block down and there is a normal force acting on the block from the surface. The normal force and the force of gravity have the same magnitude in this situation. The force diagram for this situation is:



The length of the arrows are the same to indicate that the forces have the same magnitude.

Another example could be a the same block on a surface but with an applied force, $\vec{F}_L$, to the left of 10 N and an applied force, $\vec{F}_R$, to the right of 20 N. The weight and normal also have magnitudes of 10 N.



It is important to keep the following in mind when you draw force diagrams:

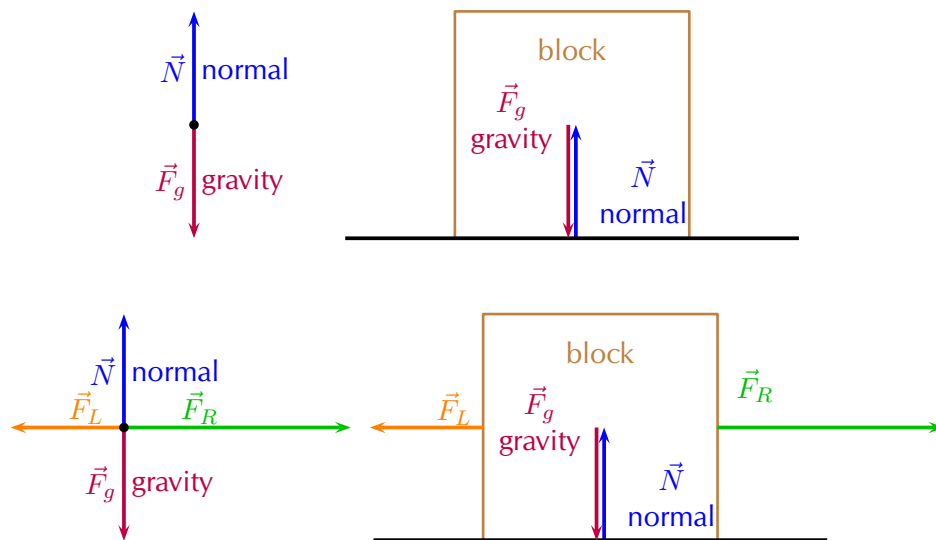
- Make your drawing large and clear.
- You must use arrows and the direction of the arrow will show the direction of the force.
- The length of the arrow will indicate the size of the force, in other words, the longer arrows in the diagram (F_R for example) indicates a bigger force than a shorter arrow (F_L). Arrows of the same length indicate forces of equal size (F_N and F_g). Use “little lines” like in maths to show this.

- Draw neat lines using a ruler. The arrows must touch the system or object.
- All arrows must have labels. Use letters with a key on the side if you do not have enough space on your drawing.
- The labels must indicate what is applying the force (the force of the car) on what the force is applied (on the trailer) and in which direction (to the right)
- If the values of the forces are known, these values can be added to the diagram or key.

Free body diagrams

ESBKN

In a free-body diagram, the object of interest is drawn as a dot and all the forces acting on it are drawn as arrows pointing away from the dot. We can redraw the two force diagrams above as free body diagrams:

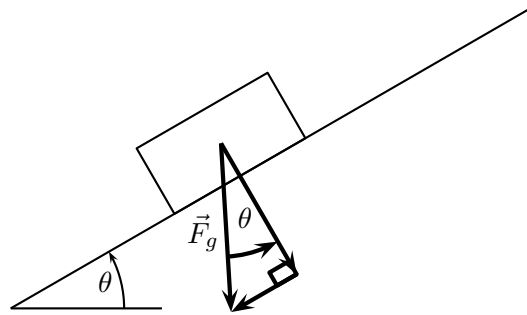


Resolving forces into components

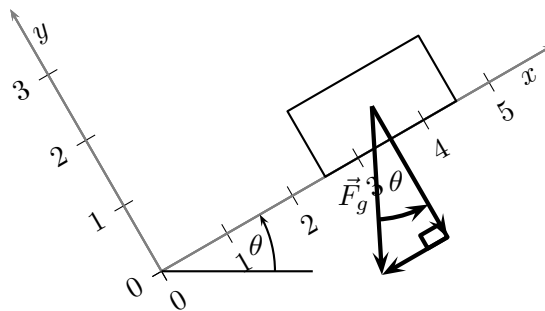
ESBKP

We have looked at resolving forces into components. There is one situation we will consider where this is particularly useful, problems involving an inclined plane. It is important because the normal force depends on the component of the gravitational force that is perpendicular to the slope.

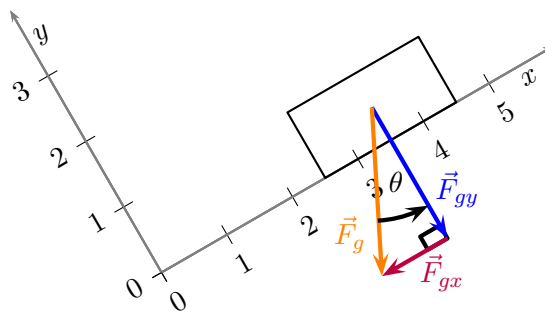
Let us consider a block on an inclined plane. The plane is inclined at an angle θ to the horizontal. It feels a gravitational force, $\vec{F}_g$, directly downwards. This force can be broken into components that are perpendicular to the plane and parallel to it. This is shown here:



We can use any coordinate system to describe the situation and the simplest thing to do is to make the x -axis of the Cartesian coordinate system align with the inclined plane. Here is the same physical situation with the coordinate system drawn in:



This means that the components of gravitational force are aligned (parallel) with one of the axes of the coordinate system. We have shown in the figures that the angle between the horizontal and the incline is also the angle between the gravitational force and its component perpendicular to the inclined plane (this is normal to the plane). Using this angle and the fact that the components form part of a right-angled triangle we can calculate the components using trigonometry:



Using trigonometry the components are given by:

$$F_{gx} = F_g \sin(\theta)$$

$$F_{gy} = F_g \cos(\theta)$$

Worked example 6: Components of force due to gravity

QUESTION

A block on an inclined plane experiences a force due to gravity, $\vec{F}_g$ of 137 N straight down. If the slope is inclined at 37° to the horizontal, what is the component of the force due to gravity perpendicular and parallel to the slope?

SOLUTION

Step 1: Components

We know that for a block on a slope we can resolve the force due to gravity, $\vec{F}_g$ into components parallel and perpendicular to the slope.

$$F_{gx} = F_g \sin(\theta)$$

$$F_{gy} = F_g \cos(\theta)$$

Step 2: Calculations

This problem is straightforward as we know that the slope is inclined at an angle of 37° . This is the same angle we need to use to calculate the components, therefore:

$$\begin{aligned} F_{gx} &= F_g \sin(\theta) \\ &= (137) \sin(37^\circ) \\ &= 82,45 \text{ N} \end{aligned}$$

$$\begin{aligned} F_{gy} &= F_g \cos(\theta) \\ &= (137) \cos(37^\circ) \\ &= 109,41 \text{ N} \end{aligned}$$

Step 3: Final answer

The component of $\vec{F}_g$ that is perpendicular to the slope is $\vec{F}_{gy} = 109,41 \text{ N}$ in the negative y -direction.

The component of $\vec{F}_g$ that is perpendicular to the slope is $\vec{F}_{gx} = 82,45 \text{ N}$ in the negative x -direction.

Exercise 2 – 2:

1. A block on an inclined plane experiences a force due to gravity, $\vec{F}_g$ of 456 N straight down. If the slope is inclined at $67,8^\circ$ to the horizontal, what is the component of the force due to gravity perpendicular and parallel to the slope?
2. A block on an inclined plane is subjected to a force due to gravity, $\vec{F}_g$ of 456 N straight down. If the component of the gravitational force parallel to the slope is $\vec{F}_{gx} = 308,7 \text{ N}$ in the negative x -direction (down the slope), what is the incline of the slope?

Think you got it? Get this answer and more practice on our Intelligent Practice Service

1. [23H2](#) 2. [23H3](#)



Finding the resultant force

ESBKQ

The easiest way to determine a resultant force is to draw a free body diagram. Remember from [Chapter 1](#) that we use the length of the arrow to indicate the vector's magnitude and the direction of the arrow to show which direction it acts in.

After we have done this, we have a diagram of vectors and we simply find the sum of the vectors to get the resultant force.

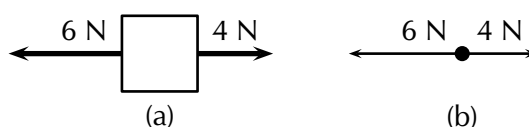


Figure 2.4: (a) Force diagram of 2 forces acting on a box. (b) Free body diagram of the box.

For example, two people push on a box from opposite sides with forces of 4 N and 6 N respectively as shown in Figure 2.4 (a). The free body diagram in Figure 2.4 (b) shows the object represented by a dot and the two forces are represented by arrows with their tails on the dot.

As you can see, the arrows point in opposite directions and have different lengths. The resultant force is 2 N to the left. This result can be obtained algebraically too, since the two forces act along the same line. First, as in motion in one direction, choose a frame of reference. Secondly, add the two vectors taking their directions into account.

For the example, assume that the positive direction is to the right, then:

$$\begin{aligned} F_R &= (4) + (-6) \\ &= -2 \\ &= -2 \text{ N to the left.} \end{aligned}$$

Remember that a negative answer means that the force acts in the *opposite* direction to the one that you chose to be positive. You can *choose* the positive direction to be any way you want, but once you have chosen it you *must* use it consistently for that problem.

As you work with more force diagrams in which the forces exactly balance, you may notice that you get a zero answer (e.g. 0 N). This simply means that the forces are balanced and the resultant is zero.

Once a force diagram has been drawn the techniques of vector addition introduced in [Chapter 1](#) can be used. Depending on the situation you might choose to use a graphical technique such as the tail-to-head method or the parallelogram method, or else an algebraic approach to determine the resultant. Since force is a vector quantity all of these methods apply.

A good strategy is:

- resolve all forces into components parallel to the x - and y -directions;
- calculate the resultant in each direction, $\vec{R}_x$ and $\vec{R}_y$, using co-linear vectors; and
- use $\vec{R}_x$ and $\vec{R}_y$ to calculate the resultant, $\vec{R}$.

Worked example 7: Finding the resultant force

QUESTION

A car (experiencing a gravitational force of magnitude 12 000 N and a normal force of the same magnitude) applies a force of 2000 N on a trailer (experiencing a gravitational force of magnitude 2500 N and normal force of the same magnitude). A constant frictional force of magnitude 200 N is acting on the trailer, and a constant frictional force of magnitude 300 N is acting on the car.

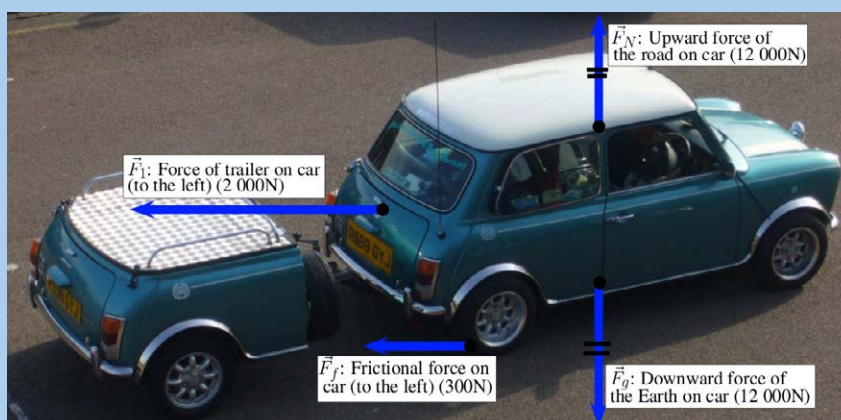


1. Draw a force diagram of all the forces acting on the car.
2. Draw a free body diagram of all the forces acting on the trailer.
3. Use the force diagram to determine the resultant force on the trailer.

SOLUTION

Step 1: Draw the force diagram for the car.

The question asks us to draw all the forces on the car. This means that we must include horizontal and vertical forces.



Step 2: Determine the resultant force on the trailer.

To find the resultant force we need to add all the horizontal forces together. We do not add vertical forces as the movement of the car and trailer will be in a horizontal direction, and not up or down. $F_R = 2000 + (-200) = 1800\text{N}$ to the right.

► See simulation: 23H4 at www.everythingscience.co.za

Exercise 2 – 3: Forces and motion

1. A boy pushes a shopping trolley (weight due to gravity of 150 N) with a constant force of 75 N. A constant frictional force of 20 N is present.
 - a) Draw a labelled force diagram to identify all the forces acting on the shopping trolley.
 - b) Draw a free body diagram of all the forces acting on the trolley.
 - c) Determine the resultant force on the trolley.
2. A donkey (experiencing a gravitational force of 2500 N) is trying to pull a cart (force due to gravity of 800 N) with a force of 400 N. The rope between the donkey and the cart makes an angle of 30° with the cart. The cart does not move.
 - a) Draw a free body diagram of all the forces acting on the donkey.
 - b) Draw a force diagram of all the forces acting on the cart.
 - c) Find the magnitude and direction of the frictional force preventing the cart from moving.

Think you got it? Get this answer and more practice on our Intelligent Practice Service

1. 23H5 2. 23H6



www.everythingscience.co.za



m.everythingscience.co.za

2.3 Newton's laws

ESBKR

In this section we will look at the effect of forces on objects and how we can make things move. This will link together what you have learnt about motion and what you have learnt about forces.

Sir Isaac Newton was a scientist who lived in England (1642-1727) who was interested in the motion of objects under various conditions. He suggested that a stationary object will remain stationary unless a force acts on it and that a moving object will continue moving unless a force slows it down, speeds it up or changes its direction of motion. From this he formulated what is known as Newton's first law of motion:

DEFINITION: *Newton's first law of motion*

An object continues in a state of rest or uniform motion (motion with a constant velocity) unless it is acted on by an unbalanced (net or resultant) force.

This property of an object, to continue in its current state of motion unless acted upon by a net force, is called *inertia*.

Let us consider the following situations:



An ice skater pushes herself away from the side of the ice rink and skates across the ice. She will continue to move in a straight line across the ice unless something stops her. Objects are also like that. If we kick a soccer ball across a soccer field, according to Newton's first law, the soccer ball should keep on moving forever! However, in real life this does not happen.

Is Newton's Law wrong? Not really. Newton's first law applies to situations where there aren't any external forces present. This means that friction is not present. In the case of the ice skater, the friction between the skates and the ice is very little and she will continue moving for quite a distance. In the case of the soccer ball, air resistance (friction between the air and the ball) and friction between the grass and the ball is present and this will slow the ball down.



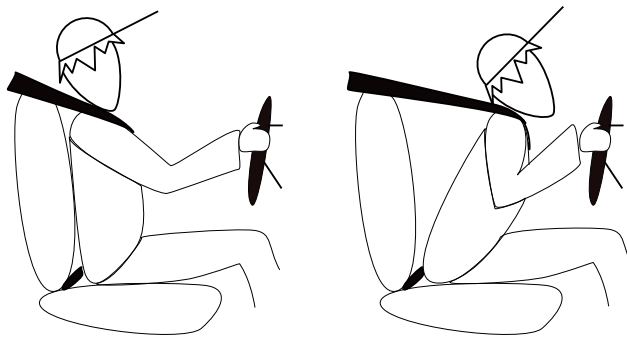
▶ See video: 23H7 at www.everythingscience.co.za

Newton's first law in action

Let us look at the following two examples. We will start with a familiar example:

Seat belts:

We wear seat belts in cars. This is to protect us when the car is involved in an accident. If a car is travelling at $120 \text{ km}\cdot\text{h}^{-1}$, the passengers in the car are also travelling at $120 \text{ km}\cdot\text{h}^{-1}$. When the car suddenly stops a force is exerted on the car (making it slow down), but not on the passengers. The passengers will carry on moving forward at $120 \text{ km}\cdot\text{h}^{-1}$ according to Newton's first law. If they are wearing seat belts, the seat belts will stop them by exerting a force on them and so prevent them from getting hurt.



▶ See video: 23H8 at www.everythingscience.co.za

Rockets:

A spaceship is launched into space. The force of the exploding gases pushes the rocket through the air into space. Once it is in space, the engines are switched off and it will keep on moving at a constant velocity. If the astronauts want to change the direction of the spaceship they need to fire an engine. This will then apply a force on the rocket and it will change its direction.

Worked example 8: Newton's first law in action

QUESTION

Why do passengers get thrown to the side when the car they are driving in goes around a corner?

SOLUTION

Step 1: What happens before the car turns

Before the car starts turning both the passengers and the car are travelling at the same velocity. (picture A)

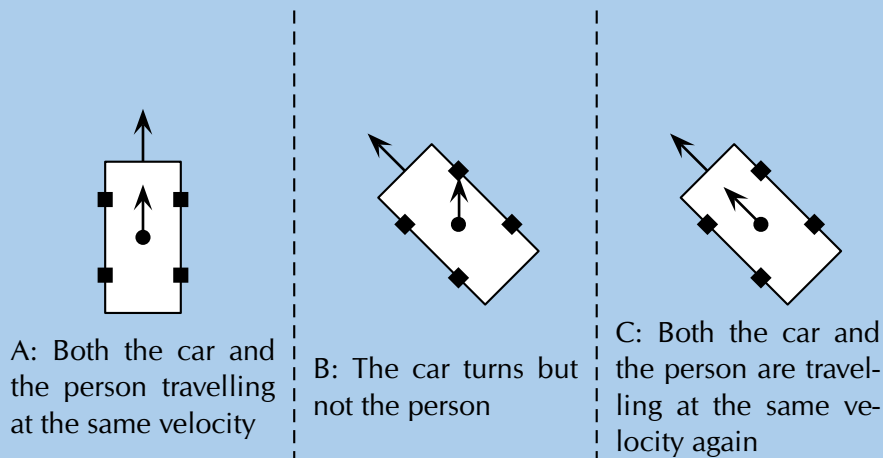
Step 2: What happens while the car turns

The driver turns the wheels of the car, which then exert a force on the car and the car turns. This force acts on the car but not the passengers, hence (by Newton's first law)

the passengers continue moving with the same original velocity. (picture B)

Step 3: Why passengers get thrown to the side?

If the passengers are wearing seat belts they will exert a force on the passengers until the passengers' velocity is the same as that of the car (picture C). Without a seat belt the passenger may hit the side of the car.



Exercise 2 – 4:

1. If a passenger is sitting in a car and the car turns round a bend to the right, what happens to the passenger? What happens if the car turns to the left?
2. Helium is less dense than the air we breathe. Discuss why a helium balloon in a car driving around a corner appears to violate Newton's first law and moves towards the inside of the turn not the outside like a passenger.

Think you got it? Get this answer and more practice on our Intelligent Practice Service

1. [23H9](#) 2. [23HB](#)



www.everythingscience.co.za



m.everythingscience.co.za

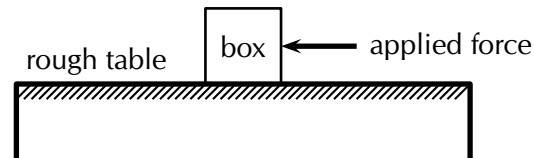
Newton's second law of motion

ESBKT

According to Newton's first law, things 'like to keep on doing what they are doing'. In other words, if an object is moving, it tends to continue moving (in a straight line and at the same speed) and if an object is stationary, it tends to remain stationary. So how do objects start moving?

Let us look at the example of a 10 kg box on a rough table. If we push lightly on the

box as indicated in the diagram, the box won't move. Let's say we applied a force of 100 N, yet the box remains stationary. At this point a frictional force of 100 N is acting on the box, preventing the box from moving. If we increase the force, let's say to 150 N and the box almost starts to move, the frictional force is 150 N. To be able to move the box, we need to push hard enough to overcome the friction and then move the box. If we therefore apply a force of 200 N remembering that a frictional force of 150 N is present, the 'first' 150 N will be used to overcome or 'cancel' the friction and the other 50 N will be used to move (accelerate) the block. In order to accelerate an object we must have a resultant force acting on the block.



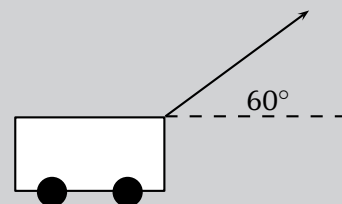
Now, what do you think will happen if we pushed harder, let's say 300 N? Or, what do you think will happen if the mass of the block was more, say 20 kg, or what if it was less? Let us investigate how the motion of an object is affected by mass and force.

Formal experiment: Newton's second law of motion

Aim:

To investigate the relation between the acceleration of objects and the application of a constant resultant force.

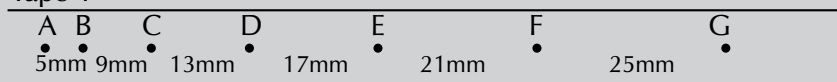
Method:



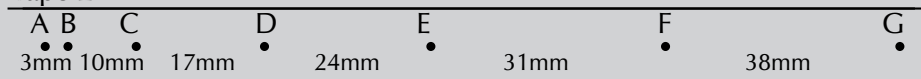
1. A constant force of 20 N, acting at an angle of 60° to the horizontal, is applied to a dynamics trolley.
2. Ticker tape attached to the trolley runs through a ticker timer of frequency 20 Hz as the trolley is moving on the frictionless surface.
3. The above procedure is repeated 4 times, each time using the same force, but varying the mass of the trolley as follows:
 - Case 1: 6,25 kg
 - Case 2: 3,57 kg
 - Case 3: 2,27 kg
 - Case 4: 1,67 kg

4. Shown below are sections of the four ticker tapes obtained. The tapes are marked with the letters A, B, C, D, etc. A is the first dot, B is the second dot and so on. The distance between each dot is also shown.

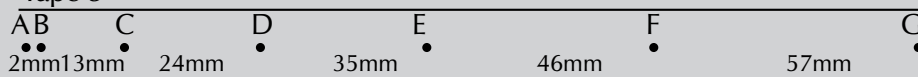
Tape 1



Tape 2



Tape 3



Tape 4



Tapes are not drawn to scale.

Instructions:

1. Use each tape to calculate the instantaneous velocity (in $\text{m}\cdot\text{s}^{-1}$) of the trolley at points B and F (remember to convert the distances to m first!). Use these velocities to calculate the trolley's acceleration in each case.
2. Tabulate the mass and corresponding acceleration values as calculated in each case. Ensure that each column and row in your table is appropriately labelled.
3. Draw a graph of acceleration vs. mass, using a scale of $1\text{ cm} = 1\text{ m}\cdot\text{s}^{-2}$ on the y-axis and $1\text{ cm} = 1\text{ kg}$ on the x-axis.
4. Use your graph to read off the acceleration of the trolley if its mass is 5 kg.
5. Write down a conclusion for the experiment.

You will have noted in the investigation above that the heavier the trolley is, the slower it moved when the force was constant. The acceleration is *inversely* proportional to the mass. In mathematical terms: $a \propto \frac{1}{m}$

In a similar investigation where the mass is kept constant, but the applied force is varied, you will find that the bigger the force is, the faster the object will move. The acceleration of the trolley is therefore *directly* proportional to the resultant force. In mathematical terms: $a \propto F$.

Rearranging the above equations, we get $a \propto \frac{F}{m}$ **or** $F = ma$.

Remember that both force and acceleration are vectors quantities. The acceleration is

in the same direction as the force that is being applied. If multiple forces are acting simultaneously then we only need to work with the resultant force or net force.

DEFINITION: *Newton's second law of motion*

If a resultant force acts on a body, it will cause the body to accelerate in the direction of the resultant force. The acceleration of the body will be directly proportional to the resultant force and inversely proportional to the mass of the body. The mathematical representation is:

$$\vec{F}_{net} = m\vec{a}$$

Force is a vector quantity. Newton's second law of motion should be applied to the y - and x -directions separately. You can use the resulting y - and x -direction resultants to calculate the overall resultant as we saw in the previous chapter.

► See video: 23HC at www.everythingscience.co.za

Applying Newton's second law of motion

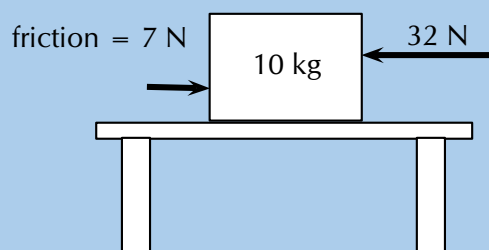
Newton's second law can be applied to a variety of situations. We will look at the main types of examples that you need to study.

Worked example 9: Newton's second law: Box on a surface

QUESTION

A 10 kg box is placed on a table. A horizontal force of magnitude 32 N is applied to the box. A frictional force of magnitude 7 N is present between the surface and the box.

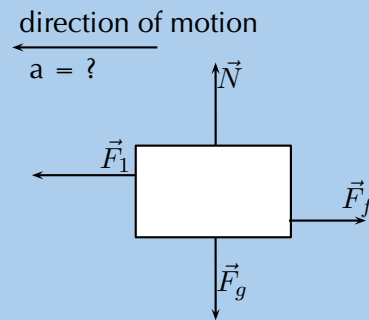
1. Draw a force diagram indicating all of the forces acting on the box.
2. Calculate the acceleration of the box.



SOLUTION

Step 1: Identify the horizontal forces and draw a force diagram

We only look at the forces acting in a horizontal direction (left-right) and not vertical (up-down) forces. The applied force and the force of friction will be included. The force of gravity, which is a vertical force, will not be included.



Step 2: Calculate the acceleration of the box

Remember that we consider the y - and x -directions separately. In this problem we can ignore the y -direction because the box is resting on a table with the gravitational force balanced by the normal force.

We have been given:

Applied force $F_1 = 32 \text{ N}$

Frictional force $F_f = -7 \text{ N}$

Mass $m = 10 \text{ kg}$

To calculate the acceleration of the box we will be using the equation $\vec{F}_R = m\vec{a}$. Therefore:

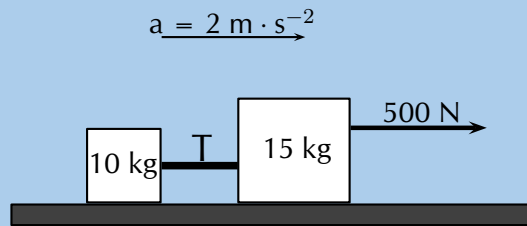
$$\begin{aligned}\vec{F}_R &= m\vec{a} \\ \vec{F}_1 + \vec{F}_f &= (10)\vec{a} \\ (32 - 7) &= (10)\vec{a} \\ 25 &= (10)\vec{a} \\ \vec{a} &= 0,25 \text{ m}\cdot\text{s}^{-2} \text{ to the left.}\end{aligned}$$

Worked example 10: Newton's second law: box on a surface

QUESTION

Two crates, 10 kg and 15 kg respectively, are connected with a thick rope according to the diagram. A force, to the right, of 500 N is applied. The boxes move with an acceleration of $2 \text{ m}\cdot\text{s}^{-2}$ to the right. One third of the total frictional force is acting on the 10 kg block and two thirds on the 15 kg block. Calculate:

1. the magnitude and direction of the total frictional force present.
2. the magnitude of the tension in the rope at T.



SOLUTION

Step 1: Evaluate what is given

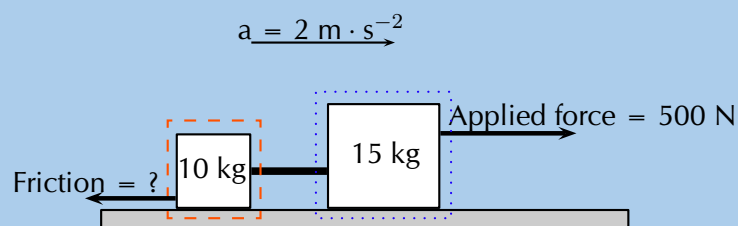
To make things easier let's give the two crates labels, let us call the 10 kg crate number 2 and the 15 kg crate number 1.

We have two crates that have overall has an acceleration that is given. The fact that the crates are tied together with a rope means that they will both have the same acceleration. They will also both feel the same force due to the tension in the rope.

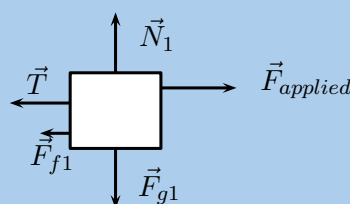
We are told that there is friction but we are only given the relationship between the total frictional force both crates experience and the fraction each one experiences. The total friction, $\vec{F}_{fT}$ will be the sum of the friction on crate 1, $\vec{F}_{f1}$, and the friction on crate 2, $\vec{F}_{f2}$. We are told that $\vec{F}_{f1} = \frac{2}{3}\vec{F}_{fT}$ and $\vec{F}_{f2} = \frac{1}{3}\vec{F}_{fT}$. We know the blocks are accelerating to the right and we know that friction will be in the direction opposite to the direction of motion and parallel to the surface.

Step 2: Draw force diagrams

The diagram for the crates is:



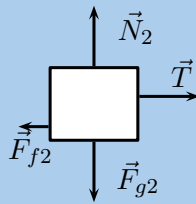
The diagram for crate 1 (indicated by blue dotted lines) will be:



Where:

- $\vec{F}_{g1}$ is the force due to gravity on the first crate
- $\vec{N}_1$ is the normal force from the surface on the first crate
- $\vec{T}$ is the force of tension in the rope
- $\vec{F}_{applied}$ is the external force being applied to the crate
- $\vec{F}_{f1}$ is the force of friction on the first crate

The diagram for crate 2 (indicated by orange dashed lines) will be:



Where:

- $\vec{F}_{g2}$ is the force due to gravity on the second crate
- $\vec{N}_2$ is the normal force from the surface on the second crate
- $\vec{T}$ is the force of tension in the rope
- $\vec{F}_{f2}$ is the force of friction on the second crate

Step 3: Apply Newton's second law of motion

The problem tells us that the crates are accelerating along the x -direction which means that the forces in the y -direction do not result in a net force. We can treat the different directions separately so we only need to consider the x -direction.

We are working with one dimension and can choose a sign convention to indicate the direction of the vectors. We choose vectors to the right (or in the positive x -direction) to be positive.

We can now apply Newton's second law of motion to the first crate because we know the acceleration and we know all the forces acting on the crate. Using positive to indicate a force to the right we know that $F_{res1} = F_{applied} - F_{f1} - T$

$$\begin{aligned}
 \vec{F}_{res1} &= m_1 \vec{a} \\
 F_{applied} - F_{f1} - T &= m_1 a \\
 F_{applied} - \frac{2}{3} F_{fT} - T &= m_1 a \\
 (500) - \frac{2}{3} F_{fT} - T &= (15)(2) \\
 -T &= (15)(2) - (500) + \frac{2}{3} F_{fT}
 \end{aligned}$$

Now apply Newton's second law of motion to the second crate because we know the acceleration and we know all the forces acting on the crate. We know that $F_{res2} = T - F_{f2}$. **Note that tension is in the opposite direction.**

$$\begin{aligned}\vec{F}_{res2} &= m_2 \vec{a} \\ T - F_{f2} &= m_2 a \\ T - \frac{1}{3} F_{fT} &= m_2 a \\ T &= (10)(2) + \frac{1}{3} F_{fT}\end{aligned}$$

Step 4: Solve simultaneously

We have used Newton's second law of motion to create two equations with two unknowns, this means we can solve simultaneously. We solved for T in the equations above but one carries a negative sign so if we add the two equations we will subtract out the value of the tension allowing us to solve for F_{fT} :

$$\begin{aligned}(T) + (-T) &= ((10)(2) + \frac{1}{3} F_{fT}) + ((15)(2) - (500) + \frac{2}{3} F_{fT}) \\ 0 &= 20 + 30 - 500 + \frac{1}{3} F_{fT} + \frac{2}{3} F_{fT} \\ 0 &= -450 + F_{fT} \\ F_{fT} &= 450 \text{ N}\end{aligned}$$

We can substitute the magnitude of F_{fT} into the equation for crate 2 to determine the magnitude of the tension:

$$\begin{aligned}T &= (10)(2) + \frac{1}{3} F_{fT} \\ T &= (10)(2) + \frac{1}{3} (450) \\ T &= 20 + 150 \\ T &= 170 \text{ N}\end{aligned}$$

Step 5: Quote final answers

The total force due to friction is 450 N to the left. The magnitude of the force of tension is 170 N.

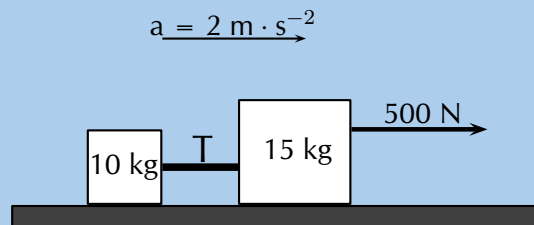
Worked example 11: Newton's second law: box on a surface (Alternative Method)

QUESTION

Two crates, 10 kg and 15 kg respectively, are connected with a thick rope according to the diagram. A force, to the right, of 500 N is applied. The boxes move with an

acceleration of $2 \text{ m} \cdot \text{s}^{-2}$ to the right. One third of the total frictional force is acting on the 10 kg block and two thirds on the 15 kg block. Calculate:

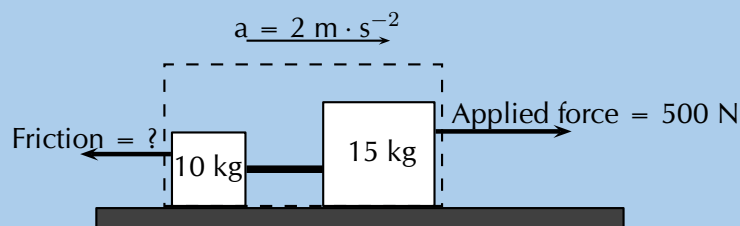
1. the magnitude and direction of the total frictional force present.
2. the magnitude of the tension in the rope at T.



SOLUTION

Step 1: Draw a force diagram

Always draw a force diagram although the question might not ask for it. The acceleration of the whole system is given, therefore a force diagram of the whole system will be drawn. Because the two crates are seen as a unit, the force diagram will look like this:



Step 2: Calculate the frictional force

To find the frictional force we will apply Newton's second law. We are given the mass ($10 + 15 \text{ kg}$) and the acceleration ($2 \text{ m} \cdot \text{s}^{-2}$). Choose the direction of motion to be the positive direction (to the right is positive).

$$\begin{aligned}
 F_R &= ma \\
 F_{\text{applied}} + F_f &= ma \\
 500 + F_f &= (10 + 15)(2) \\
 F_f &= 50 - 500 \\
 F_f &= -450 \text{ N}
 \end{aligned}$$

The frictional force is 450 N opposite to the direction of motion (to the left).

Step 3: Find the tension in the rope

To find the tension in the rope we need to look at one of the two crates on their own. Let's choose the 10 kg crate. Firstly, we need to draw a force diagram:

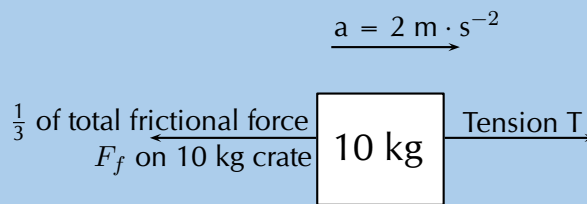


Figure 2.5: Force diagram of 10 kg crate.

The frictional force on the 10 kg block is one third of the total, therefore:

$$F_f = \frac{1}{3} \times 450$$

$$F_f = 150 \text{ N}$$

If we apply Newton's second law:

$$F_R = ma$$

$$T + F_f = (10)(2)$$

$$T + (-150) = 20$$

$$T = 170 \text{ N}$$

Note: If we had used the same principle and applied it to 15 kg crate, our calculations would have been the following:

$$F_R = ma$$

$$F_{\text{applied}} + T + F_f = (15)(2)$$

$$500 + T + (-300) = 30$$

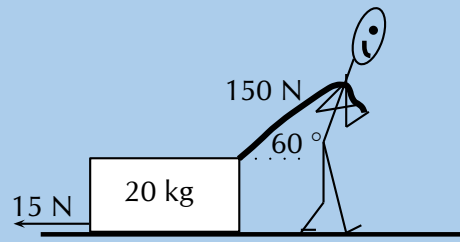
$$T = -170 \text{ N}$$

The negative answer here means that the force is in the direction opposite to the motion, in other words to the left, which is correct. However, the question asks for the magnitude of the force and your answer will be quoted as 170 N.

Worked example 12: Newton's second law: man pulling a box

QUESTION

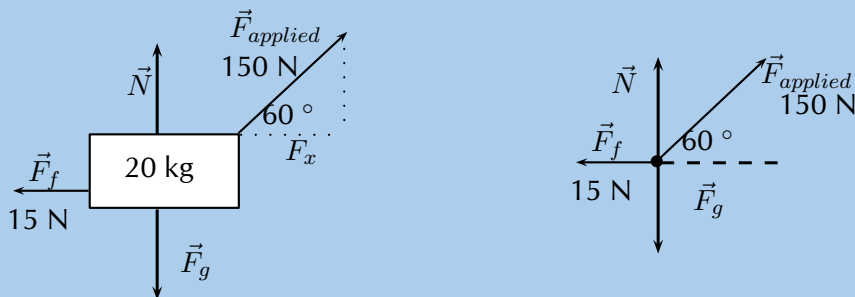
A man is pulling a 20 kg box with a rope that makes an angle of 60° with the horizontal. If he applies a force of magnitude 150 N and a frictional force of magnitude 15 N is present, calculate the acceleration of the box.



SOLUTION

Step 1: Draw a force diagram or free body diagram

The motion is horizontal and therefore we will only consider the forces in a horizontal direction. Remember that vertical forces do not influence horizontal motion and vice versa.



Step 2: Calculate the horizontal component of the applied force

We first need to choose a direction to be the positive direction in this problem. We choose the positive x -direction (to the right) to be positive.

The applied force is acting at an angle of 60° to the horizontal. We can only consider forces that are parallel to the motion. The horizontal component of the applied force needs to be calculated before we can continue:

$$\begin{aligned} F_x &= F_{applied} \cos(\theta) \\ &= 150 \cos(60^\circ) \\ &= 75 \text{ N} \end{aligned}$$

Step 3: Calculate the acceleration

To find the acceleration we apply Newton's second law:

$$\begin{aligned} F_R &= ma \\ F_x + F_f &= (20)a \\ (75) + (-15) &= (20)a \\ a &= \frac{60}{20} \\ a &= 3 \text{ m}\cdot\text{s}^{-2} \end{aligned}$$

The acceleration is 3 m·s⁻² to the right.

QUESTION

A 2000 kg truck pulls a 500 kg trailer with a constant acceleration. The engine of the truck produces a thrust of 10 000 N. Ignore the effect of friction. Calculate the:

1. acceleration of the truck; and
2. tension in the tow bar T between the truck and the trailer, if the tow bar makes an angle of 25° with the horizontal.

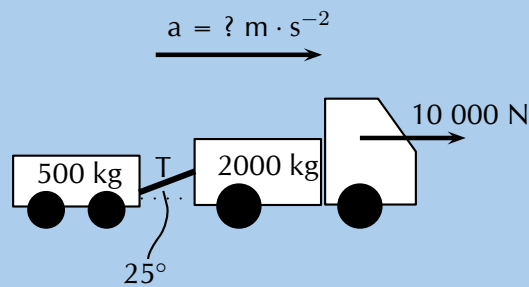


Figure 2.6: Truck pulling a trailer.

SOLUTION

Step 1: Draw a force diagram

Draw a force diagram indicating all the forces on the system as a whole:

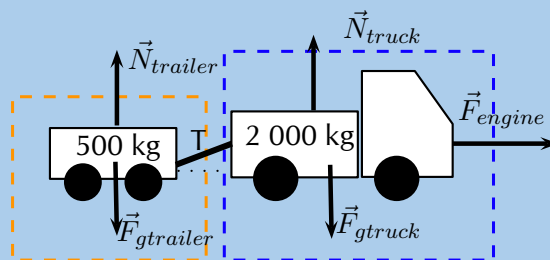
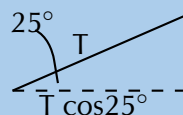


Figure 2.7: Free body diagrams for truck pulling a trailer.

Step 2: Apply Newton's second law of motion

We choose the positive x -direction to be the positive direction. We only need to consider the horizontal forces. Using only the horizontal forces means that we first need to note that the tension acts at an angle to the horizontal and we need to use the horizontal component of the tension in our calculations.



The horizontal component has a magnitude of $T \cos(25^\circ)$.

In the absence of friction, the only force that causes the system to accelerate is the thrust of the engine. If we now apply Newton's second law of motion to the truck we have:

$$\begin{aligned}\vec{F}_{Rtruck} &= m_{truck}\vec{a} \text{ (we use signs to indicate direction)} \\ F_{engine} - T \cos(25^\circ) &= (2000)a \\ (10\,000) - T \cos(25^\circ) &= (2000)a \\ a &= \frac{(10\,000) - T \cos(25^\circ)}{(2000)}\end{aligned}$$

We now apply the same principle to the trailer (remember that the direction of the tension will be opposite to the case of the truck):

$$\begin{aligned}\vec{F}_{Rtrailer} &= m_{trailer}\vec{a} \text{ (we use signs to indicate direction)} \\ T \cos(25^\circ) &= (500)a \\ a &= \frac{T \cos(25^\circ)}{(500)}\end{aligned}$$

We now have two equations and two unknowns so we can solve simultaneously. We subtract the second equation from the first to get:

$$\begin{aligned}(a) - (a) &= \left(\frac{(10\,000) - T \cos(25^\circ)}{(2000)} \right) - \left(\frac{T \cos(25^\circ)}{(500)} \right) \\ 0 &= \left(\frac{(10\,000) - T \cos(25^\circ)}{(2000)} \right) - \left(\frac{T \cos(25^\circ)}{(500)} \right) \\ &\text{(multiply through by 2000)} \\ 0 &= (10\,000) - T \cos(25^\circ) - 4T \cos(25^\circ) \\ 5T \cos(25^\circ) &= (10\,000) \\ T &= \frac{(10\,000)}{5 \cos(25^\circ)} \\ T &= 2206,76 \text{ N}\end{aligned}$$

Now substitute this result back into the second equation to solve for the magnitude of a

$$\begin{aligned}a &= \frac{T \cos(25^\circ)}{(500)} \\ &= \frac{(2206,76) \cos(25^\circ)}{(500)} \\ &= 4,00 \text{ m}\cdot\text{s}^{-2}\end{aligned}$$

TIP

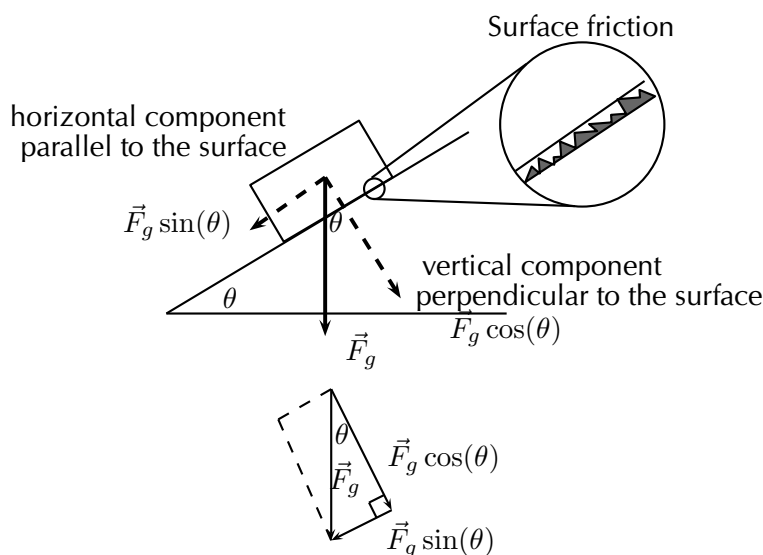
Do not use the abbreviation W for weight as it is used to abbreviate 'work'. Rather use the force of gravity F_g for weight.

Object on an inclined plane

In an earlier section we looked at the components of the gravitational force parallel and perpendicular to the slope for objects on an inclined plane. When we look at problems on an inclined plane we need to include the component of the gravitational force parallel to the slope.

Think back to the pictures of the book on a table, as one side of the table is lifted higher the book starts to slide. Why? The book starts to slide because the component of the gravitational force parallel to the surface of the table gets larger for the larger angle of inclination. This is like the applied force and it eventually becomes larger than the frictional force and the book accelerates down the table or inclined plane.

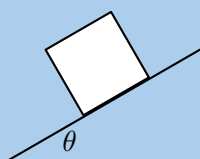
The force of gravity will also tend to push an object 'into' the slope. This is the component of the force perpendicular to the slope. There is no movement in this direction as this force is balanced by the slope pushing up against the object. This "pushing force" is the normal force (N) which we have already learnt about and is equal in magnitude to the perpendicular component of the gravitational force, but opposite in direction.



► See simulation: [23HD](#) at www.everythingscience.co.za

Worked example 14: Newton's second law: box on inclined plane**QUESTION**

A body of mass M is at rest on an inclined plane due to friction.



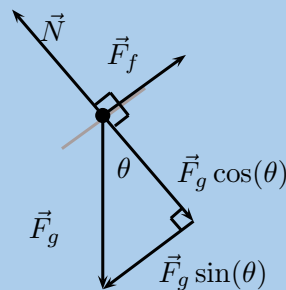
What of the following option is the magnitude of the frictional force acting on the body?

1. F_g
2. $F_g \cos(\theta)$
3. $F_g \sin(\theta)$
4. $F_g \tan(\theta)$

SOLUTION

Step 1: Analyse the situation

The question asks us to determine the magnitude of the frictional force. The body is said to be at rest on the plane, which means that it is not moving and therefore the acceleration is zero. We know that the frictional force will act parallel to the slope. If there were no friction the box would slide down the slope so friction must be acting up the slope. We also know that there will be a component of gravity perpendicular to the slope and parallel to the slope. The free body diagram for the forces acting on the block is:



Step 2: Determine the magnitude of the frictional force

We can apply Newton's second law to this problem. We know that the object is not moving so the resultant acceleration is zero. We choose up the slope to be the positive direction. Therefore:

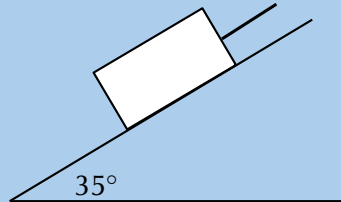
$$\begin{aligned}\vec{F}_R &= m\vec{a} \text{ using signs for direction} \\ F_f - F_g \sin(\theta) &= m(0) \\ F_f - F_g \sin(\theta) &= m(0) \\ F_f &= F_g \sin(\theta)\end{aligned}$$

Step 3: Quote your final answer

The force of friction has the same magnitude as the component of the force of gravitation parallel to the slope, $F_g \sin(\theta)$.

QUESTION

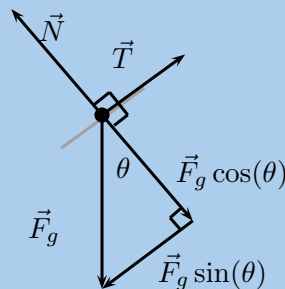
A force of magnitude $T = 312\text{N}$ up an incline is required to keep a body at rest on a frictionless inclined plane which makes an angle of 35° with the horizontal. Calculate the magnitudes of the force due to gravity and the normal force, giving your answers to three significant figures.



SOLUTION

Step 1: Find the magnitude of $\vec{F}_g$

We are usually asked to find the magnitude of $\vec{T}$, but in this case $\vec{T}$ is given and we are asked to find $\vec{F}_g$. We can use the same equation. T is the force that balances the component of $\vec{F}_g$ parallel to the plane (F_{gx}) and therefore it has the same magnitude.



We can apply Newton's second law to this problem. We know that the object is not moving so the resultant acceleration is zero. We choose up the slope to be the positive direction. Therefore:

$$\begin{aligned}\vec{F}_R &= m\vec{a} \text{ using signs for direction} \\ T - F_g \sin(\theta) &= m(0) \\ F_g &= \frac{T}{\sin(\theta)} \\ &= \frac{312}{\sin(35^\circ)} \\ &= 543,955 \text{ N}\end{aligned}$$

Step 2: Find the magnitude of $\vec{N}$

We treat the forces parallel and perpendicular to the slope separately. The block is stationary so the acceleration perpendicular to the slope is zero. Once again we can

apply Newton's second law of motion. We choose the direction of the normal force as the positive direction.

$$\begin{aligned}\vec{F}_R &= m\vec{a} \text{ using signs for direction} \\ N - F_g \cos(\theta) &= m(0) \\ N &= F_g \cos(\theta)\end{aligned}$$

We could substitute in the value of F_g calculated earlier. We would like to illustrate that there is another approach to adopt to ensure you get the correct answer even if you made a mistake calculating F_g . $F_g \cos(\theta)$ can also be determined with the use of trigonometric ratios. We know from the previous part of the question that $T = F_g \sin(\theta)$. We also know that

$$\begin{aligned}\tan(\theta) &= \frac{F_g \sin(\theta)}{F_g \cos(\theta)} \\ &= \frac{T}{N} \\ N &= \frac{T}{\tan(\theta)} \\ &= \frac{312}{\tan(35^\circ)} \\ &= 445,58 \text{ N}\end{aligned}$$

Note that the question asks that the answers be given to 3 significant figures. We therefore round $\vec{N}$ from 445,58 N up to 446 N perpendicular to the surface upwards and $\vec{T}$ from 543,955 N up to 544 N parallel to the plane, up the slope.

Lifts and rockets

So far we have looked at objects being pulled or pushed across a surface, in other words motion parallel to the surface the object rests on. Here we only considered forces parallel to the surface, but we can also lift objects up or let them fall. This is vertical motion where only vertical forces are being considered.

Let us consider a 500 kg lift, with no passengers, hanging on a cable. The purpose of the cable is to pull the lift upwards so that it can reach the next floor or lower the lift so that it can move downwards to the floor below. We will look at five possible stages during the motion of the lift and apply our knowledge of Newton's second law of motion to the situation. The 5 stages are:

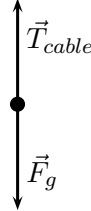
1. A stationary lift suspended above the ground.
2. A lift accelerating upwards.
3. A lift moving at a constant velocity.
4. A lift decelerating (slowing down).
5. A lift accelerating downwards (the cable snaps!).

We choose the upwards direction to be the positive direction for this discussion.

Stage 1:

The 500 kg lift is stationary at the second floor of a tall building.

The lift is not accelerating. There must be a tension $\vec{T}$ from the cable acting on the lift and there must be a force due to gravity, $\vec{F}_g$. There are no other forces present and we can draw the free body diagram:



We apply Newton's second law to the vertical direction:

$$\begin{aligned}\vec{F}_R &= m_{\text{lift}}\vec{a} \text{ (we use signs to indicate direction)} \\ T - F_g &= m_{\text{lift}}(0) \\ T &= F_g\end{aligned}$$

The forces are equal in magnitude and opposite in direction.

Stage 2:

The lift moves upwards at an acceleration of $1 \text{ m}\cdot\text{s}^{-2}$.

If the lift is accelerating, it means that there is a resultant force in the direction of the motion. This means that the force acting upwards is now greater than the force due to gravity $\vec{F}_g$ (down). To find the magnitude of the $\vec{T}$ applied by the cable we can do the following calculation: (Remember we have chosen upwards as positive.)

We apply Newton's second law to the vertical direction:

$$\begin{aligned}\vec{F}_R &= m_{\text{lift}}\vec{a} \text{ (we use signs to indicate direction)} \\ T - F_g &= m_{\text{lift}}(1) \\ T &= F_g + m_{\text{lift}}(1)\end{aligned}$$

The answer makes sense as we need a bigger force upwards to cancel the effect of gravity as well as have a positive resultant force.

Stage 3:

The lift moves at a constant velocity.

When the lift moves at a constant velocity, the acceleration is zero,

$$\begin{aligned}\vec{F}_R &= m_{\text{lift}}\vec{a} \text{ (we use signs to indicate direction)} \\ T - F_g &= m_{\text{lift}}(0) \\ T &= F_g\end{aligned}$$

The forces are equal in magnitude and opposite in direction. It is common **mistake** to think that because the lift is moving there is a net force acting on it. It is only if it is **accelerating** that there is a net force acting.

Stage 4:

The lift slows down at a rate of $2 \text{ m}\cdot\text{s}^{-2}$. The lift was moving upwards so this means that it is decelerating or accelerating in the direction opposite to the direction of motion. This means that the acceleration is in the negative direction.

$$\begin{aligned}\vec{F}_R &= m_{\text{lift}}\vec{a} \text{ (we use signs to indicate direction)} \\ T - F_g &= m_{\text{lift}}(-2) \\ T &= F_g - 2m_{\text{lift}}\end{aligned}$$

As the lift is now slowing down there is a resultant force downwards. This means that the force acting downwards is greater than the force acting upwards.

This makes sense as we need a smaller force upwards to ensure that the resultant force is downward. The force of gravity is now greater than the upward pull of the cable and the lift will slow down.

Stage 5:

The cable snaps.

When the cable snaps, the force that used to be acting upwards is no longer present. The only force that is present would be the force of gravity. The lift will fall freely and its acceleration.

Apparent weight

Your weight is the magnitude of the gravitational force acting on your body. When you stand in a lift that is stationary and then starts to accelerate upwards you feel you are pressed into the floor while the lift accelerates. You feel like you are heavier and your weight is more. When you are in a stationary lift that starts to accelerate downwards you feel lighter on your feet. You feel like your weight is less.

Weight is measured through normal forces. When the lift accelerates upwards you feel a greater normal force acting on you as the force required to accelerate you upwards in addition to balancing out the gravitational force.

When the lift accelerates downwards you feel a smaller normal force acting on you. This is because a net force downwards is required to accelerate you downwards. This phenomenon is called **apparent weight** because your weight didn't actually change.

Rockets

As with lifts, rockets are also examples of objects in vertical motion. The force of gravity pulls the rocket down while the thrust of the engine pushes the rocket upwards. The force that the engine exerts must overcome the force of gravity so that the rocket can accelerate upwards. The worked example below looks at the application of Newton's second law in launching a rocket.

Worked example 16: Newton's second law: rocket

QUESTION

A rocket (of mass 5000 kg) is launched vertically upwards into the sky at an acceleration of $20 \text{ m}\cdot\text{s}^{-2}$. If the magnitude of the force due to gravity on the rocket is 49 000 N, calculate the magnitude and direction of the thrust of the rocket's engines.

SOLUTION

Step 1: Analyse what is given and what is asked

We have the following:

$$m = 5000 \text{ kg}$$

$$\vec{a} = 20 \text{ m}\cdot\text{s}^{-2} \text{ upwards.}$$

$$\vec{F}_g = 49\,000 \text{ N downwards.}$$

We are asked to find the thrust of the rocket engine $\vec{F}$.

Step 2: Find the thrust of the engine

We will apply Newton's second law:

$$\vec{F}_R = m\vec{a} \text{ (using signs to indicate direction)}$$

$$F - F_g = (5000)(20)$$

$$F - (49\,000) = (5000)(20)$$

$$F = 149\,000 \text{ N}$$

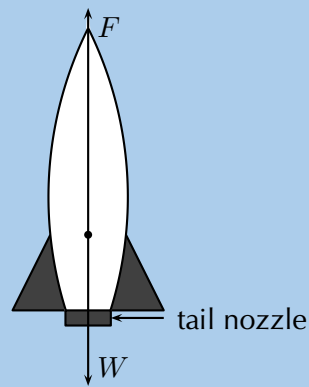
Step 3: Quote your final answer

The force due to the thrust is 149 000 N upwards.

Worked example 17: Rockets

QUESTION

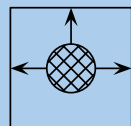
How do rockets accelerate in space?



SOLUTION

Step 1:

- Gas explodes inside the rocket.
- This exploding gas exerts a force on each side of the rocket (as shown in the picture below of the explosion chamber inside the rocket).



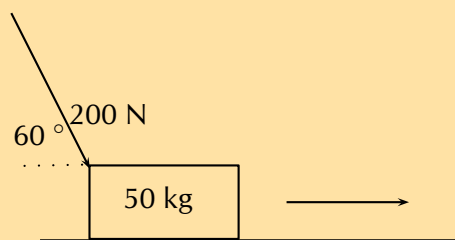
Note that the forces shown in this picture are representative. With an explosion there will be forces in all directions.

- Due to the symmetry of the situation, all the forces exerted on the rocket are balanced by forces on the opposite side, except for the force opposite the open side. This force on the upper surface is unbalanced.
- This is therefore the resultant force acting on the rocket and it makes the rocket accelerate forwards.

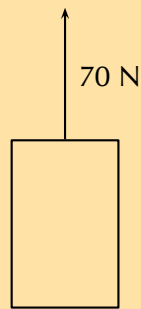
Exercise 2 – 5:

1. A tug is capable of pulling a ship with a force of 100 kN. If two such tugs are pulling on one ship, they can produce any force ranging from a minimum of 0 kN to a maximum of 200 kN. Give a detailed explanation of how this is possible. Use diagrams to support your result.
2. A car of mass 850 kg accelerates at $2 \text{ m}\cdot\text{s}^{-2}$. Calculate the magnitude of the resultant force that is causing the acceleration.
3. Find the force needed to accelerate a 3 kg object at $4 \text{ m}\cdot\text{s}^{-2}$.
4. Calculate the acceleration of an object of mass 1000 kg accelerated by a force of magnitude 100 N.

5. An object of mass 7 kg is accelerating at $2,5 \text{ m}\cdot\text{s}^{-2}$. What resultant force acts on it?
6. Find the mass of an object if a force of 40 N gives it an acceleration of $2 \text{ m}\cdot\text{s}^{-2}$.
7. Find the acceleration of a body of mass 1000 kg that has a force with a magnitude of 150 N acting on it.
8. Find the mass of an object which is accelerated at $3 \text{ m}\cdot\text{s}^{-2}$ by a force of magnitude 25 N.
9. Determine the acceleration of a mass of 24 kg when a force of magnitude 6 N acts on it. What is the acceleration if the force were doubled and the mass was halved?
10. A mass of 8 kg is accelerating at $5 \text{ m}\cdot\text{s}^{-2}$.
 - a) Determine the resultant force that is causing the acceleration.
 - b) What acceleration would be produced if we doubled the force and reduced the mass by half?
11. A motorcycle of mass 100 kg is accelerated by a resultant force of 500 N. If the motorcycle starts from rest:
 - a) What is its acceleration?
 - b) How fast will it be travelling after 20 s?
 - c) How long will it take to reach a speed of $35 \text{ m}\cdot\text{s}^{-1}$?
 - d) How far will it travel from its starting point in 15 s?
12. A force of 200 N, acting at 60° to the horizontal, accelerates a block of mass 50 kg along a horizontal plane as shown.



- a) Calculate the component of the 200 N force that accelerates the block horizontally.
 - b) If the acceleration of the block is $1,5 \text{ m}\cdot\text{s}^{-2}$, calculate the magnitude of the frictional force on the block.
 - c) Calculate the vertical force exerted by the block on the plane.
13. A toy rocket experiences a force due to gravity of magnitude 4,5 N is supported vertically by placing it in a bottle. The rocket is then ignited. Calculate the force that is required to accelerate the rocket vertically upwards at $8 \text{ m}\cdot\text{s}^{-2}$.
14. A constant force of magnitude 70 N is applied vertically to a block as shown. The block experiences a force due to gravity of 49 N. Calculate the acceleration of the block.



15. A student experiences a gravitational force of magnitude 686 N investigates the motion of a lift. While he stands in the lift on a bathroom scale (calibrated in newton), he notes three stages of his journey.

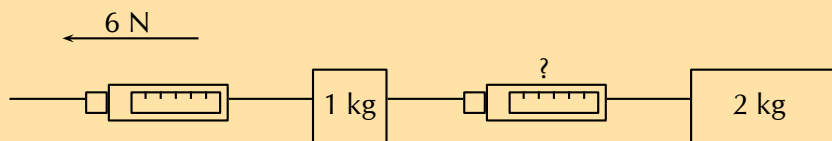
- a) For 2 s immediately after the lift starts, the scale reads 574 N.
- b) For a further 6 s it reads 686 N.
- c) For the final 2 s it reads 854 N.

Answer the following questions:

- a) Is the motion of the lift upward or downward? Give a reason for your answer.
- b) Write down the magnitude and the direction of the resultant force acting on the student for each of the stages 1, 2 and 3.

16. A car of mass 800 kg accelerates along a level road at $4 \text{ m}\cdot\text{s}^{-2}$. A frictional force of 700 N opposes its motion. What force is produced by the car's engine?

17. Two objects, with masses of 1 kg and 2 kg respectively, are placed on a smooth surface and connected with a piece of string. A horizontal force of 6 N is applied with the help of a spring balance to the 1 kg object. Ignoring friction, what will the force acting on the 2 kg mass, as measured by a second spring balance, be?



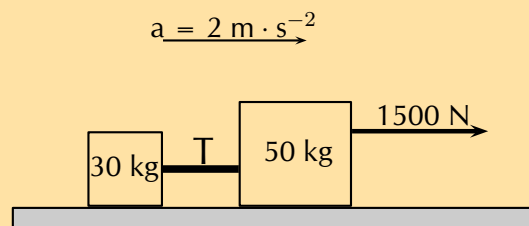
18. A rocket of mass 200 kg has a resultant force of 4000 N upwards on it.

- a) What is its acceleration on the Earth, where it experiences a gravitational force of 1960 N?
- b) What driving force does the rocket engine need to exert on the back of the rocket on the Earth?

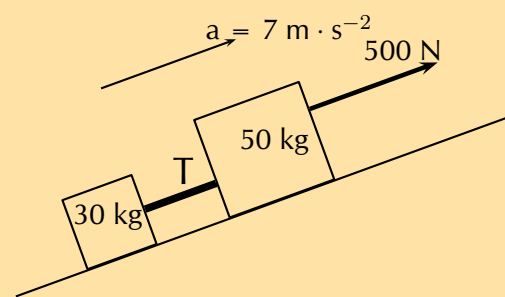
19. A car going at $20 \text{ m}\cdot\text{s}^{-1}$ accelerates uniformly and comes to a stop in a distance of 20 m.

- a) What is its acceleration?
- b) If the car is 1000 kg how much force do the brakes exert?

20. A block on an inclined plane experiences a force due to gravity, $\vec{F}_g$ of 300 N straight down. If the slope is inclined at $67,8^\circ$ to the horizontal, what is the component of the force due to gravity perpendicular and parallel to the slope? At what angle would the perpendicular and parallel components of the force due to gravity be equal?
21. A block on an inclined plane is subjected to a force due to gravity, $\vec{F}_g$ of 287 N straight down. If the component of the gravitational force parallel to the slope is $\vec{F}_{gx} = 123,7$ N in the negative x -direction (down the slope), what is the incline of the slope?
22. A block on an inclined plane experiences a force due to gravity, $\vec{F}_g$ of 98 N straight down. If the slope is inclined at an unknown angle to the horizontal but we are told that the ratio of the components of the force due to gravity perpendicular and parallel to the slope is 7:4. What is the angle the incline makes with the horizontal?
23. Two crates, 30 kg and 50 kg respectively, are connected with a thick rope according to the diagram. A force, to the right, of 1500 N is applied. The boxes move with an acceleration of $7 \text{ m} \cdot \text{s}^{-2}$ to the right. The ratio of the frictional forces on the two crates is the same as the ratio of their masses. Calculate:



- a) the magnitude and direction of the total frictional force present.
- b) the magnitude of the tension in the rope at T.
24. Two crates, 30 kg and 50 kg respectively, are connected with a thick rope according to the diagram. If they are dragged up an incline such that the ratio of the parallel and perpendicular components of the gravitational force on each block are 3 : 5. The boxes move with an acceleration of $7 \text{ m} \cdot \text{s}^{-2}$ up the slope. The ratio of the frictional forces on the two crates is the same as the ratio of their masses. The magnitude of the force due to gravity on the 30 kg crate is 294 N and on the 50 kg crate is 490 N. Calculate:



- a) the magnitude and direction of the total frictional force present.
- b) the magnitude of the tension in the rope at T.

Think you got it? Get this answer and more practice on our Intelligent Practice Service

- | | | | | | |
|----------|----------|----------|----------|----------|----------|
| 1. 23HF | 2. 23HG | 3. 23HH | 4. 23HJ | 5. 23HK | 6. 23HM |
| 7. 23HN | 8. 23HP | 9. 23HQ | 10. 23HR | 11. 23HS | 12. 23HT |
| 13. 23HV | 14. 23HW | 15. 23HX | 16. 23HY | 17. 23HZ | 18. 23J2 |
| 19. 23J3 | 20. 23J4 | 21. 23J5 | 22. 23J6 | 23. 23J7 | 24. 23J8 |



www.everythingscience.co.za



m.everythingscience.co.za

Newton's third law of motion

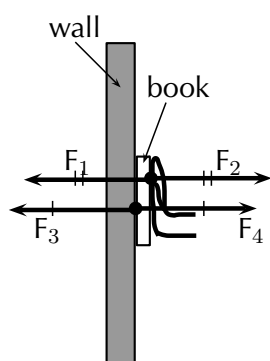
ESBKV



Figure 2.8: Newton's action-reaction pairs.

Newton's third law of Motion deals with the interaction between pairs of objects. For example, if you hold a book up against a wall you are exerting a force on the book (to keep it there) and the book is exerting a force back at you (to keep you from falling through the book). This may sound strange, but if the book was not pushing back at you, your hand would push through the book! These two forces (the force of the hand on the book (F_1) and the force of the book on the hand (F_2)) are called an action-reaction pair of forces. They have the same magnitude, but act in opposite directions and act on different objects (the one force is onto the book and the other is onto your hand).

There is another action-reaction pair of forces present in this situation. The book is pushing against the wall (action force) and the wall is pushing back at the book (reaction). The force of the book on the wall (F_3) and the force of the wall on the book (F_4) are shown in the diagram.



- F_1 : force of hand on book
- F_2 : force of book on hand
- F_3 : force of book on wall
- F_4 : force of wall on book

DEFINITION: *Newton's third law of motion*

If body A exerts a force on body B, then body B exerts a force of equal magnitude on body A, but in the opposite direction.

▶ See video: 23J9 at www.everythingscience.co.za

These action-reaction pairs have several properties:

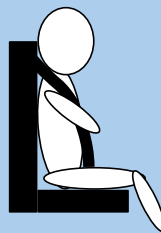
- the same type of force acts on the objects,
- the forces have the same magnitude but opposite direction, and
- the forces act on different objects.

Newton's action-reaction pairs can be found everywhere in life where two objects interact with one another. The following worked examples will illustrate this:

Worked example 18: Newton's third law - seat belt

QUESTION

Dineo is seated in the passenger seat of a car with the seat belt on. The car suddenly stops and he moves forwards (Newton's first law - he continues in his state of motion) until the seat belt stops him. Draw a labelled force diagram identifying two action-reaction pairs in this situation.



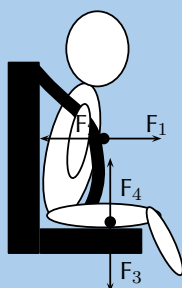
SOLUTION

Step 1: Draw a force diagram

Start by drawing the picture. You will be using arrows to indicate the forces so make your picture large enough so that detailed labels can also be added. The picture needs to be accurate, but not artistic! Use stick-men if you have to.

Step 2: Label the diagram

Take one pair at a time and label them carefully. If there is not enough space on the drawing, then use a key on the side.



F_1 : The force of Dineo on the seat belt
 F_2 : The force of the seat belt on Dineo
 F_3 : The force of Dineo on the seat (downwards)
 F_4 : The force of the seat on Dineo (upwards)

Worked example 19: Newton's third law: forces in a lift

QUESTION

Tammy travels from the ground floor to the fifth floor of a hotel in a lift moving at constant velocity. Which ONE of the following statements is TRUE about the magnitude of the force exerted by the floor of the lift on Tammy's feet? Use Newton's third law to justify your answer.

1. It is greater than the magnitude of Tammy's weight.
2. It is equal in magnitude to the force Tammy's feet exert on the floor of the lift.
3. It is equal to what it would be in a stationary lift.
4. It is greater than what it would be in a stationary lift.

SOLUTION

Step 1: Analyse the situation

This is a Newton's third law question and not Newton's second law. We need to focus on the action-reaction pairs of forces and not the motion of the lift. The following diagram will show the action-reaction pairs that are present when a person is standing on a scale in a lift.

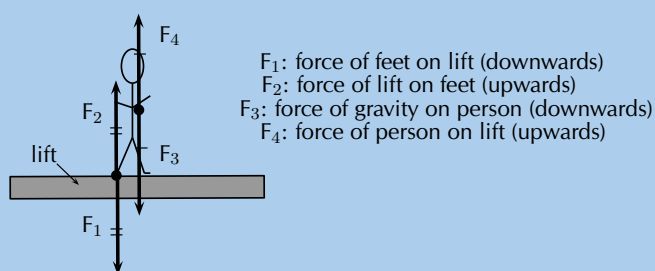


Figure 2.9: Newton's action-reaction pairs in a lift.

In this question statements are made about the force of the floor (lift) on Tammy's feet. This force corresponds to F_2 in our diagram. The reaction force that pairs up with

this one is F_1 , which is the force that Tammy's feet exerts on the floor of the lift. The magnitude of these two forces are the same, but they act in opposite directions.

Step 2: Choose the correct answer

It is important to analyse the question first, before looking at the answers. The answers might confuse you if you look at them first. Make sure that you understand the situation and know what is asked before you look at the options.

The correct answer is number 2.

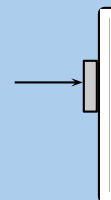
Worked example 20: Newton's third law: book and wall

QUESTION

Bridget presses a book against a vertical wall as shown in the photograph.



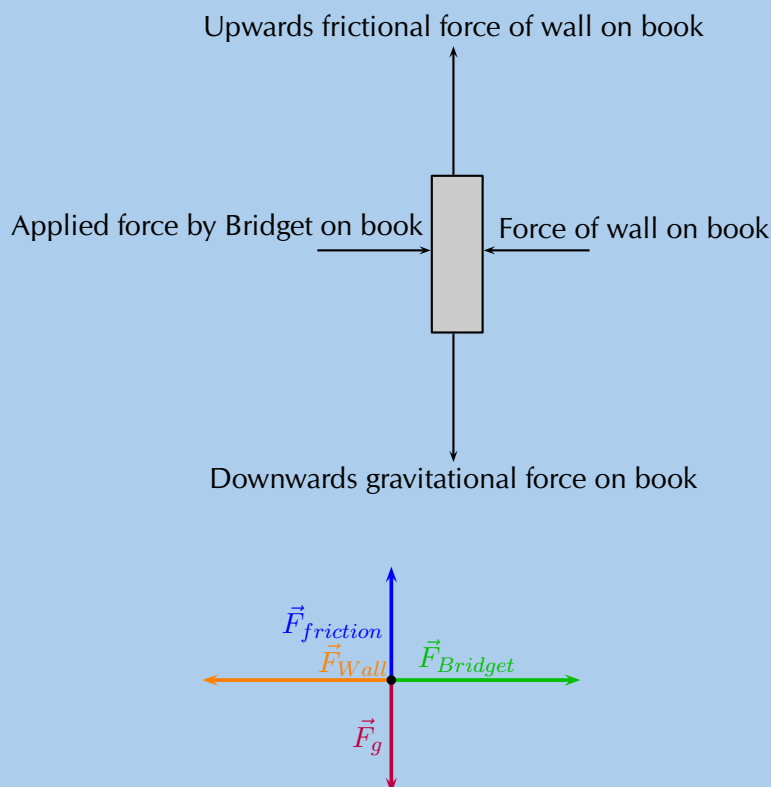
1. Draw a labelled force diagram indicating all the forces acting on the book.
2. State, in words, Newton's third law of Motion.
3. Name the action-reaction pairs of forces acting in the horizontal plane.



SOLUTION

Step 1: Draw a force diagram

A force diagram will look like this:



Note that we had to draw all the forces acting on the book and not the action-reaction pairs. None of the forces drawn are action-reaction pairs, because they all act on the same object (the book). When you label forces, be as specific as possible, including the direction of the force and both objects involved, for example, do not say gravity (which is an incomplete answer) but rather say 'Downward (direction) gravitational force of the Earth (object) on the book (object)'.

Step 2: State Newton's third law

If body A exerts a force onto body B, then body B will exert a force equal in magnitude, but opposite in direction, onto body A.

Step 3: Name the action-reaction pairs

The question only asks for action-reaction forces in the horizontal plane. Therefore:

Pair 1: Action: Applied force of Bridget on the book; Reaction: The force of the book on the girl.

Pair 2: Action: Force of the book on the wall; Reaction: Force of the wall on the book.

Note that a Newton's third law pair will always involve the same combination of words, like 'book on wall' and 'wall on book'. The objects are 'swapped around' in naming the pairs.

General experiment: Balloon rocket

Aim:

In this experiment for the entire class, you will use a balloon rocket to investigate Newton's third law. A fishing line will be used as a track and a plastic straw taped to the balloon will help attach the balloon to the track.

Apparatus:

You will need the following items for this experiment:

1. balloons (one for each team)
2. plastic straws (one for each team)
3. tape (cellophane or masking)
4. fishing line, 10 metres in length
5. a stopwatch - optional (a cell phone can also be used)
6. a measuring tape - optional



Method:

1. Divide into groups of at least five.
2. Attach one end of the fishing line to the blackboard with tape. Have one teammate hold the other end of the fishing line so that it is taut and roughly horizontal. The line must be held steady and **must not** be moved up or down during the experiment.
3. Have one teammate blow up a balloon and hold it shut with his or her fingers. Have another teammate tape the straw along the side of the balloon. Thread the fishing line through the straw and hold the balloon at the far end of the line.
4. Let go of the rocket and observe how the rocket moves forward.
5. Optionally, the rockets of each group can be timed to determine a winner of the fastest rocket.
 - a) Assign one teammate to time the event. The balloon should be let go when the time keeper yells "Go!" Observe how your rocket moves toward the blackboard.
 - b) Have another teammate stand right next to the blackboard and yell "Stop!" when the rocket hits its target. If the balloon does not make it all the way to the blackboard, "Stop!" should be called when the balloon stops moving. The timekeeper should record the flight time.
 - c) Measure the exact distance the rocket travelled. Calculate the average speed at which the balloon travelled. To do this, divide the distance travelled by the time the balloon was "in flight." Fill in your results for Trial 1 in the Table below.
 - d) Each team should conduct two more races and complete the sections in the Table for Trials 2 and 3. Then calculate the average speed for the three trials to determine your team's race entry time.

Results:

	Distance (m)	Time (s)	Speed ($\text{m}\cdot\text{s}^{-1}$)
Trial 1			
Trial 2			
Trial 3			
		Average:	

Conclusions:

The winner of this race is the team with the fastest average balloon speed.

▶ See video: 23JB at www.everythingscience.co.za

While doing the experiment, you should think about,

1. What made your rocket move?
2. How is Newton's third law of Motion demonstrated by this activity?
3. Draw pictures using labelled arrows to show the forces acting on the inside of the balloon before it was released and after it was released.

▶ See video: 23JC at www.everythingscience.co.za

▶ See video: 23JD at www.everythingscience.co.za

The Saturn V (pronounced "Saturn Five") was an American human-rated expendable rocket used by NASA's Apollo and Skylab programs from 1967 until 1973. A multistage liquid-fuelled launch vehicle, NASA launched 13 Saturn Vs from the Kennedy Space Center, Florida with no loss of crew or payload. It remains the tallest, heaviest, and most powerful rocket ever brought to operational status and still holds the record for the heaviest launch vehicle payload.

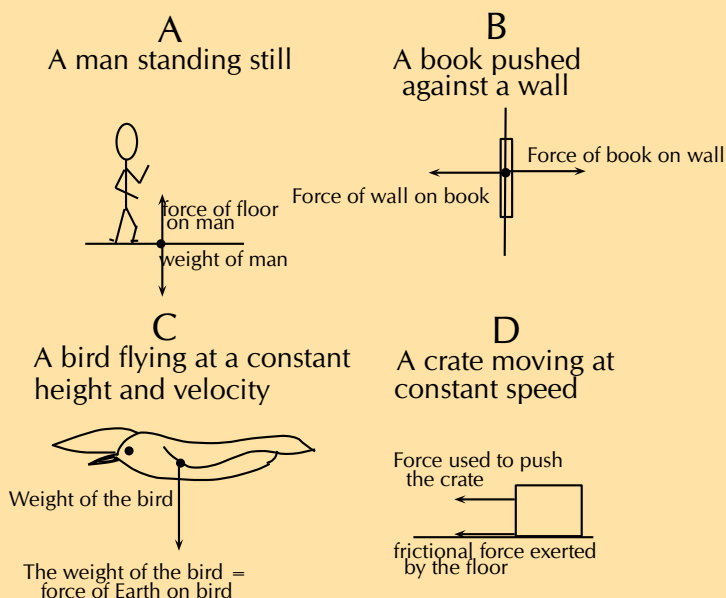


Exercise 2 – 6:

1. A fly hits the front windscreen of a moving car. Compared to the magnitude of the force the fly exerts on the windscreen, the magnitude of the force the windscreen exerts on the fly during the collision, is:

- a) zero.
- b) smaller, but not zero.
- c) bigger.
- d) the same.

2. Which of the following pairs of forces correctly illustrates Newton's third law?



Think you got it? Get this answer and more practice on our Intelligent Practice Service

1. 23JF 2. 23JG



www.everythingscience.co.za



m.everythingscience.co.za

Forces in equilibrium

ESBKW

At the beginning of this chapter it was mentioned that resultant forces cause objects to accelerate in a straight line. If an object is stationary or moving at constant velocity then either,

- no forces are acting on the object, or
- the forces acting on that object are exactly balanced.

In other words, for stationary objects or objects moving with constant velocity, the resultant force acting on the object is zero.

DEFINITION: *Equilibrium*

An object in equilibrium has both the sum of the forces acting on it equal to zero.

Gravity is arguably the first force that people really learn about. People don't really think of it as learning about gravity because it is such a big part of our everyday lives. Babies learning to crawl or walk are struggling against gravity, games involving jumping, climbing or balls all give people a sense of the effect of gravity. The saying "everything that goes up, must come down" is all about gravity. Rain falls from the sky because of gravity and so much more. We all know that these things happen but we don't often stop to ask what is gravity, what causes it and how can we describe it more accurately than "everything that goes up, must come down"?

All of the examples mentioned involve objects with mass falling to the ground, they are being attracted to the Earth. Gravity is the name we give to the force that arises between objects because of their mass. It is always an attractive force. It is a non-contact force, therefore it acts at a distance. The Earth is responsible for a gravitational force on the moon which keeps it in its orbit around the Earth and the gravitational force the moon exerts on the Earth is the dominant cause of ocean tides.

Sir Isaac Newton was the first scientist to precisely define the gravitational force, and to show that it could explain both falling bodies and astronomical motions. The gravitational force is relatively simple. It is **always attractive**, and it **depends only on the masses** involved and the **distance between them**. Stated in modern language, Newton's universal law of gravitation states that every particle in the universe attracts every other particle with a force along a line joining them. The force is directly proportional to the product of their masses and inversely proportional to the square of the distance between them.

Newton's law of universal gravitation

ESBKY

DEFINITION: *Newton's law of universal gravitation*

Every point mass attracts every other point mass by a force directed along the line connecting the two. This force is proportional to the product of the masses and inversely proportional to the square of the distance between them.

The magnitude of the attractive gravitational force between the two point masses, F is given by:

$$F = G \frac{m_1 m_2}{d^2}$$

where: F is in newtons (N), G is the gravitational constant $6,67 \times 10^{-11} \text{ N} \cdot \text{m}^2 \cdot \text{kg}^{-2}$, m_1 is the mass of the first point mass in kilograms (kg), m_2 is the mass of the second point mass in kilograms (kg) and d is the distance between the two point masses in metres (m). For any large objects (not point masses) we use the distance from the **centre of the object(s)** to do the calculation. This is very important when dealing with very large objects like planets. The distance from the centre of the planet and from the surface of the planet differ by a large amount. **Remember** that this is a force of attraction and should be described by a vector. We use Newton's law of universal gravitation to determine the magnitude of the force and then analyse the problem to determine the direction.

For example, consider a man of mass 80 kg standing 10 m from a woman with a mass

of 65 kg. The attractive gravitational force between them would be:

$$\begin{aligned} F &= G \frac{m_1 m_2}{d^2} \\ &= \left(6,67 \times 10^{-11} \right) \left(\frac{(80)(65)}{(10)^2} \right) \\ &= 3,47 \times 10^{-9} \text{ N} \end{aligned}$$

If the man and woman move to 1 m apart, then the force is:

$$\begin{aligned} F &= G \frac{m_1 m_2}{d^2} \\ &= \left(6,67 \times 10^{-11} \right) \left(\frac{(80)(65)}{(1)^2} \right) \\ &= 3,47 \times 10^{-7} \text{ N} \end{aligned}$$

As you can see, these forces are very small.

Now consider the gravitational force between the Earth and the Moon. The mass of the Earth is $5,98 \times 10^{24}$ kg, the mass of the Moon is $7,35 \times 10^{22}$ kg and the Earth and Moon are $3,8 \times 10^8$ m apart. The gravitational force between the Earth and Moon is:

$$\begin{aligned} F &= G \frac{m_1 m_2}{d^2} \\ &= \left(6,67 \times 10^{-11} \right) \left(\frac{(5,98 \times 10^{24})(7,35 \times 10^{22})}{(3,8 \times 10^8)^2} \right) \\ &= 2,03 \times 10^{20} \text{ N} \end{aligned}$$

From this example you can see that the force is very large.

These two examples demonstrate that the greater the masses, the greater the force between them. The $1/d^2$ factor tells us that the distance between the two bodies plays a role as well. The closer two bodies are, the stronger the gravitational force between them is. We feel the gravitational attraction of the Earth most at the surface since that is the closest we can get to it, but if we were in outer-space, we would barely feel the effect of the Earth's gravity!

Remember that $\vec{F} = m\vec{a}$ which means that every object on Earth feels the same gravitational acceleration! That means whether you drop a pen or a book (from the same height), they will both take the same length of time to hit the ground... in fact they will be head to head for the entire fall if you drop them at the same time. We can show this easily by using Newton's second law and the equation for the gravitational force. The force between the Earth (which has the mass M_{Earth}) and an object of mass m_o is $F = \frac{Gm_o M_{\text{Earth}}}{d^2}$ and the acceleration of an object of mass m_o (in terms of the force acting on it) is $a_o = \frac{F}{m_o}$.

So we equate them and we find that:

$$a_o = G \frac{M_{\text{Earth}}}{d_{\text{Earth}}^2}$$

Since it doesn't depend on the mass of the object, m_o , the acceleration on a body (due to the Earth's gravity) does not depend on the mass of the body. Thus all objects

experience the same gravitational acceleration. The force on different bodies will be different but the acceleration will be the same. Due to the fact that this acceleration caused by gravity is the same on all objects we label it differently, instead of using a we use g_{Earth} which we call the gravitational acceleration and it has a magnitude of approximately $9,8 \text{ m}\cdot\text{s}^{-2}$.

The fact that gravitational acceleration is independent of the mass of the object holds for any planet, not just Earth, but each planet will have a different magnitude of gravitational acceleration.

Exercise 2 – 7:

1. When the planet Jupiter is closest to Earth it is $6,28 \times 10^8 \text{ km}$ away. If Jupiter has a mass of $1,9 \times 10^{27} \text{ kg}$, what is the magnitude of the gravitational force between Jupiter and the Earth?
2. When the planet Jupiter is furthest from the Earth it is $9,28 \times 10^8 \text{ km}$ away. If Jupiter has a mass of $1,9 \times 10^{27} \text{ kg}$, what is the magnitude of the gravitational force between Jupiter and the Earth?
3. What distance must a satellite with a mass of 80 kg be away from the Earth to feel a force of 1000 N ? How far from Jupiter to feel the same force?
4. The radius of Jupiter is $71,5 \times 10^3 \text{ km}$ and the radius of the moon is $1,7 \times 10^3 \text{ km}$, if the moon has a mass of $7,35 \times 10^{22} \text{ kg}$ work out the gravitational acceleration on Jupiter and on the moon.
5. Astrology, NOT astronomy, makes much of the position of the planets at the moment of one's birth. The only known force a planet exerts on Earth is gravitational. Calculate:
 - a) the gravitational force exerted on a $4,20 \text{ kg}$ baby by a 100 kg father $0,200 \text{ m}$ away at birth
 - b) the force on the baby due to Jupiter if it is at its closest distance to Earth, some $6,29 \times 10^{11} \text{ m}$ away.
 - c) How does the force of Jupiter on the baby compare to the force of the father on the baby?
6. The existence of the dwarf planet Pluto was proposed based on irregularities in Neptune's orbit. Pluto was subsequently discovered near its predicted position. But it now appears that the discovery was fortuitous, because Pluto is small and the irregularities in Neptune's orbit were not well known. To illustrate that Pluto has a minor effect on the orbit of Neptune compared with the closest planet to Neptune:
 - a) Calculate the acceleration due to gravity at Neptune due to Pluto when they are $4,50 \times 10^{12} \text{ m}$ apart, as they are at present. The mass of Pluto is $1,4 \times 10^{22} \text{ kg}$ and the mass of Neptune is: $1,02 \times 10^{26} \text{ kg}$.
 - b) Calculate the acceleration due to gravity at Neptune due to Uranus, presently about $2,50 \times 10^{12} \text{ m}$ apart, and compare it with that due to Pluto. The mass of Uranus is $8,62 \times 10^{25} \text{ kg}$.

Think you got it? Get this answer and more practice on our Intelligent Practice Service

1. 23JH 2. 23JJ 3. 23JK 4. 23JM 5. 23JN 6. 23JP



www.everythingscience.co.za



m.everythingscience.co.za

Weight and mass

ESBKZ

In everyday discussion many people use weight and mass to mean the same thing which is not true.

Mass is a scalar and weight is a vector. Mass is a measurement of how much matter is in an object; weight is a measurement of how hard gravity is pulling on that object. Your mass is the same wherever you are, on Earth; on the moon; floating in space, because the amount of stuff you're made of doesn't change. Your **weight** depends on how strong a gravitational force is acting on you at the moment; you'd weigh less on the moon than on Earth, and in space you'd weigh almost nothing at all. Mass is measured in kilograms, kg, and weight is a force and measured in newtons, N.

When you stand on a scale you are trying measure how much of you there is. People who are trying to reduce their mass hope to see the reading on the scale get smaller but they talk about losing weight. Their weight will decrease but it is because their mass is decreasing. A scale uses the persons weight to determine their mass.

You can use $\vec{F}_g = m\vec{g}$ to calculate weight.

Worked example 21: Newton's second law: lifts and g

QUESTION

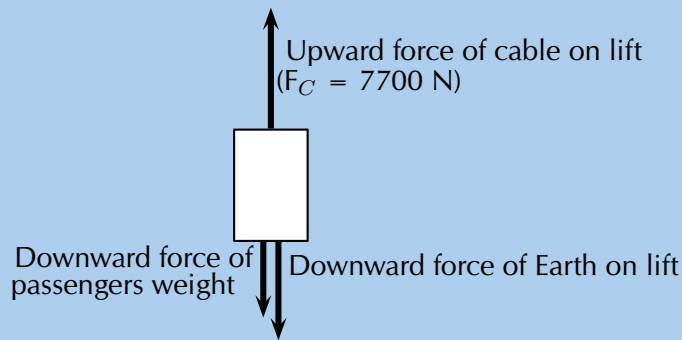
A lift, with a mass of 250 kg, is initially at rest on the ground floor of a tall building. Passengers with an unknown total mass, m , climb into the lift. The lift accelerates upwards at $1,6 \text{ m}\cdot\text{s}^{-2}$. The cable supporting the lift exerts a constant upward force of 7700 N.

1. Draw a labelled force diagram indicating all the forces acting on the lift while it accelerates upwards.
2. What is the maximum mass, m , of the passengers the lift can carry in order to achieve a constant upward acceleration of $1,6 \text{ m}\cdot\text{s}^{-2}$.

SOLUTION

Step 1: Draw a force diagram.

We choose upwards as the positive direction.



Step 2: Gravitational force

We know that the gravitational acceleration on any object on Earth, due to the Earth, is $\vec{g} = 9,8 \text{ m}\cdot\text{s}^{-2}$ towards the centre of the Earth (downwards). We know that the force due to gravity on a lift or the passengers in the lift will be $\vec{F}_g = m\vec{g}$.

Step 3: Find the mass, m .

Let us look at the lift with its passengers as a unit. The mass of this unit will be $(250 \text{ kg} + m)$ and the force of the Earth pulling downwards (F_g) will be $(250 + m) \times 9,8 \text{ m}\cdot\text{s}^{-2}$. If we apply Newton's second law to the situation we get:

$$\begin{aligned}
 F_{\text{net}} &= ma \\
 F_C - F_g &= ma \\
 7700 - (250 + m)(9,8) &= (250 + m)(1,6) \\
 7700 - 2500 - 9,8 m &= 400 + 1,6 m \\
 4800 &= 11,4 m \\
 m &= 421,05 \text{ kg}
 \end{aligned}$$

Step 4: Quote your final answer

The mass of the passengers is 421,05 kg. If the mass were larger then the total downward force would be greater and the cable would need to exert a larger force in the positive direction to maintain the same acceleration.

In everyday use we often talk about weighing things. We also refer to how much something weighs. It is important to remember that when someone asks how much you weigh or how much an apple weighs they are actually wanting to know your mass or the apples mass, not the force due to gravity acting on you or the apple.

Weightlessness is not because there is no gravitational force or that there is no weight anymore. Weightless is an extreme case of apparent weight. Think about the lift accelerating downwards when you feel a little lighter. If the lift accelerated downwards with the same magnitude as gravitational acceleration there would be no normal force acting on you, the lift and you would be falling with exactly the same acceleration and you would feel weightless. Eventually the lift has to come to a stop.

In a space shuttle in space it is almost exactly the same case. The astronauts and space shuttle feel exactly the same gravitational acceleration so their apparent weight is

zero. The one difference is that they are not falling downwards, they have a very large velocity perpendicular to the direction of the gravitational force that is pulling them towards the earth. They are falling but in a circle around the earth. The gravitational force and their velocity are perfectly balanced that they orbit the earth.

In a weightless environment, defining up and down doesn't make as much sense as in our every day life. In space this affects all sorts of things, for example, when a candle burns the hot gas can't go up because the usual up is defined by which way gravity acts. This has actually been tested.

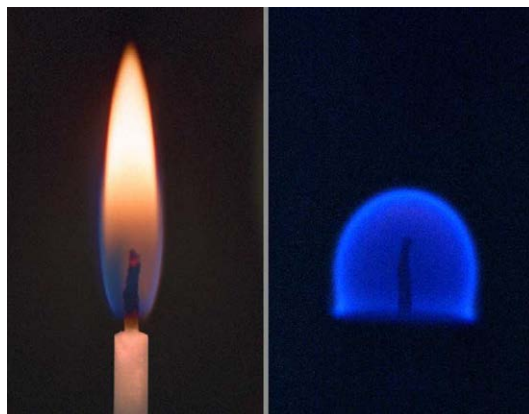


Figure 2.10: A candle burning on earth (left) and one burning in space (right).

Exercise 2 – 8:

1. Jojo has a mass of 87,5 kg, what is his weight on the following planets:
 - a) Mercury (radius of $2,440 \times 10^3$ km and mass of $3,3 \times 10^{23}$ kg)
 - b) Mars (radius of $3,39 \times 10^3$ km and mass of $6,42 \times 10^{23}$ kg)
 - c) Neptune (radius of $24,76 \times 10^3$ km and mass of $1,03 \times 10^{26}$ kg)?
2. If object 1 has a weight of $1,78 \times 10^3$ N on Neptune and object 2 has a weight of $3,63 \times 10^5$ N on Mars, which has the greater mass?

Think you got it? Get this answer and more practice on our Intelligent Practice Service

1. [23JQ](#) 2. [23JR](#)



www.everythingscience.co.za



m.everythingscience.co.za

Comparative problems

ESBM2

Comparative problems involve calculation of something in terms of something else that we know. For example, if you weigh 490 N on Earth and the gravitational acceleration

on Venus is 0,903 that of the gravitational acceleration on the Earth, then you would weigh $0,903 \times 490\text{N} = 442,5\text{N}$ on Venus.

Method for answering comparative problems

- Write out equations and calculate all quantities for the given situation
- Write out all relationships between variable from first and second case
- Write out second case
- Substitute all first case variables into second case
- Write second case in terms of first case

Worked example 22: Comparative problem

QUESTION

A man has a mass of 70 kg. The planet Zirgon is the same size as the Earth but has twice the mass of the Earth. What would the man weigh on Zirgon, if the gravitational acceleration on Earth is $9,8 \text{ m}\cdot\text{s}^{-2}$?

SOLUTION

Step 1: Determine what information has been given

The following has been provided:

- the mass of the man, m
- the mass of the planet Zirgon (m_Z) in terms of the mass of the Earth (M_{Earth}),
 $m_Z = 2M_{\text{Earth}}$
- the radius of the planet Zirgon (r_Z) in terms of the radius of the Earth (r_E),
 $r_Z = r_{\text{Earth}}$

Step 2: Determine how to approach the problem

We are required to determine the man's weight on Zirgon (w_Z). We can do this by using:

$$F_g = mg = G \frac{m_1 \cdot m_2}{d^2}$$

to calculate the weight of the man on Earth and then use this value to determine the weight of the man on Zirgon.

Step 3: Situation on Earth

$$\begin{aligned}
 F_{\text{Earth}} &= mg_E = G \frac{M_{\text{Earth}} \cdot m}{d_E^2} \\
 &= (70 \text{ kg}) (9,8 \text{ m}\cdot\text{s}^{-2}) \\
 &= 686 \text{ N}
 \end{aligned}$$

Step 4: Situation on Zirgon in terms of situation on Earth

Write the equation for the gravitational force on Zirgon and then substitute the values for m_Z and r_Z , in terms of the values for the Earth.

$$\begin{aligned}
 w_Z &= mg_Z = G \frac{m_Z \cdot m}{r_Z^2} \\
 &= G \frac{2M_{\text{Earth}} \cdot m}{r_{\text{Earth}}^2} \\
 &= 2 \left(G \frac{M_{\text{Earth}} \cdot m}{r_E^2} \right) \\
 &= 2F_{\text{Earth}} \\
 &= 2(686 \text{ N}) \\
 &= 1372 \text{ N}
 \end{aligned}$$

Step 5: Quote the final answer

The man weighs 1372 N on Zirgon.

Worked example 23: Comparative problem

QUESTION

A man has a mass of 70 kg. On the planet Beeble how much will he weigh if Beeble has a mass half of that of the Earth and a radius one quarter that of the Earth. Gravitational acceleration on Earth is $9,8 \text{ m}\cdot\text{s}^{-2}$.

SOLUTION

Step 1: Determine what information has been given

The following has been provided:

- the mass of the man on Earth, m
- the mass of the planet Beeble (m_B) in terms of the mass of the Earth (M_{Earth}), $m_B = \frac{1}{2}M_{\text{Earth}}$
- the radius of the planet Beeble (r_B) in terms of the radius of the Earth (r_E), $r_B = \frac{1}{4}r_E$

Step 2: Determine how to approach the problem

We are required to determine the man's weight on Beeble (w_B). We can do this by using:

$$F_g = mg = G \frac{m_1 m_2}{d^2}$$

to calculate the weight of the man on Earth and then use this value to determine the weight of the man on Beeble.

Step 3: Situation on Earth

$$\begin{aligned} F_{Earth} &= mg_{Earth} = G \frac{M_{Earth}}{r_E^2} \\ &= (70)(9,8) \\ &= 686 \text{ N} \end{aligned}$$

Step 4: Situation on Beeble in terms of situation on Earth

Write the equation for the gravitational force on Beeble and then substitute the values for m_B and r_B , in terms of the values for the Earth.

$$\begin{aligned} F_{Beeble} &= mg_{Beeble} = G \frac{M_{Beeble}}{r_B^2} \\ &= G \frac{\frac{1}{2} M_{Earth}}{\frac{1}{4} r_E^2} \\ &= 8 \left(G \frac{M_{Earth}}{r_E^2} \right) \\ &= 8(686) \\ &= 5488 \text{ N} \end{aligned}$$

Step 5: Quote the final answer

The man weighs 5488 N on Beeble.

Exercise 2 – 9:

1. Two objects of mass $2X$ and $3X$ respectively, where X is an unknown quantity, exert a force F on each other when they are a certain distance apart. What will be the force between two objects situated the same distance apart but having a mass of $5X$ and $6X$ respectively?

- a) 0,2 F
 - b) 1,2 F
 - c) 2,2 F
 - d) 5 F
2. As the distance of an object above the surface of the Earth is greatly increased, the weight of the object would
- a) increase
 - b) decrease
 - c) increase and then suddenly decrease
 - d) remain the same
3. A satellite circles around the Earth at a height where the gravitational force is a factor 4 less than at the surface of the Earth. If the Earth's radius is R, then the height of the satellite above the surface is:
- a) R
 - b) 2 R
 - c) 4 R
 - d) 16 R
4. A satellite experiences a force F when at the surface of the Earth. What will be the force on the satellite if it orbits at a height equal to the diameter of the Earth:
- a) $\frac{1}{F}$
 - b) $\frac{1}{2} F$
 - c) $\frac{1}{3} F$
 - d) $\frac{1}{9} F$
5. The weight of a rock lying on surface of the Moon is W. The radius of the Moon is R. On planet Alpha, the same rock has weight 8W. If the radius of planet Alpha is half that of the Moon, and the mass of the Moon is M, then the mass, in kg, of planet Alpha is:
- a) $\frac{M}{2}$
 - b) $\frac{M}{4}$
 - c) 2 M
 - d) 4 M
6. Consider the symbols of the two physical quantities g and G used in Physics.
- a) Name the physical quantities represented by g and G .
 - b) Derive a formula for calculating g near the Earth's surface using Newton's law of universal gravitation. M and R represent the mass and radius of the Earth respectively.
7. Two spheres of mass 800 g and 500 g respectively are situated so that their centres are 200 cm apart. Calculate the gravitational force between them.

8. Two spheres of mass 2 kg and 3 kg respectively are situated so that the gravitational force between them is $2,5 \times 10^{-8}$ N. Calculate the distance between them.
9. Two identical spheres are placed 10 cm apart. A force of $1,6675 \times 10^{-9}$ N exists between them. Find the masses of the spheres.
10. Halley's comet, of approximate mass 1×10^{15} kg was $1,3 \times 10^8$ km from the Earth, at its point of closest approach during its last sighting in 1986.
 - a) Name the force through which the Earth and the comet interact.
 - b) Is the magnitude of the force experienced by the comet the same, greater than or less than the force experienced by the Earth? Explain.
 - c) Does the acceleration of the comet increase, decrease or remain the same as it moves closer to the Earth? Explain.
 - d) If the mass of the Earth is 6×10^{24} kg, calculate the magnitude of the force exerted by the Earth on Halley's comet at its point of closest approach.

Think you got it? Get this answer and more practice on our Intelligent Practice Service

1. [23JS](#) 2. [23JT](#) 3. [23JV](#) 4. [23JW](#) 5. [23JX](#) 6. [23JY](#)
 7. [23JZ](#) 8. [23K2](#) 9. [23K3](#) 10. [23K4](#)



www.everythingscience.co.za



m.everythingscience.co.za

2.5 Chapter summary

ESBM3

▶ See presentation: [23K5](#) at www.everythingscience.co.za

- The normal force, $\vec{N}$, is the force exerted by a surface on an object in contact with it. The normal force is perpendicular to the surface.
- Frictional force is the force that opposes the motion of an object in contact with a surface and it acts parallel to the surface the object is in contact with. The magnitude of friction is proportional to the normal force.
- For every surface we can determine a constant factor, the coefficient of friction, that allows us to calculate what the frictional force would be if we know the magnitude of the normal force. We know that static friction and kinetic friction have different magnitudes so we have different coefficients for the two types of friction:
 - μ_s is the coefficient of **static** friction
 - μ_k is the coefficient of **kinetic** friction
- The components of the force due to gravity, $\vec{F}_g$, parallel (x -direction) and perpendicular (y -direction) to a slope are given by:

$$F_{gx} = F_g \sin(\theta)$$

$$F_{gy} = F_g \cos(\theta)$$

- **Newton's first law:** An object continues in a state of rest or uniform motion (motion with a constant velocity) unless it is acted on by an unbalanced (net or resultant) force.
- **Newton's second law:** If a resultant force acts on a body, it will cause the body to accelerate in the direction of the resultant force. The acceleration of the body will be directly proportional to the resultant force and inversely proportional to the mass of the body. The mathematical representation is:

$$\vec{F}_{net} = m\vec{a}$$

- **Newton's third law:** If body A exerts a force on body B, then body B exerts a force of equal magnitude on body A, but in the opposite direction.
- **Newton's law of universal gravitation:** Every point mass attracts every other point mass by a force directed along the line connecting the two. This force is proportional to the product of the masses and inversely proportional to the square of the distance between them.

$$F = G \frac{m_1 m_2}{d^2}$$

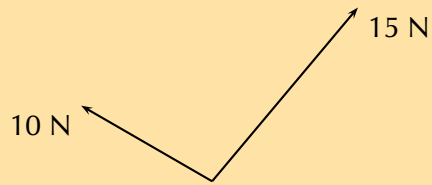
Physical Quantities		
Quantity	Unit name	Unit symbol
Acceleration (a)	metres per second squared	$\text{m}\cdot\text{s}^{-1}$
Force (F)	newton	N
Mass (m)	kilogram	kg
Tension (T)	newton	N
Weight (N)	newton	N

Exercise 2 – 10: Forces and Newton's Laws

1. A force acts on an object. Name three effects that the force can have on the object.
2. Identify each of the following forces as contact or non-contact forces.
 - a) The force between the north pole of a magnet and a paper clip.
 - b) The force required to open the door of a taxi.
 - c) The force required to stop a soccer ball.
 - d) The force causing a ball, dropped from a height of 2 m, to fall to the floor.
3. A book of mass 2 kg is lying on a table. Draw a labelled force diagram indicating all the forces on the book.
4. A constant, resultant force acts on a body which can move freely in a straight line. Which physical quantity will remain constant?
 - a) acceleration
 - b) velocity
 - c) momentum
 - d) kinetic energy

[SC 2003/11]

5. Two forces, 10 N and 15 N, act at an angle at the same point.

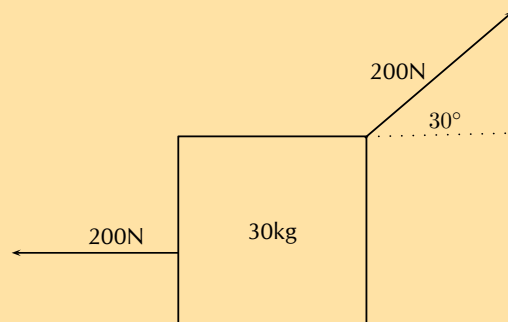


Which of the following **cannot** be the resultant of these two forces?

- a) 2 N
- b) 5 N
- c) 8 N
- d) 20 N

[SC 2005/11 SG1]

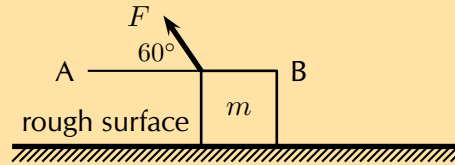
6. A concrete block weighing 250 N is at rest on an inclined surface at an angle of 20° . The magnitude of the normal force, in newtons, is
- a) 250
 - b) $250 \cos 20^\circ$
 - c) $250 \sin 20^\circ$
 - d) $2500 \cos 20^\circ$
7. A 30 kg box sits on a flat frictionless surface. Two forces of 200 N each are applied to the box as shown in the diagram. Which statement best describes the motion of the box?
- a) The box is lifted off the surface.
 - b) The box moves to the right.
 - c) The box does not move.
 - d) The box moves to the left.



8. A concrete block weighing 200 N is at rest on an inclined surface at an angle of 20° . The normal force, in newtons, is
- a) 200
 - b) $200 \cos 20^\circ$

- c) $200 \sin 20^\circ$
- d) $2000 \cos 20^\circ$

9. A box, mass m , is at rest on a rough horizontal surface. A force of constant magnitude F is then applied on the box at an angle of 60° to the horizontal, as shown.



If the box has a uniform horizontal acceleration of magnitude, a , the frictional force acting on the box is...

- a) $F \cos 60^\circ - ma$ in the direction of A
- b) $F \cos 60^\circ - ma$ in the direction of B
- c) $F \sin 60^\circ - ma$ in the direction of A
- d) $F \sin 60^\circ - ma$ in the direction of B

[SC 2003/11]

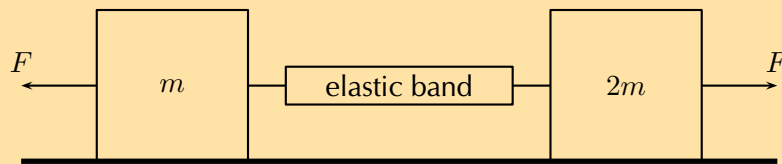
10. Thabo stands in a train carriage which is moving eastwards. The train suddenly brakes. Thabo continues to move eastwards due to the effect of:
- a) his inertia.
 - b) the inertia of the train.
 - c) the braking force on him.
 - d) a resultant force acting on him.

[SC 2002/11 SG]

11. A 100 kg crate is placed on a slope that makes an angle of 45° with the horizontal. The gravitational force on the box is 98 N. The box does not slide down the slope. Calculate the magnitude and direction of the frictional force and the normal force present in this situation.
12. A body moving at a *CONSTANT VELOCITY* on a horizontal plane, has a number of unequal forces acting on it. Which one of the following statements is TRUE?
- a) At least two of the forces must be acting in the same direction.
 - b) The resultant of the forces is zero.
 - c) Friction between the body and the plane causes a resultant force.
 - d) The vector sum of the forces causes a resultant force which acts in the direction of motion.

[SC 2002/11 HG1]

13. Two masses of m and $2m$ respectively are connected by an elastic band on a frictionless surface. The masses are pulled in opposite directions by two forces each of magnitude F , stretching the elastic band and holding the masses stationary.

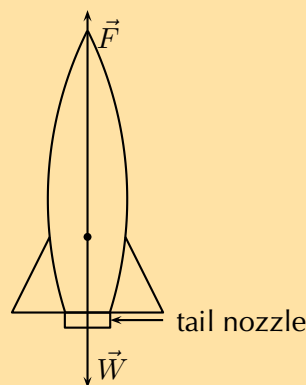


Which of the following gives the magnitude of the tension in the elastic band?

- a) zero
- b) $\frac{1}{2}F$
- c) F
- d) $2F$

[IEB 2005/11 HG]

14. A rocket takes off from its launching pad, accelerating up into the air.

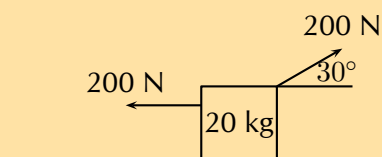


The rocket accelerates because the magnitude of the upward force, F is greater than the magnitude of the rocket's weight, W . Which of the following statements **best** describes how force F arises?

- a) F is the force of the air acting on the base of the rocket.
- b) F is the force of the rocket's gas jet *pushing down* on the air.
- c) F is the force of the rocket's gas jet *pushing down* on the ground.
- d) F is the reaction to the force that the rocket exerts on the gases which escape out through the tail nozzle.

[IEB 2005/11 HG]

15. A box of mass 20 kg rests on a smooth horizontal surface. What will happen to the box if two forces each of magnitude 200 N are applied simultaneously to the box as shown in the diagram.



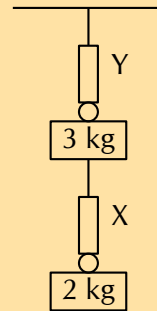
The box will:

- a) be lifted off the surface.

- b) move to the left.
- c) move to the right.
- d) remain at rest.

[SC 2001/11 HG1]

16. A 2 kg mass is suspended from spring balance X, while a 3 kg mass is suspended from spring balance Y. Balance X is in turn suspended from the 3 kg mass. Ignore the weights of the two spring balances.

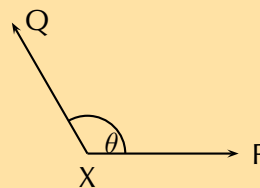


The readings (in N) on balances X and Y are as follows:

	X	Y
a)	19,6	29,4
b)	19,6	49
c)	24,5	24,5
d)	49	49

[SC 2001/11 HG1]

17. P and Q are two forces of equal magnitude applied simultaneously to a body at X.

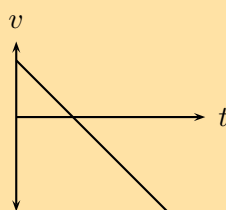


As the angle θ between the forces is **decreased** from 180° to 0° , the magnitude of the resultant of the two forces will

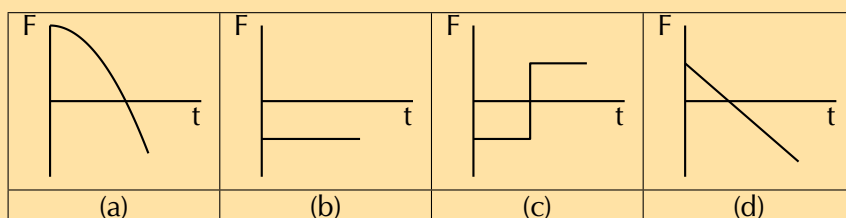
- a) initially increase and then decrease.
- b) initially decrease and then increase.
- c) increase only.
- d) decrease only.

[SC 2002/03 HG1]

18. The graph below shows the velocity-time graph for a moving object:

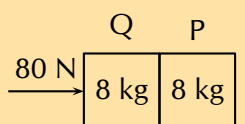


Which of the following graphs could best represent the relationship between the resultant force applied to the object and time?



[SC 2002/03 HG1]

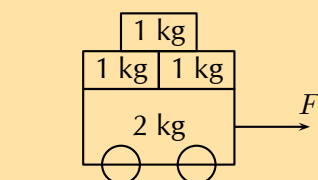
19. Two blocks each of mass 8 kg are in contact with each other and are accelerated along a frictionless surface by a force of 80 N as shown in the diagram. The force which block Q will exert on block P is equal to ...



- a) 0 N
- b) 40 N
- c) 60 N
- d) 80 N

[SC 2002/03 HG1]

20. A 12 kg box is placed on a rough surface. A force of 20 N applied at an angle of 30° to the horizontal cannot move the box. Calculate the magnitude and direction of the normal and friction forces.
21. Three 1 kg mass pieces are placed on top of a 2 kg trolley. When a force of magnitude F is applied to the trolley, it experiences an acceleration a .



If one of the 1 kg mass pieces falls off while F is still being applied, the trolley will accelerate at ...

- a) $\frac{1}{5}a$
- b) $\frac{4}{5}a$

- c) $\frac{5}{4}a$
- d) $5a$

[SC 2002/03 HG1]

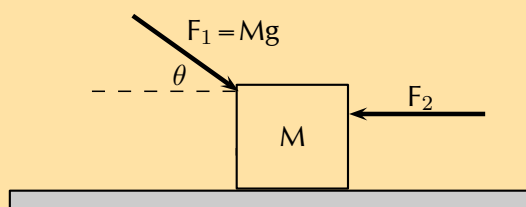
22. A car moves along a horizontal road at constant velocity. Which of the following statements is true?
- a) The car is not in equilibrium.
 - b) There are no forces acting on the car.
 - c) There is zero resultant force.
 - d) There is no frictional force.

[IEB 2004/11 HG1]

23. A crane lifts a load vertically upwards at constant speed. The upward force exerted on the load is F . Which of the following statements is correct?
- a) The acceleration of the load is $9,8 \text{ m}\cdot\text{s}^{-1}$ downwards.
 - b) The resultant force on the load is F .
 - c) The load has a weight equal in magnitude to F .
 - d) The forces of the crane on the load, and the weight of the load, are an example of a Newton's third law 'action-reaction' pair.

[IEB 2004/11 HG1]

24. A body of mass M is at rest on a smooth horizontal surface with two forces applied to it as in the diagram below. Force F_1 is equal to Mg . The force F_1 is applied to the right at an angle θ to the horizontal, and a force of F_2 is applied horizontally to the left.



How is the body affected when the angle θ is increased?

- a) It remains at rest.
- b) It lifts up off the surface, and accelerates towards the right.
- c) It lifts up off the surface, and accelerates towards the left.
- d) It accelerates to the left, moving along the smooth horizontal surface.

[IEB 2004/11 HG1]

25. Which of the following statements correctly explains why a passenger in a car, who is not restrained by the seat belt, continues to move forward when the brakes are applied suddenly?
- a) The braking force applied to the car exerts an equal and opposite force on the passenger.

- b) A forward force (called inertia) acts on the passenger.
- c) A resultant forward force acts on the passenger.
- d) A zero resultant force acts on the passenger.

[IEB 2003/11 HG1]

26. A rocket (mass 20 000 kg) accelerates from rest to $40 \text{ m}\cdot\text{s}^{-1}$ in the first 1,6 seconds of its journey upwards into space.

The rocket's propulsion system consists of exhaust gases, which are pushed out of an outlet at its base.

- a) Explain, with reference to the appropriate law of Newton, how the escaping exhaust gases exert an upwards force (thrust) on the rocket.
- b) What is the magnitude of the total thrust exerted on the rocket during the first 1,6 s?
- c) An astronaut of mass 80 kg is carried in the space capsule. Determine the resultant force acting on him during the first 1,6 s.
- d) Explain why the astronaut, seated in his chair, feels "heavier" while the rocket is launched.

[IEB 2004/11 HG1]

27. a) State Newton's second law of Motion.
- b) A sports car (mass 1000 kg) is able to accelerate uniformly from rest to $30 \text{ m}\cdot\text{s}^{-1}$ in a minimum time of 6 s.
- i. Calculate the magnitude of the acceleration of the car.
 - ii. What is the magnitude of the resultant force acting on the car during these 6 s?
- c) The magnitude of the force that the wheels of the vehicle exert on the road surface as it accelerates is 7500 N. What is the magnitude of the retarding forces acting on this car?
- d) By reference to a suitable Law of Motion, explain why a headrest is important in a car with such a rapid acceleration.

[IEB 2003/11 HG1 - Sports Car]

28. A child (mass 18 kg) is strapped in his car seat as the car moves to the right at constant velocity along a straight level road. A tool box rests on the seat beside him.



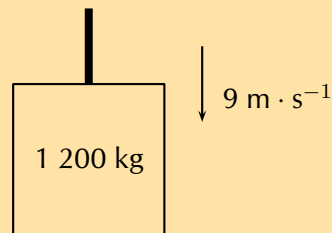
The driver brakes suddenly, bringing the car rapidly to a halt. There is negligible friction between the car seat and the box.

- Draw a labelled free-body diagram of **the forces acting on the child** during the time that the car is being braked.
- Draw a labelled free-body diagram of **the forces acting on the box** during the time that the car is being braked.
- Modern cars are designed with safety features (besides seat belts) to protect drivers and passengers during collisions e.g. the crumple zones on the car's body. Rather than remaining rigid during a collision, the crumple zones allow the car's body to collapse steadily.*

State Newton's second law of motion.

[IEB 2005/11 HG1]

29. The total mass of a lift together with its load is 1200 kg. It is moving downwards at a constant velocity of $9 \text{ m} \cdot \text{s}^{-1}$.



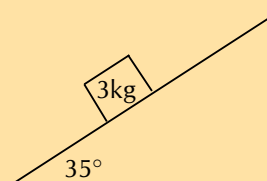
- What will be the magnitude of the force exerted by the cable on the lift while it is moving downwards at constant velocity? Give an explanation for your answer.
- The lift is now uniformly brought to rest over a distance of 18 m.* Calculate the magnitude of the acceleration of the lift.
- Calculate the magnitude of the force exerted by the cable while the lift is being brought to rest.

[SC 2003/11 HG1]

30. A driving force of 800 N acts on a car of mass 600 kg.

- Calculate the car's acceleration.
- Calculate the car's speed after 20 s.
- Calculate the new acceleration if a frictional force of 50 N starts to act on the car after 20 s.
- Calculate the speed of the car after another 20 s (i.e. a total of 40 s after the start).

31. A stationary block of mass 3 kg is on top of a plane inclined at 35° to the horizontal.



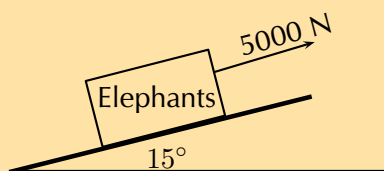
- Draw a force diagram (not to scale). Include the weight of the block as well as the components of the weight that are perpendicular and parallel to the inclined plane.

- b) Determine the values of the weight's perpendicular and parallel components.

32. A crate on an inclined plane

Elephants are being moved from the Kruger National Park to the Eastern Cape. They are loaded into crates that are pulled up a ramp (an inclined plane) on frictionless rollers.

The diagram shows a crate being held stationary on the ramp by means of a rope parallel to the ramp. The tension in the rope is 5000 N.



- Explain how one can deduce the following: "The forces acting on the crate are in equilibrium".
- Draw a labelled free-body diagram of the forces acting on the elephant. (Regard the crate and elephant as one object, and represent them as a dot. Also show the relevant angles between the forces.)
- The crate has a mass of 800 kg. Determine the mass of the elephant.
- The crate is now pulled up the ramp at a constant speed. How does the crate being pulled up the ramp at a constant speed affect the forces acting on the crate and elephant? Justify your answer, mentioning any law or principle that applies to this situation.

[IEB 2002/11 HG1]

33. Car in Tow

Car A is towing Car B with a light tow rope. The cars move along a straight, horizontal road.

- Write down a statement of Newton's second law of Motion (in words).
- As they start off, Car A exerts a forwards force of 600 N at its end of the tow rope. The force of friction on Car B when it starts to move is 200 N. The mass of Car B is 1200 kg. Calculate the acceleration of Car B.
- After a while, the cars travel at constant velocity. The force exerted on the tow rope is now 300 N while the force of friction on Car B increases. What is the magnitude and direction of the force of friction on Car B now?
- Towing with a rope is very dangerous. A solid bar should be used in preference to a tow rope. This is especially true should Car A suddenly apply brakes. What would be the advantage of the solid bar over the tow rope in such a situation?
- The mass of Car A is also 1200 kg. Car A and Car B are now joined by a solid tow bar and the total braking force is 9600 N. Over what distance could the cars stop from a velocity of $20 \text{ m}\cdot\text{s}^{-1}$?

[IEB 2002/11 HG1]

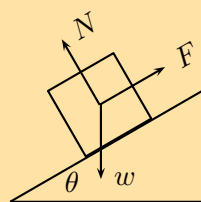
34. Testing the Brakes of a Car

A braking test is carried out on a car travelling at $20 \text{ m}\cdot\text{s}^{-1}$. A braking distance of 30 m is measured when a braking force of 6000 N is applied to stop the car.

- Calculate the acceleration of the car when a braking force of 6000 N is applied.
- Show** that the mass of this car is 900 kg.
- How long (in s) does it take for this car to stop from $20 \text{ m}\cdot\text{s}^{-1}$ under the braking action described above?
- A trailer of mass 600 kg is attached to the car and the braking test is repeated from $20 \text{ m}\cdot\text{s}^{-1}$ using the same braking force of 6000 N. How much longer will it take to stop the car with the trailer in tow?

[IEB 2001/11 HG1]

35. A box is held stationary on a smooth plane that is inclined at angle θ to the horizontal.

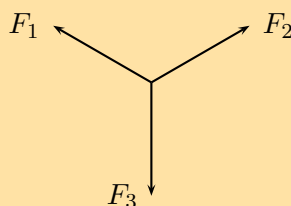


F is the force exerted by a rope on the box. w is the weight of the box and N is the normal force of the plane on the box. Which of the following statements is correct?

- $\tan \theta = \frac{F}{w}$
- $\tan \theta = \frac{F}{N}$
- $\cos \theta = \frac{F}{w}$
- $\sin \theta = \frac{N}{w}$

[IEB 2005/11 HG]

36. As a result of three forces F_1 , F_2 and F_3 acting on it, an object at point P is in equilibrium.

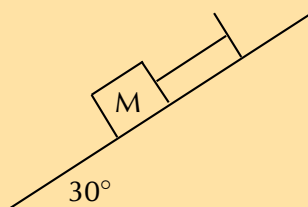


Which of the following statements is **not true** with reference to the three forces?

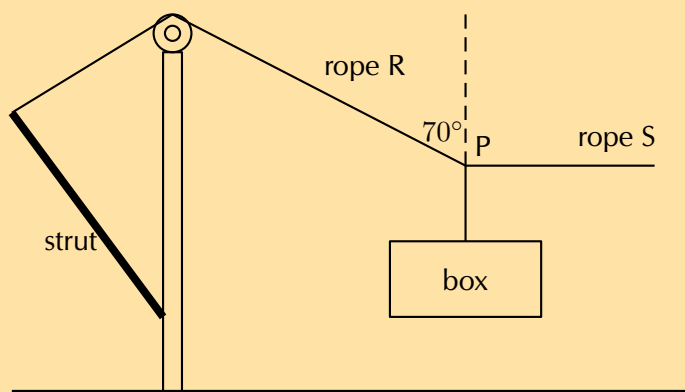
- The resultant of forces F_1 , F_2 and F_3 is zero.
- Force F_1 , F_2 and F_3 lie in the same plane.
- Force F_3 is the resultant of forces F_1 and F_2 .
- The sum of the components of all the forces in any chosen direction is zero.

[SC 2001/11 HG1]

37. A block of mass M is held stationary by a rope of negligible mass. The block rests on a frictionless plane which is inclined at 30° to the horizontal.



- a) Draw a labelled force diagram which shows all the forces acting on the block.
 - b) Resolve the force due to gravity into components that are parallel and perpendicular to the plane.
 - c) Calculate the weight of the block when the force in the rope is 8 N.
38. A heavy box, mass m , is lifted by means of a rope R which passes over a pulley fixed to a pole. A second rope S, tied to rope R at point P, exerts a horizontal force and pulls the box to the right. After lifting the box to a certain height, the box is held stationary as shown in the sketch below. Ignore the masses of the ropes. The tension in rope R is 5850 N.



- a) Draw a diagram (with labels) of all the forces acting at the point P, when P is in equilibrium.
 - b) By resolving the force exerted by rope R into components, calculate the...
 - i. magnitude of the force exerted by rope S.
 - ii. mass, m , of the box.
- [SC 2003/11]
39. A tow truck attempts to tow a broken down car of mass 400 kg. The coefficient of static friction is 0,60 and the coefficient of kinetic (dynamic) friction is 0,4. A rope connects the tow truck to the car. Calculate the force required:
- a) to just move the car if the rope is parallel to the road.
 - b) to keep the car moving at constant speed if the rope is parallel to the road.
 - c) to just move the car if the rope makes an angle of 30° to the road.
 - d) to keep the car moving at constant speed if the rope makes an angle of 30° to the road.

Think you got it? Get this answer and more practice on our Intelligent Practice Service

- | | | | | | |
|----------|----------|----------|----------|----------|----------|
| 1. 23K6 | 2a. 23K7 | 2b. 23K8 | 2c. 23K9 | 2d. 23KB | 3. 23KC |
| 4. 23KD | 5. 23KF | 6. 23KG | 7. 23KH | 8. 23KJ | 9. 23KK |
| 10. 23KM | 11. 23KN | 12. 23KP | 13. 23KQ | 14. 23KR | 15. 23KS |
| 16. 23KT | 17. 23KV | 18. 23KW | 19. 23KX | 20. 23KY | 21. 23KZ |
| 22. 23M2 | 23. 23M3 | 24. 23M4 | 25. 23M5 | 26. 23M6 | 27. 23M7 |
| 28. 23M8 | 29. 23M9 | 30. 23MB | 31. 23MC | 32. 23MD | 33. 23MF |
| 34. 23MG | 35. 23MJ | 36. 23MK | 37. 23MM | 38. 23MN | 39. 23MP |



www.everythingscience.co.za



m.everythingscience.co.za

Exercise 2 – 11: Gravitation

1. An object attracts another with a gravitational force F . If the distance between the centres of the two objects is now decreased to a third ($\frac{1}{3}$) of the original distance, the force of attraction that the one object would exert on the other would become...

- a) $\frac{1}{9}F$
- b) $\frac{1}{3}F$
- c) $3F$
- d) $9F$

[SC 2003/11]

2. An object is dropped from a height of 1 km above the Earth. If air resistance is ignored, the acceleration of the object is dependent on the...

- a) mass of the object
- b) radius of the earth
- c) mass of the earth
- d) weight of the object

[SC 2003/11]

3. A man has a mass of 70 kg on Earth. He is walking on a new planet that has a mass four times that of the Earth and the radius is the same as that of the Earth ($M_E = 6 \times 10^{24}$ kg, $r_E = 6 \times 10^6$ m)

- a) Calculate the force between the man and the Earth.
- b) What is the man's weight on the new planet?
- c) Would his weight be bigger or smaller on the new planet? Explain how you arrived at your answer.

4. Calculate the distance between two objects, 5000 kg and 6×10^{12} kg respectively, if the magnitude of the force between them is 3×10^8 N

5. An astronaut in a satellite 1600 km above the Earth experiences a gravitational force of magnitude 700 N on Earth. The Earth's radius is 6400 km. Calculate:

- a) The magnitude of the gravitational force which the astronaut experiences in the satellite.
 - b) The magnitude of the gravitational force on an object in the satellite which weighs 300 N on Earth.
6. An astronaut of mass 70 kg on Earth lands on a planet which has half the Earth's radius and twice its mass. Calculate the magnitude of the force of gravity which is exerted on him on the planet.
7. Calculate the magnitude of the gravitational force of attraction between two spheres of lead with a mass of 10 kg and 6 kg respectively if they are placed 50 mm apart.
8. The gravitational force between two objects is 1200 N. What is the gravitational force between the objects if the mass of each is doubled and the distance between them halved?
9. Calculate the gravitational force between the Sun with a mass of 2×10^{30} kg and the Earth with a mass of 6×10^{24} kg if the distance between them is $1,4 \times 10^8$ km.
10. How does the gravitational force of attraction between two objects change when
 - a) the mass of each object is doubled.
 - b) the distance between the centres of the objects is doubled.
 - c) the mass of one object is halved, and the distance between the centres of the objects is halved.
11. Read each of the following statements and say whether you agree or not. Give reasons for your answer and rewrite the statement if necessary:
 - a) The gravitational acceleration g is constant.
 - b) The weight of an object is independent of its mass.
 - c) G is dependent on the mass of the object that is being accelerated.
12. An astronaut weighs 750 N on the surface of the Earth.
 - a) What will his weight be on the surface of Saturn, which has a mass 100 times greater than the Earth, and a radius 5 times greater than the Earth?
 - b) What is his mass on Saturn?
13. Your mass is 60 kg in Paris at ground level. How much less would you weigh after taking a lift to the top of the Eiffel Tower, which is 405 m high? Assume the Earth's mass is $6,0 \times 10^{24}$ kg and the Earth's radius is 6400 km.
14.
 - a) State Newton's law of universal gravitation.
 - b) Use Newton's law of universal gravitation to determine the magnitude of the acceleration due to gravity on the Moon.
The mass of the Moon is $7,4 \times 10^{22}$ kg.
The radius of the Moon is $1,74 \times 10^6$ m.
 - c) Will an astronaut, kitted out in his space suit, jump higher on the Moon or on the Earth? Give a reason for your answer.

Think you got it? Get this answer and more practice on our Intelligent Practice Service

1. [23MS](#) 2. [23MT](#) 3. [23MV](#) 4. [23MW](#) 5. [23MY](#) 6. [23MZ](#)
7. [23N2](#) 8. [23N3](#) 9. [23N4](#) 10. [23N5](#) 11a. [23N6](#) 11b. [23N7](#)
11c. [23N8](#) 12. [23N9](#) 13. [23NC](#) 14. [23ND](#)



www.everythingscience.co.za



m.everythingscience.co.za

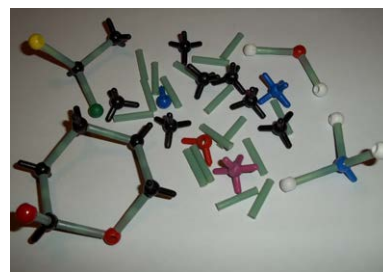
Atomic combinations

3.1	<i>Chemical bonds</i>	136
3.2	<i>Molecular shape</i>	148
3.3	<i>Electronegativity</i>	153
3.4	<i>Energy and bonding</i>	159
3.5	<i>Chapter summary</i>	160

We live in a world that is made up of many complex compounds. All around us we see evidence of chemical bonding from the chair you are sitting on, to the book you are holding, to the air you are breathing. Imagine if all the elements on the periodic table did not form bonds but rather remained on their own. Our world would be pretty boring with only 100 or so elements to use.

Imagine you were painting a picture and wanted to show the colours around you. The only paints you have are red, green, yellow, blue, white and black. Yet you are able to make pink, purple, orange and many other colours by mixing these paints. In the same way, the elements can be thought of as nature's paint box. The elements can be joined together in many different ways to make new compounds and so create the world around you.

In Grade 10 we started exploring chemical bonding. In this chapter we will go on to explain more about chemical bonding and why chemical bonding occurs. We looked at the three types of bonding: covalent, ionic and metallic. In this chapter we will focus mainly on covalent bonding and on the molecules that form as a result of covalent bonding.



NOTE:

In this chapter we will use the term molecule to mean a covalent molecular structure. This is a covalent compound that interacts and exists as a single entity.

3.1 Chemical bonds

ESBM4

Why do atoms bond?

ESBM5

As we begin this section, it's important to remember that what we will go on to discuss is a *model* of bonding, that is based on a particular *model* of the atom. You will remember from the discussion on atoms (in Grade 10) that a model is a *representation* of what is happening in reality. In the model of the atom that you are learnt in Grade 10, the atom is made up of a central nucleus, surrounded by electrons that are arranged in fixed energy levels (sometimes called *shells*). Within each energy level, electrons move in *orbitals* of different shapes. The electrons in the outermost energy level of an atom are called the **valence electrons**. This model of the atom is useful in trying to understand how different types of bonding take place between atoms.

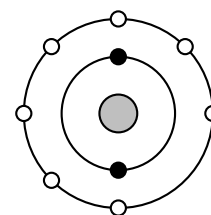


Figure 3.1:

Electron arrangement of a fluorine atom. The black electrons (small circles on the inner ring) are the core electrons and the white electrons (small circles on the outer ring) are the valence electrons.

The following points were made in these earlier discussions on electrons and energy levels:

- Electrons always try to occupy the lowest possible energy level.
- The noble gases have a full valence electron orbital. For example neon has the following electronic configuration: $1s^2 2s^2 2p^6$. The second energy level is the outermost (valence) shell and is full.
- Atoms form bonds to try to achieve the same electron configuration as the noble gases.
- Atoms with a full valence electron orbital are less reactive.

TIP

A model takes what we see in the world around us and uses that to make certain predictions about what we cannot see.

Energy and bonding

ESBM6

There are two cases that we need to consider when two atoms come close together. The first case is where the two atoms come close together and form a bond. The second case is where the two atoms come close together but do not form a bond. We will use hydrogen as an example of the first case and helium as an example of the second case.

Case 1: A bond forms

Let's start by imagining that there are two hydrogen atoms approaching one another. As they move closer together, there are three forces that act on the atoms at the same time. These forces are described below:

1. **repulsive force** between the electrons of the atoms, since like charges repel

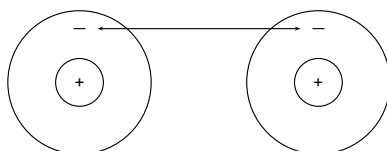


Figure 3.2: Repulsion between electrons

2. **attractive force** between the nucleus of one atom and the electrons of another

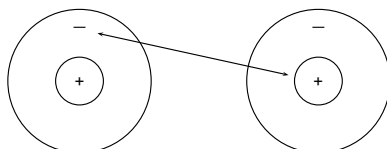


Figure 3.3: Attraction between electrons and protons.

3. **repulsive force** between the two positively-charged nuclei

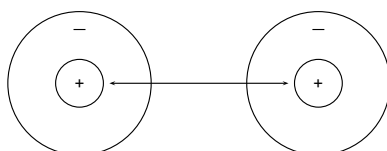


Figure 3.4: Repulsion between protons

These three forces work together when two atoms come close together. As the total force experienced by the atoms changes, the amount of energy in the system also changes.

Now look at Figure 3.5 to understand the energy changes that take place when the two atoms move towards each other.

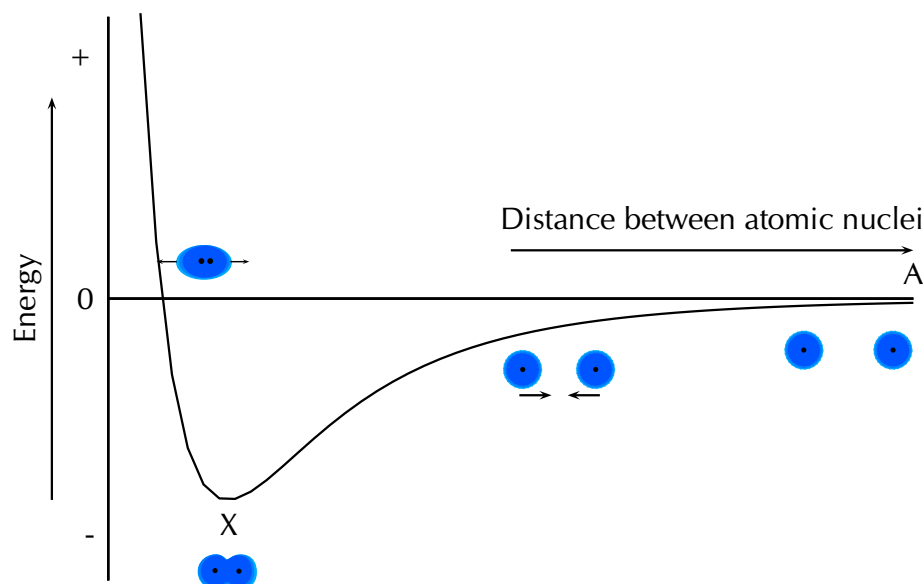


Figure 3.5: Graph showing the change in energy that takes place as two hydrogen atoms move closer together.

Let us imagine that we have fixed the one atom and we will move the other atom closer to the first atom. As we move the second hydrogen atom closer to the first (from point A to point X) the energy of the system decreases. Attractive forces dominate this part of the interaction. As the second atom approaches the first one and gets closer to point X, more energy is needed to pull the atoms apart. This gives a negative potential energy.

At point X, the attractive and repulsive forces acting on the two hydrogen atoms are balanced. The energy of the system is at a minimum.

Further to the left of point X, the repulsive forces are stronger than the attractive forces and the energy of the system increases.

For hydrogen the energy at point X is low enough that the two atoms stay together and do not break apart again. This is why when we draw the Lewis diagram for a hydrogen molecule we draw two hydrogen atoms next to each other with an electron pair between them.



We also note that this arrangement gives both hydrogen atoms a full outermost energy level (through the sharing of electrons or covalent bonding).

Case 2: A bond does not form

Now if we look at helium we see that each helium atom has a filled outer energy level. Looking at Figure 3.6 we find that the energy minimum for two helium atoms is very close to zero. This means that the two atoms can come together and move apart very easily and never actually stick together.

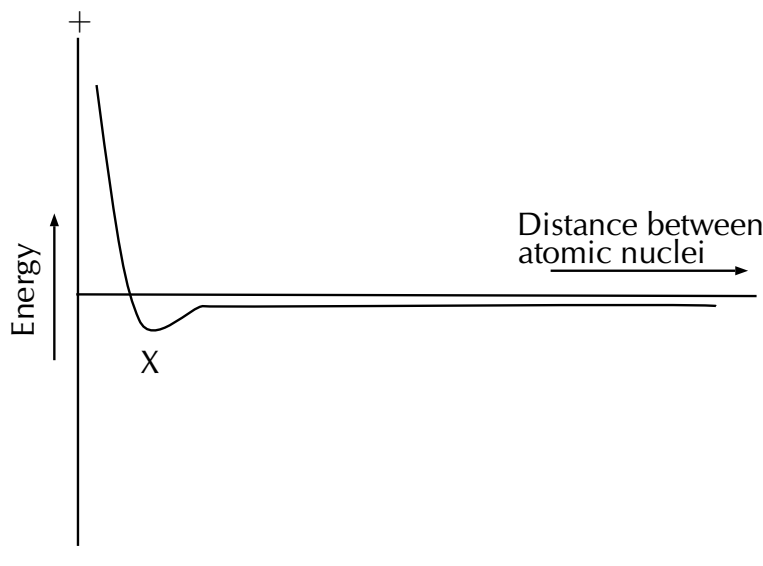


Figure 3.6: Graph showing the change in energy that takes place as two helium atoms move closer together.

For helium the energy minimum at point X is not low enough that the two atoms stay together and so they move apart again. This is why when we draw the Lewis diagram for helium we draw one helium atom on its own. There is no bond.

We also see that helium already has a full outermost energy level and so no compound forms.



► See simulation: 23NF at www.everythingscience.co.za

Valence electrons and Lewis diagrams

ESBM7

Now that we understand a bit more about bonding we need to refresh the concept of Lewis diagrams that you learnt about in Grade 10. With the knowledge of why atoms bond and the knowledge of how to draw Lewis diagrams we will have all the tools that we need to try to predict which atoms will bond and what shape the molecule will be.

In grade 10 we learnt how to write the electronic structure for any element. For drawing Lewis diagrams the one that you should be familiar with is the spectroscopic notation. For example the electron configuration of chlorine in spectroscopic notation is: $1s^2 2s^2 2p^5$. Or if we use the condensed form: $[\text{He}] 2s^2 2p^5$. The condensed spectroscopic notation quickly shows you the valence electrons for the element.

TIP

A **Lewis diagram** uses dots or crosses to represent the **valence electrons** on different atoms. The chemical symbol of the element is used to represent the nucleus and the core electrons of the atom.

TIP

You can place the unpaired electrons anywhere (top, bottom, left or right). The exact ordering in a Lewis diagram does not matter.

Using the number of valence electrons we can easily draw Lewis diagrams for any element. In Grade 10 you learnt how to draw Lewis diagrams. We will refresh the concepts here as they will aid us in our discussion of bonding.

Lewis diagrams for the elements in period 2 are shown below:

Element	Group number	Valence electrons	Spectroscopic notation	Lewis diagram
Lithium	1	1	[He]2s ¹	Li •
Beryllium	2	2	[He]2s ²	Be ••
Boron	13	3	[He]2s ² 2p ¹	• B ••
Carbon	14	4	[He]2s ² 2p ²	•• • C ••
Nitrogen	15	5	[He]2s ² 2p ³	•• • N ••
Oxygen	16	6	[He]2s ² 2p ⁴	•• • O ••
Fluorine	17	7	[He]2s ² 2p ⁵	•• • F ••
Neon	18	8	[He]2s ² 2p ⁶	•• • Ne ••

Exercise 3 – 1: Lewis diagrams

Give the spectroscopic notation and draw the Lewis diagram for:

1. magnesium
2. sodium
3. chlorine
4. aluminium
5. argon

Think you got it? Get this answer and more practice on our Intelligent Practice Service

1. [23NG](#) 2. [23NH](#) 3. [23NJ](#) 4. [23NK](#) 5. [23NM](#)



www.everythingscience.co.za



m.everythingscience.co.za

Covalent bonds and bond formation

ESBM8

Covalent bonding involves the sharing of electrons to form a chemical bond. The outermost orbitals of the atoms overlap so that unpaired electrons in each of the bonding atoms can be shared. By overlapping orbitals, the outer energy shells of all the bonding atoms are filled. The shared electrons move in the orbitals around *both* atoms. As

they move, there is an attraction between these negatively charged electrons and the positively charged nuclei. This attractive force holds the atoms together in a covalent bond.

DEFINITION: *Covalent bond*

A form of chemical bond where pairs of electrons are shared between atoms.

We will look at a few simple cases to deduce some rules about covalent bonds.

Case 1: Two atoms that each have an unpaired electron

For this case we will look at hydrogen chloride and methane.

TIP

Covalent bonds are examples of interatomic forces.

TIP

Remember that it is only the *valence electrons* that are involved in bonding, and so when diagrams are drawn to show what is happening during bonding, it is only these electrons that are shown. Dots or crosses represent electrons in different atoms.

Worked example 1: Lewis diagrams: Simple molecules

QUESTION

Represent hydrogen chloride (HCl) using a Lewis diagram.

SOLUTION

Step 1: For each atom, determine the number of valence electrons in the atom, and represent these using dots and crosses.

The electron configuration of hydrogen is $1s^1$ and the electron configuration for chlorine is $[\text{He}]2s^22p^5$. The hydrogen atom has 1 valence electron and the chlorine atom has 7 valence electrons.

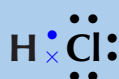
The Lewis diagrams for hydrogen and chlorine are:



Notice the single unpaired electron (highlighted in blue) on each atom. This does not mean this electron is different, we use highlighting here to help you see the unpaired electron.

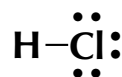
Step 2: Arrange the electrons so that the outermost energy level of each atom is full.

Hydrogen chloride is represented below.



Notice how the two unpaired electrons (one from each atom) form the covalent bond.

The dot and cross in between the two atoms, represent the pair of electrons that are shared in the covalent bond. We can also show this bond using a single line:



Note how we still show the other electron pairs around chlorine.

From this we can conclude that any electron on its own will try to pair up with another electron. So in practise atoms that have at least one unpaired electron can form bonds with any other atom that also has an unpaired electron. This is not restricted to just two atoms.

Worked example 2: Lewis diagrams: Simple molecules

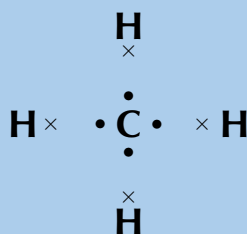
QUESTION

Represent methane (CH_4) using a Lewis diagram

SOLUTION

Step 1: For each atom, determine the number of valence electrons in the atom, and represent these using dots and crosses.

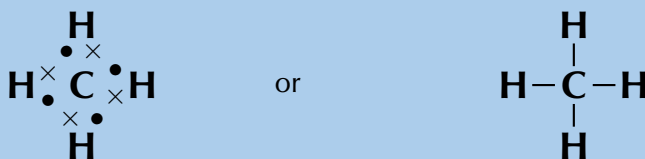
The electron configuration of hydrogen is $1s^1$ and the electron configuration for carbon is $[\text{He}]2s^2 2p^2$. Each hydrogen atom has 1 valence electron and the carbon atom has 4 valence electrons.



Remember that we said we can place unpaired electrons at any position (top, bottom, left, right) around the elements symbol.

Step 2: Arrange the electrons so that the outermost energy level of each atom is full.

The methane molecule is represented below.



Exercise 3 – 2:

Represent the following molecules using Lewis diagram:

1. chlorine (Cl_2)

2. boron trifluoride (BF_3)

Think you got it? Get this answer and more practice on our Intelligent Practice Service

1. 23NN 2. 23NP



www.everythingscience.co.za



m.everythingscience.co.za

TIP

Notice how in this example we wrote a 2 in front of the hydrogen? Instead of writing the Lewis diagram for hydrogen twice, we simply write it once and use the 2 in front of it to indicate that two hydrogens are needed for each oxygen.

Case 2: Atoms with lone pairs

We will use water as an example. Water is made up of one oxygen and two hydrogen atoms. Hydrogen has one unpaired electron. Oxygen has two unpaired electrons and two electron pairs. From what we learnt in the first examples we see that the unpaired electrons can pair up. But what happens to the two pairs? Can these form bonds?

Worked example 3: Lewis diagrams: Simple molecules

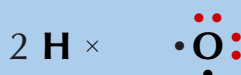
QUESTION

Represent water (H_2O) using a Lewis diagram

SOLUTION

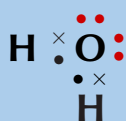
Step 1: For each atom, determine the number of valence electrons in the atom, and represent these using dots and crosses.

The electron configuration of hydrogen is $1s^1$ and the electron configuration for oxygen is $[\text{He}]2s^22p^4$. Each hydrogen atom has 1 valence electron and the oxygen atom has 6 valence electrons.

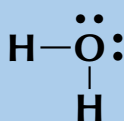


Step 2: Arrange the electrons so that the outermost energy level of each atom is full.

The water molecule is represented below.



or



TIP

A lone pair can be used to form a dative covalent bond.

And now we can answer the questions that we asked before the worked example. We see that oxygen forms two bonds, one with each hydrogen atom. Oxygen however keeps its electron pairs and does not share them. We can generalise this to any atom. If an atom has an electron pair it will normally not share that electron pair.

A **lone pair** is an unshared electron pair. A lone pair stays on the atom that it belongs to.

In the example above the lone pairs on oxygen are highlighted in red. When we draw the bonding pairs using lines it is much easier to see the lone pairs on oxygen.

Exercise 3 – 3:

Represent the following molecules using Lewis diagrams:

1. ammonia (NH_3)

2. oxygen difluoride (OF_2)

Think you got it? Get this answer and more practice on our Intelligent Practice Service

1. [23NQ](#) 2. [23NR](#)



www.everythingscience.co.za



m.everythingscience.co.za

Case 3: Atoms with multiple bonds

We will use oxygen and hydrogen cyanide as examples.

Worked example 4: Lewis diagrams: Molecules with multiple bonds**QUESTION**

Represent oxygen (O_2) using a Lewis diagram

SOLUTION

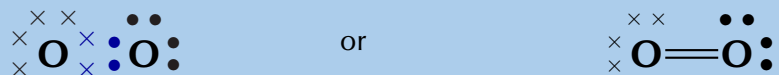
Step 1: For each atom, determine the number of valence electrons that the atom has from its electron configuration.

The electron configuration of oxygen is $[\text{He}]2s^22p^4$. Oxygen has 6 valence electrons.



Step 2: Arrange the electrons in the O₂ molecule so that the outermost energy level in each atom is full.

The O₂ molecule is represented below. Notice the two electron pairs between the two oxygen atoms (highlighted in blue). Because these two covalent bonds are between the same two atoms, this is a *double bond*.



Each oxygen atom uses its two unpaired electrons to form two bonds. This forms a double covalent bond (which is shown by a double line between the two oxygen atoms).

Worked example 5: Lewis diagrams: Molecules with multiple bonds

QUESTION

Represent hydrogen cyanide (HCN) using a Lewis diagram

SOLUTION

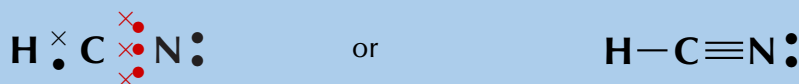
Step 1: For each atom, determine the number of valence electrons that the atom has from its electron configuration.

The electron configuration of hydrogen is 1s¹, the electron configuration of nitrogen is [He]2s²2p³ and for carbon is [He]2s²2p². Hydrogen has 1 valence electron, carbon has 4 valence electrons and nitrogen has 5 valence electrons.



Step 2: Arrange the electrons in the HCN molecule so that the outermost energy level in each atom is full.

The HCN molecule is represented below. Notice the three electron pairs (highlighted in red) between the nitrogen and carbon atom. Because these three covalent bonds are between the same two atoms, this is a *triple bond*.



As we have just seen carbon shares one electron with hydrogen and three with nitrogen. Nitrogen keeps its electron pair and shares its three unpaired electrons with carbon.

Exercise 3 – 4:

Represent the following molecules using Lewis diagrams:

1. acetylene (C_2H_2)

2. formaldehyde (CH_2O)

Think you got it? Get this answer and more practice on our Intelligent Practice Service

1. [23NS](#) 2. [23NT](#)



www.everythingscience.co.za



m.everythingscience.co.za

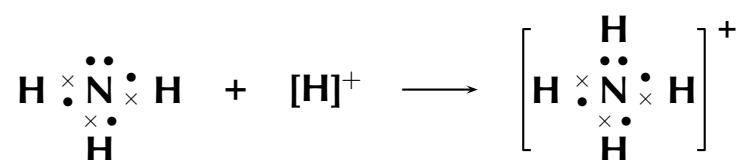
Case 4: Co-ordinate or dative covalent bonds

DEFINITION: *Dative covalent bond*

This type of bond is a description of covalent bonding that occurs between two atoms in which both electrons shared in the bond come from the same atom.

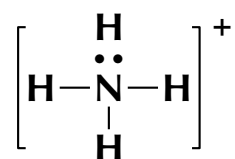
A dative covalent bond is also known as a coordinate covalent bond. Earlier we said that atoms with a pair of electrons will normally not share that pair to form a bond. But now we will see how an electron pair can be used by atoms to form a covalent bond.

One example of a molecule that contains a dative covalent bond is the ammonium ion (NH_4^+) shown in the figure below. The hydrogen ion H^+ does not contain any electrons, and therefore the electrons that are in the bond that forms between this ion and the nitrogen atom, come only from the nitrogen.



Notice that the hydrogen ion is charged and that this charge is shown on the ammonium ion using square brackets and a plus sign outside the square brackets.

We can also show this as:



Note that we do not use a line for the dative covalent bond.

Another example of this is the hydronium ion (H_3O^+).

To summarise what we have learnt:

- Any electron on its own will try to pair up with another electron. So in theory atoms that have at least one unpaired electron can form bonds with any other atom that also has an unpaired electron. This is not restricted to just two atoms.
- If an atom has an electron pair it will normally not share that pair to form a bond. This electron pair is known as a lone pair.
- If an atom has more than one unpaired electron it can form multiple bonds to another atom. In this way double and triple bonds are formed.
- A dative covalent bond can be formed between an atom with no electrons and an atom with a lone pair.

▶ See simulation: 23NV at www.everythingscience.co.za

Exercise 3 – 5: Atomic bonding and Lewis diagrams

1. Represent each of the following *atoms* using Lewis diagrams:

- | | | |
|------------|----------------|------------|
| a) calcium | c) phosphorous | e) silicon |
| b) lithium | d) potassium | f) sulfur |

2. Represent each of the following *molecules* using Lewis diagrams:

- | | |
|-------------------------------------|---|
| a) bromine (Br_2) | d) hydronium ion (H_3O^+) |
| b) carbon dioxide (CO_2) | |
| c) nitrogen (N_2) | e) sulfur dioxide (SO_2) |

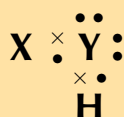
3. Two chemical reactions are described below.

- nitrogen and hydrogen react to form NH_3
- carbon and hydrogen bond to form CH_4

For each reaction, give:

- the number of valence electrons for each of the atoms involved in the reaction
- the Lewis diagram of the product that is formed
- the name of the product

4. A chemical compound has the following Lewis diagram:



- How many valence electrons does element Y have?
- How many valence electrons does element X have?
- How many covalent bonds are in the molecule?
- Suggest a name for the elements X and Y.

5. Complete the following table:

Compound	CO ₂	CF ₄	HI	C ₂ H ₂
Lewis diagram				
Total number of bonding pairs				
Total number of non-bonding pairs				
Single, double or triple bonds				

Think you got it? Get this answer and more practice on our Intelligent Practice Service

- 1a. [23NW](#) 1b. [23NX](#) 1c. [23NY](#) 1d. [23NZ](#) 1e. [23P2](#) 1f. [23P3](#)
 2a. [23P4](#) 2b. [23P5](#) 2c. [23P6](#) 2d. [23P7](#) 2e. [23P8](#) 3. [23P9](#)
 4. [23PB](#) 5. [23PC](#)



www.everythingscience.co.za



m.everythingscience.co.za

3.2 Molecular shape

ESBM9

Molecular shape (the shape that a single molecule has) is important in determining how the molecule interacts and reacts with other molecules. Molecular shape also influences the boiling point and melting point of molecules. If all molecules were linear then life as we know it would not exist. Many of the properties of molecules come from the particular shape that a molecule has. For example if the water molecule was linear, it would be non-polar and so would not have all the special properties it has.

Valence shell electron pair repulsion (VSEPR) theory

ESBMB

The shape of a covalent molecule can be predicted using the Valence Shell Electron Pair Repulsion (VSEPR) theory. Very simply, VSEPR theory says that the valence electron pairs in a molecule will arrange themselves around the central atom(s) of the molecule so that the repulsion between their negative charges is as small as possible.

In other words, the valence electron pairs arrange themselves so that they are as **far apart** as they can be.

DEFINITION: *Valence Shell Electron Pair Repulsion Theory*

Valence shell electron pair repulsion (VSEPR) theory is a model in chemistry, which is used to predict the shape of individual molecules. VSEPR is based upon minimising the extent of the electron-pair repulsion around the central atom being considered.

VSEPR theory is based on the idea that the geometry (shape) of a molecule is mostly determined by repulsion among the pairs of electrons around a central atom. The pairs of electrons may be bonding or non-bonding (also called lone pairs). Only valence electrons of the central atom influence the molecular shape in a meaningful way.

TIP

The central atom is the atom around which the other atoms are arranged. So in a molecule of water, the central atom is oxygen. In a molecule of ammonia, the central atom is nitrogen.

To predict the shape of a covalent molecule, follow these steps:

1. Draw the molecule using a Lewis diagram. Make sure that you draw *all* the valence electrons around the molecule's central atom.
2. Count the number of electron pairs around the central atom.
3. Determine the basic geometry of the molecule using the table below. For example, a molecule with two electron pairs (and no lone pairs) around the central atom has a *linear* shape, and one with four electron pairs (and no lone pairs) around the central atom would have a *tetrahedral* shape.

The table below gives the common molecular shapes. In this table we use **A** to represent the central atom, **X** to represent the terminal atoms (i.e. the atoms around the central atom) and **E** to represent any lone pairs.

Number of bonding electron pairs	Number of lone pairs	Geometry	General formula
1 or 2	0	linear	AX or AX ₂
2	2	<i>bent or angular</i>	AX ₂ E ₂
3	0	trigonal planar	AX ₃
3	1	<i>trigonal pyramidal</i>	AX ₃ E
4	0	tetrahedral	AX ₄
5	0	trigonal bipyramidal	AX ₅
6	0	octahedral	AX ₆

Table 3.1: The effect of electron pairs in determining the shape of molecules. Note that in the general example A is the central atom and X represents the terminal atoms.

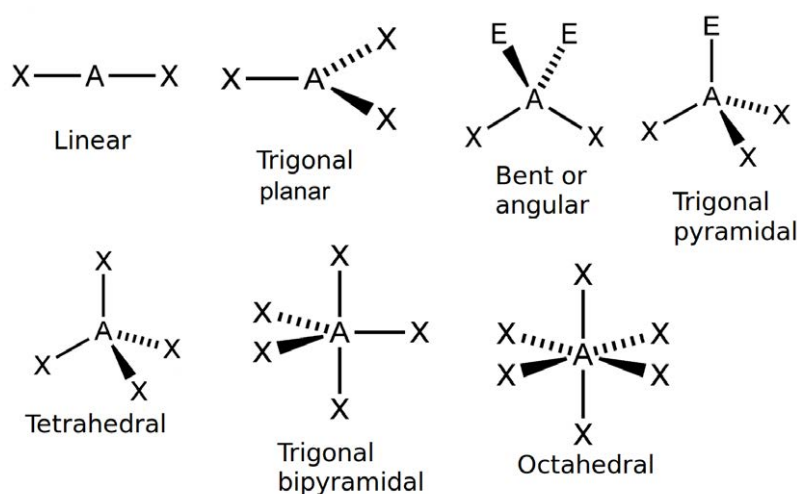


Figure 3.7: The common molecular shapes.

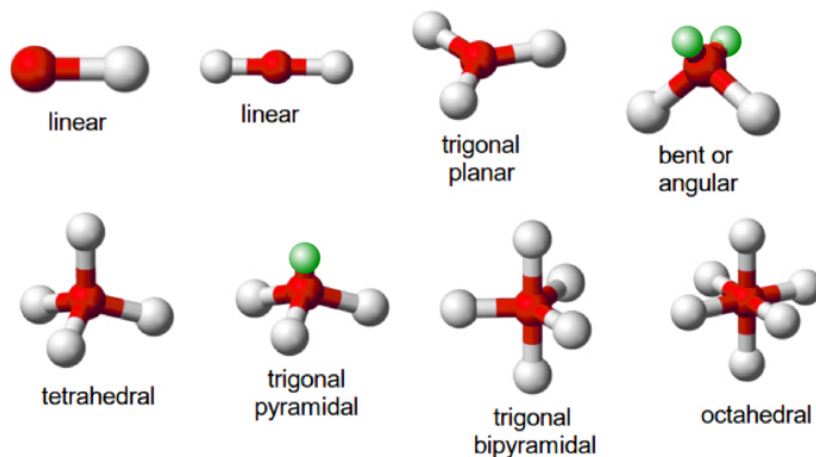


Figure 3.8: The common molecular shapes in 3-D.

In Figure 3.8 the green balls represent the lone pairs (E), the white balls (X) are the terminal atoms and the red balls (A) are the center atoms.

Of these shapes, the ones with no lone pairs are called the **ideal shapes**. The five ideal shapes are: linear, trigonal planar, tetrahedral, trigonal bipyramidal and octahedral.

One important point to note about molecular shape is that all diatomic (compounds with two atoms) compounds are **linear**. So H_2 , HCl and Cl_2 are all linear.

▶ See simulation: 23PD at www.everythingscience.co.za

Worked example 6: Molecular shape

QUESTION

Determine the shape of a molecule of BeCl_2

SOLUTION

Step 1: Draw the molecule using a Lewis diagram



The central atom is beryllium.

Step 2: Count the number of electron pairs around the central atom

There are two electron pairs.

Step 3: Determine the basic geometry of the molecule

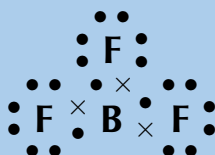
There are two electron pairs and no lone pairs around the central atom. BeCl_2 has the general formula: AX_2 . Using this information and Table 3.1 we find that the molecular shape is linear.

Step 4: Write the final answer

The molecular shape of BeCl_2 is linear.

Worked example 7: Molecular shape**QUESTION**

Determine the shape of a molecule of BF_3

SOLUTION**Step 1: Draw the molecule using a Lewis diagram**

The central atom is boron.

Step 2: Count the number of electron pairs around the central atom

There are three electron pairs.

Step 3: Determine the basic geometry of the molecule

There are three electron pairs and no lone pairs around the central atom. The molecule has the general formula AX_3 . Using this information and Table 3.1 we find that the molecular shape is trigonal planar.

Step 4: Write the final answer

The molecular shape of BF_3 is trigonal planar.

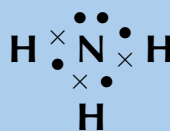
Worked example 8: Molecular shape**QUESTION**

Determine the shape of a molecule of NH_3

SOLUTION**Step 1: Draw the molecule using a Lewis diagram**

TIP

We can also work out the shape of a molecule with double or triple bonds. To do this, we count the double or triple bond as one pair of electrons.



The central atom is nitrogen.

Step 2: Count the number of electron pairs around the central atom

There are four electron pairs.

Step 3: Determine the basic geometry of the molecule

There are three bonding electron pairs and one lone pair. The molecule has the general formula AX_3E . Using this information and Table 3.1 we find that the molecular shape is trigonal pyramidal.

Step 4: Write the final answer

The molecular shape of NH_3 is trigonal pyramidal.

Group discussion: Building molecular models

In groups, you are going to build a number of molecules using jellytots to represent the atoms in the molecule, and toothpicks to represent the bonds between the atoms. In other words, the toothpicks will hold the atoms (jellytots) in the molecule together. Try to use different coloured jellytots to represent different elements.

You will need jellytots, toothpicks, labels or pieces of paper.

On each piece of paper, write the words: "lone pair".

You will build models of the following molecules:

HCl , CH_4 , H_2O , BF_3 , PCl_5 , SF_6 and NH_3 .

For each molecule, you need to:

- Determine the molecular geometry of the molecule
- Build your model so that the atoms are as far apart from each other as possible (remember that the electrons around the central atom will try to avoid the repulsions between them).
- Decide whether this shape is accurate for that molecule or whether there are any lone pairs that may influence it. If there are lone pairs then add a toothpick to the central jellytot. Stick a label (i.e. the piece of paper with "lone pair" on it) onto this toothpick.

- Adjust the position of the atoms so that the bonding pairs are further away from the lone pairs.
- How has the shape of the molecule changed?
- Draw a simple diagram to show the shape of the molecule. It doesn't matter if it is not 100% accurate. This exercise is only to help you to visualise the 3-dimensional shapes of molecules. (See Figure 3.8 to help you).

Do the models help you to have a clearer picture of what the molecules look like? Try to build some more models for other molecules you can think of.

TIP

Depending on which source you use for electronegativities you may see slightly different values.

Exercise 3 – 6: Molecular shape

Determine the shape of the following molecules.

- | | | | |
|--------------------|-------------------|------------------|-------------------------|
| 1. BeCl_2 | 3. PCl_5 | 5. CO_2 | 7. H_2O |
| 2. F_2 | 4. SF_6 | 6. CH_4 | 8. COH_2 |

Think you got it? Get this answer and more practice on our Intelligent Practice Service

1. [23PF](#) 2. [23PG](#) 3. [23PH](#) 4. [23PJ](#) 5. [23PK](#) 6. [23PM](#)
 7. [23PN](#) 8. [23PP](#)



www.everythingscience.co.za



m.everythingscience.co.za

3.3 Electronegativity

ESBMD

So far we have looked at covalent molecules. But how do we know that they are covalent? The answer comes from electronegativity. Each element (except for the noble gases) has an electronegativity value.

Electronegativity is a measure of how strongly an atom pulls a shared electron pair towards it. The table below shows the electronegativities of some of the elements.

For a full list of electronegativities see the periodic table at the front of the book. On this periodic table the electronegativity values are given in the top right corner. Do not confuse these values with the other numbers shown for the elements. Electronegativities will always be between 0 and 4 for any element. If you use a number greater than 4 then you are not using the electronegativity.

FACT

The concept of electronegativity was introduced by *Linus Pauling* in 1932, and this became very useful in explaining the nature of bonds between atoms in molecules. For this work, Pauling was awarded the Nobel Prize in Chemistry in 1954. He also received the Nobel Peace Prize in 1962 for his campaign against above-ground nuclear testing.

Element	Electronegativity	Element	Electronegativity
Hydrogen (H)	2,1	Lithium (Li)	1,0
Beryllium (Be)	1,5	Boron (B)	2,0
Carbon (C)	2,5	Nitrogen (N)	3,0
Oxygen (O)	3,5	Fluorine (F)	4,0
Sodium (Na)	0,9	Magnesium (Mg)	1,2
Aluminium (Al)	1,5	Silicon (Si)	1,8
Phosphorous (P)	2,1	Sulfur (S)	2,5
Chlorine (Cl)	3,0	Potassium (K)	0,8
Calcium (Ca)	1,0	Bromine (Br)	2,8

Table 3.2: Table of electronegativities for selected elements.

DEFINITION: *Electronegativity*

Electronegativity is a chemical property which describes the power of an atom to attract electrons towards itself.

The greater the electronegativity of an atom of an element, the stronger its attractive pull on electrons. For example, in a molecule of hydrogen bromide (HBr), the electronegativity of bromine (2,8) is higher than that of hydrogen (2,1), and so the shared electrons will spend more of their time closer to the bromine atom. Bromine will have a slightly negative charge, and hydrogen will have a slightly positive charge. In a molecule like hydrogen (H_2) where the electronegativities of the atoms in the molecule are the same, both atoms have a neutral charge.

Worked example 9: Calculating electronegativity differences**QUESTION**

Calculate the electronegativity difference between hydrogen and oxygen.

SOLUTION

Step 1: Read the electronegativity of each element off the periodic table.

From the periodic table we find that hydrogen has an electronegativity of 2,1 and oxygen has an electronegativity of 3,5.

Step 2: Calculate the electronegativity difference

$$3,5 - 2,1 = 1,4$$

Exercise 3 – 7:

1. Calculate the electronegativity difference between:
Be and C; H and C; Li and F; Al and Na; C and O.

Think you got it? Get this answer and more practice on our Intelligent Practice Service

1. 23PQ



www.everythingscience.co.za



m.everythingscience.co.za

TIP

Note that metallic bonding is not given here. Metals have low electronegativities and so the valence electrons are not drawn strongly to any one atom. Instead, the valence electrons are loosely shared by all the atoms in the metallic network.

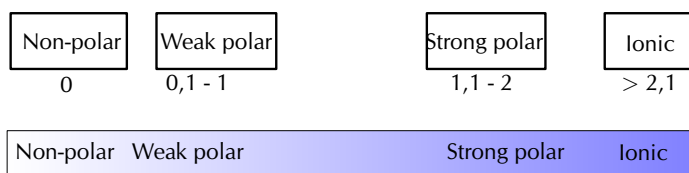
TIP

The symbol δ is read as delta.

Electronegativity and bonding

ESBMF

The electronegativity difference between two atoms can be used to determine what type of bonding exists between the atoms. The table below lists the approximate values. Although we have given ranges here bonding is more like a spectrum than a set of boxes.



Electronegativity difference	Type of bond
0	Non-polar covalent
0 - 1	Weak polar covalent
1,1 - 2	Strong polar covalent
> 2,1	Ionic

Non-polar and polar covalent bonds

ESBMG

It is important to be able to determine if a molecule is polar or non-polar since the *polarity* of molecules affects properties such as *solubility*, *melting points* and *boiling points*.

Electronegativity can be used to explain the difference between two types of covalent bonds. **Non-polar covalent bonds** occur between two identical non-metal atoms, e.g. H_2 , Cl_2 and O_2 . Because the two atoms have the same electronegativity, the electron pair in the covalent bond is shared equally between them. However, if two different non-metal atoms bond then the shared electron pair will be pulled more strongly by the atom with the higher electronegativity. As a result, a **polar covalent bond** is formed where one atom will have a slightly negative charge and the other a slightly positive charge.

This slightly positive or slightly negative charge is known as a partial charge. These partial charges are represented using the symbols δ^+ (slightly positive) and δ^- (slightly negative). So, in a molecule such as hydrogen chloride (HCl), hydrogen is $\text{H}^{\delta+}$ and chlorine is $\text{Cl}^{\delta-}$.

TIP

To determine if a molecule is symmetrical look first at the atoms around the central atom. If they are different then the molecule is not symmetrical. If they are the same then the molecule may be symmetrical and we need to look at the shape of the molecule.

Polar molecules

ESBMH

Some molecules with polar covalent bonds are **polar molecules**, e.g. H_2O . But not *all* molecules with polar covalent bonds are polar. An example is CO_2 . Although CO_2 has two polar covalent bonds (between $\text{C}^{\delta+}$ atom and the two $\text{O}^{\delta-}$ atoms), the molecule itself is not polar. The reason is that CO_2 is a linear molecule, with both terminal atoms the same, and is therefore symmetrical. So there is no difference in charge between the two ends of the molecule.

DEFINITION: *Polar molecules*

A **polar molecule** is one that has one end with a slightly positive charge, and one end with a slightly negative charge. Examples include water, ammonia and hydrogen chloride.

DEFINITION: *Non-polar molecules*

A **non-polar molecule** is one where the charge is equally spread across the molecule or a symmetrical molecule with polar bonds. Examples include carbon dioxide and oxygen.

We can easily predict which molecules are likely to be polar and which are likely to be non-polar by looking at the molecular shape. The following activity will help you determine this and will help you understand more about symmetry.

Activity: Polar and non-polar molecules

The following table lists the molecular shapes. Build the molecule given for each case using jellytots and toothpicks. Determine if the shape is symmetrical. (Does it look the same whichever way you look at it?) Now decide if the molecule is polar or non-polar.

Geometry	Molecule	Symmetrical	Polar or non-polar
Linear	HCl		
Linear	CO_2		
Linear	HCN		
Bent or angular	H_2O		
Trigonal planar	BF_3		
Trigonal planar	BF_2Cl		
Trigonal pyramidal	NH_3		
Tetrahedral	CH_4		
Tetrahedral	CH_3Cl		
Trigonal bipyramidal	PCl_5		
Trigonal bipyramidal	PCl_4F		
Octahedral	SF_6		
Octahedral	SF_5Cl		

▶ See simulation: [23PR](http://23PR.at/www.everythingscience.co.za) at www.everythingscience.co.za

Worked example 10: Polar and non-polar molecules

QUESTION

State whether hydrogen (H_2) is polar or non-polar.

SOLUTION

Step 1: Determine the shape of the molecule

The molecule is linear. There is one bonding pair of electrons and no lone pairs.

Step 2: Write down the electronegativities of each atom

Hydrogen: 2,1

Step 3: Determine the electronegativity difference for each bond

There is only one bond and the difference is 0.

Step 4: Determine the polarity of each bond

The bond is non-polar.

Step 5: Determine the polarity of the molecule

The molecule is non-polar.

Worked example 11: Polar and non-polar molecules

QUESTION

State whether methane (CH_4) is polar or non-polar.

SOLUTION

Step 1: Determine the shape of each molecule

The molecule is tetrahedral. There are four bonding pairs of electrons and no lone pairs.

Step 2: Determine the electronegativity difference for each bond

There are four bonds. Since each bond is between carbon and hydrogen, we only need to calculate one electronegativity difference. This is: $2,5 - 2,1 = 0,4$

Step 3: Determine the polarity of each bond

Each bond is polar.

Step 4: Determine the polarity of the molecule

The molecule is symmetrical and so is non-polar.

Worked example 12: Polar and non-polar molecules

QUESTION

State whether hydrogen cyanide (HCN) is polar or non-polar.

SOLUTION

Step 1: Determine the shape of the molecule

The molecule is linear. There are four bonding pairs, three of which form a triple bond and so are counted as 1. There is one lone pair on the nitrogen atom.

Step 2: Determine the electronegativity difference and polarity for each bond

There are two bonds. One between hydrogen and carbon and the other between carbon and nitrogen. The electronegativity difference between carbon and hydrogen is 0,4 and the electronegativity difference between carbon and nitrogen is 0,5. Both of the bonds are polar.

Step 3: Determine the polarity of the molecule

The molecule is not symmetrical and so is polar.

Exercise 3 – 8: Electronegativity

1. In a molecule of beryllium chloride (BeCl_2):
 - a) What is the electronegativity of beryllium?
 - b) What is the electronegativity of chlorine?
 - c) Which atom will have a slightly positive charge and which will have a slightly negative charge in the molecule? Represent this on a sketch of the molecule using partial charges.
 - d) Is the bond a non-polar or polar covalent bond?
 - e) Is the molecule polar or non-polar?

2. Complete the table below:

Molecule	Difference in electronegativity between atoms	Non-polar/polar covalent bond	Polar/non-polar molecule
H ₂ O			
HBr			
F ₂			
CH ₄			
PF ₅			
BeCl ₂			
CO			
C ₂ H ₂			
SO ₂			
BF ₃			

Think you got it? Get this answer and more practice on our Intelligent Practice Service

1. 23PS 2. 23PT



www.everythingscience.co.za



m.everythingscience.co.za

3.4 Energy and bonding

ESBMJ

As we saw earlier in the chapter we can show the energy changes that occur as atoms come together (Figure 3.5). Shown below is the same image but this time with two extra pieces of information: the bond energy and the bond length.

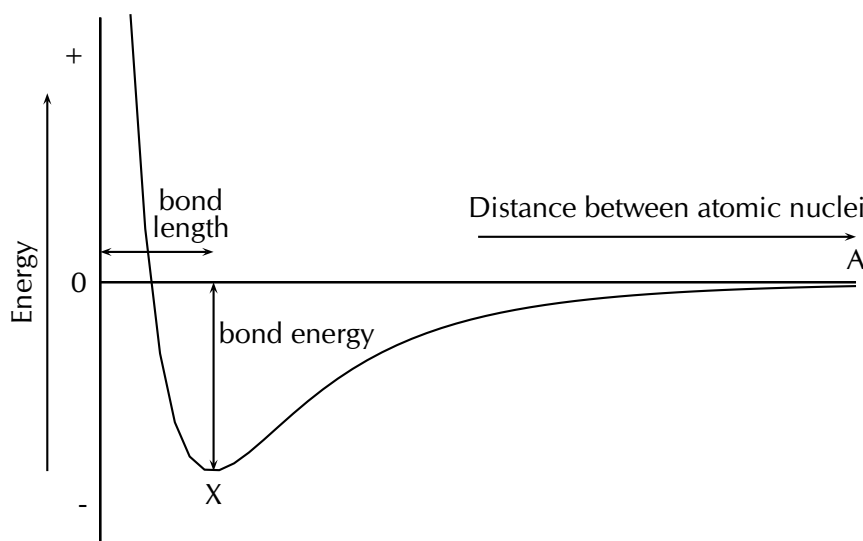


Figure 3.9: Graph showing the change in energy that takes place as atoms move closer together.

DEFINITION: *Bond length*

The distance between the nuclei of two adjacent atoms when they bond.

DEFINITION: *Bond energy*

The amount of energy that must be added to the system to break the bond that has formed.

It is important to remember that bond length is measured between two atoms that are bonded to each other. The following diagrams show the bond length for CO and for CO₂. The grey circle represents carbon and the white circle represents oxygen.

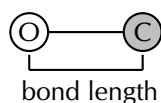


Figure 3.10:

The bond length for carbon monoxide (CO).

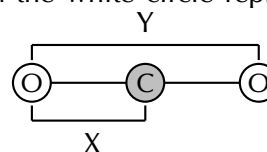


Figure 3.11:

The bond length for each C – O bond in carbon dioxide (CO₂) is indicated by X. Y is **not** the bond length.

A third property of bonds is the bond strength. **Bond strength** means how strongly one atom attracts and is held to another. The strength of a bond is related to the bond length, the size of the bonded atoms and the number of bonds between the atoms. In general:

- the shorter the bond length, the stronger the bond between the atoms.
- the smaller the atoms involved, the stronger the bond.
- the more bonds that exist between the same atoms, the stronger the bond.

3.5 Chapter summary

ESBMK

▶ See presentation: 23PV at www.everythingscience.co.za

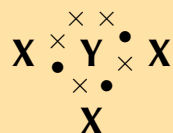
- A **chemical bond** is the physical process that causes atoms to be attracted together and to be bound in new compounds.
- The noble gases have a full valence shell. Atoms bond to try fill their outer valence shell.
- There are three **forces** that act between atoms: attractive forces between the positive nucleus of one atom and the negative electrons of another; repulsive forces between like-charged electrons, and repulsion between like-charged nuclei.
- The **energy** of a system of two atoms is at a minimum when the attractive and repulsive forces are balanced.
- **Lewis** diagrams are one way of representing molecular structure. In a Lewis diagram, dots or crosses are used to represent the valence electrons around the central atom.

- A **covalent bond** is a form of chemical bond where pairs of electrons are shared between two atoms.
- A **single bond** occurs if there is one electron pair that is shared between the same two atoms.
- A **double bond** occurs if there are two electron pairs that are shared between the same two atoms.
- A **triple bond** occurs if there are three electron pairs that are shared between the same two atoms.
- A **dative covalent bond** is a description of covalent bonding that occurs between two atoms in which both electrons shared in the bond come from the same atom.
- Dative covalent bonds occur between atoms of elements with a lone pair and atoms of elements with no electrons. Examples include the hydronium ion (H_3O^+) and the ammonium ion (NH_4^+).
- The **shape of molecules** can be predicted using the VSEPR theory.
- Valence shell electron pair repulsion (VSEPR) theory is a model in chemistry, which is used to predict the shape of individual molecules. VSEPR is based upon minimising the extent of the electron-pair repulsion around the central atom being considered.
- **Electronegativity** is a chemical property which describes the power of an atom to attract electrons towards itself in a chemical.
- Electronegativity can be used to explain the difference between two types of covalent bonds: **polar covalent bonds** (between non-identical atoms) and **non-polar covalent bonds** (between identical atoms or atoms with the same electronegativity).
- A **polar** molecule is one that has one end with a slightly positive charge, and one end with a slightly negative charge. Examples include water, ammonia and hydrogen chloride.
- A **non-polar** molecule is one where the charge is equally spread across the molecule or a symmetrical molecule with polar bonds.
- **Bond length** is the distance between the nuclei of two atoms when they bond.
- **Bond energy** is the amount of energy that must be added to the system to break the bond that has formed.
- **Bond strength** means how strongly one atom attracts and is held to another atom. Bond strength depends on the length of the bond, the size of the atoms and the number of bonds between the two atoms.

Exercise 3 – 9:

1. Give **one word/term** for each of the following descriptions.
 - a) The distance between two adjacent atoms in a molecule.
 - b) A type of chemical bond that involves the sharing of electrons between two atoms.

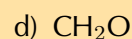
- c) A measure of an atom's ability to attract electrons to itself in a chemical bond.
2. Which ONE of the following best describes the bond formed between an H^+ ion and the NH_3 molecule?
- Covalent bond
 - Dative covalent (co-ordinate covalent) bond
 - Ionic Bond
 - Hydrogen Bond
3. Explain the meaning of each of the following terms:
- valence electrons
 - bond energy
 - covalent bond
4. Which of the following reactions will *not* take place? Explain your answer.
- $\text{H} + \text{H} \rightarrow \text{H}_2$
 - $\text{Ne} + \text{Ne} \rightarrow \text{Ne}_2$
 - $\text{Cl} + \text{Cl} \rightarrow \text{Cl}_2$
5. Draw the Lewis diagrams for each of the following:
- An atom of strontium (Sr). (Hint: Which group is it in? It will have an identical Lewis diagram to other elements in that group).
 - An atom of iodine.
 - A molecule of hydrogen bromide (HBr).
 - A molecule of nitrogen dioxide (NO_2). (Hint: There will be a single unpaired electron).
6. Given the following Lewis diagram, where X and Y each represent a different element:



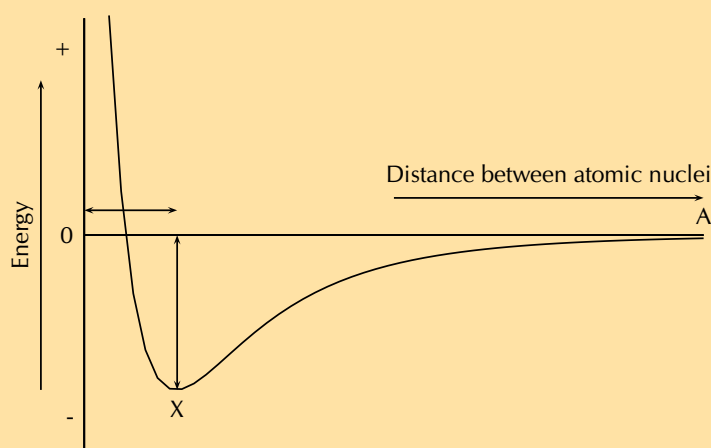
- How many valence electrons does X have?
 - How many valence electrons does Y have?
 - Which elements could X and Y represent?
7. Determine the shape of the following molecules:
- | | | | |
|-------------------|-------------------|---------------------------|---------------------------|
| a) O_2 | c) BCl_3 | e) CCl_4 | g) Br_2 |
| b) MgI_2 | d) CS_2 | f) CH_3Cl | h) SCl_5F |
8. Complete the following table:

Element pair	Electronegativity difference	Type of bond that could form
Hydrogen and lithium		
Hydrogen and boron		
Hydrogen and oxygen		
Hydrogen and sulfur		
Magnesium and nitrogen		
Magnesium and chlorine		
Boron and fluorine		
Sodium and fluorine		
Oxygen and nitrogen		
Oxygen and carbon		

9. Are the following molecules polar or non-polar?



10. Given the following graph for hydrogen:



a) The bond length for hydrogen is 74 pm. Indicate this value on the graph. (Remember that pm is a picometer and means 74×10^{-12} m).

The bond energy for hydrogen is $436 \text{ kJ}\cdot\text{mol}^{-1}$. Indicate this value on the graph.

b) What is important about point X?

11. Hydrogen chloride has a bond length of 127 pm and a bond energy of $432 \text{ kJ}\cdot\text{mol}^{-1}$. Draw a graph of energy versus distance and indicate these values on your graph. The graph does not have to be accurate, a rough sketch graph will do.

Think you got it? Get this answer and more practice on our Intelligent Practice Service

- 1a. 23PW 1b. 23PX 1c. 23PY 2. 23PZ 3a. 23Q2 3b. 23Q3
 3c. 23Q4 4. 23Q5 5a. 23Q6 5b. 23Q7 5c. 23Q8 5d. 23Q9
 6. 23QB 7a. 23QC 7b. 23QD 7c. 23QF 7d. 23QG 7e. 23QH
 7f. 23QJ 7g. 23QK 7h. 23QM 8. 23QN 9a. 23QP 9b. 23QQ
 9c. 23QR 9d. 23QS 10. 23QT 11. 23QV



www.everythingscience.co.za



m.everythingscience.co.za

Intermolecular forces

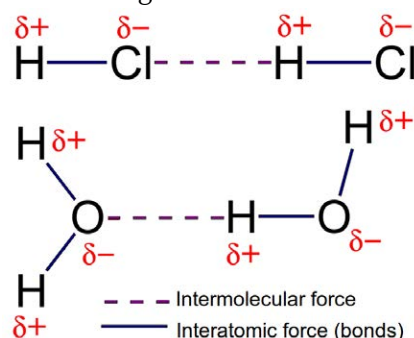
4.1	<i>Intermolecular and interatomic forces</i>	166
4.2	<i>The chemistry of water</i>	184
4.3	<i>Chapter summary</i>	189

All around us we see matter in different phases. The air we breathe is a gas, while the water you drink is a liquid and the chair you are sitting on is a solid. In this chapter we are going to look at one of the reasons that matter exists as solids and liquids.

In the previous chapter, we discussed the different forces that exist *between atoms* (interatomic forces). When atoms are joined to one another they form molecules, and these molecules in turn have forces that bind them together. These forces are known as **intermolecular forces**.

Intermolecular forces allow us to determine which substances are likely to dissolve in which other substances and what the melting and boiling points of substances are. Without intermolecular forces holding molecules together we would not exist.

Note that we will use the term molecule throughout this chapter as the compounds we are looking at are all covalently bonded and do not exist as giant networks (recall from grade 10 that there are three types of bonding: metallic, ionic and covalent). Sometimes you will see the term simple molecule. This is a covalent molecular structure.



NOTE:

Interatomic (between atoms) forces are also known as intramolecular (within molecules) forces. You can remember this by thinking of **international** which means between nations.

4.1 Intermolecular and interatomic forces

ESBMM

DEFINITION: *Intermolecular forces*

Intermolecular forces are forces that act between molecules.

You will also recall from the previous chapter, that we can describe molecules as being either **polar** or **non-polar**. A polar molecule is one in which there is a difference in electronegativity between the atoms in the molecule, such that the shared electron pair spends more time close to the atom that attracts it more strongly. The result is that one end of the molecule will have a slightly positive charge (δ^+), and the other end will have a slightly negative charge (δ^-). The molecule is said to be a **dipole**.

A dipole molecule is a molecule that has two (di) poles. One end of the molecule is slightly positive and the other is slightly negative. We can depict this very simply as an oval with one positive side and one negative. In reality however, the molecules do not look like this, they look more like the images in Figure 4.1.

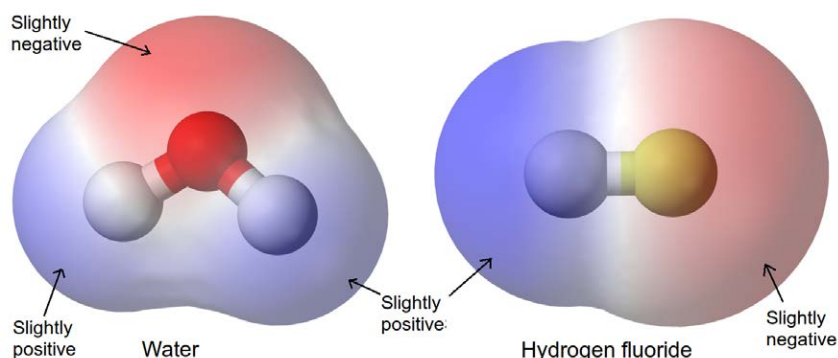
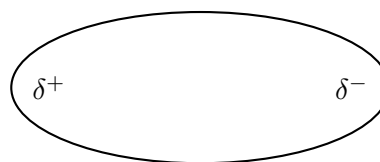


Figure 4.1: A different representation of dipole molecules. The red region is slightly negative, and the blue region is slightly positive.

It is important to remember that just because the bonds within a molecule are polar, the molecule itself may not necessarily be polar. The shape of the molecule may also affect its polarity. A few examples are shown in Table 4.1 to refresh your memory. Note that we have shown tetrahedral molecules with all the terminal atoms at 90° to each other (i.e. flat or 2-dimensional), but the shape is really 3-dimensional.

Molecule	Chemical formula	Bond between atoms	Shape of molecule	Polarity of molecule
Hydrogen	H ₂	Non-polar covalent	Linear molecule H — H	Non-polar
Hydrogen chloride	HCl	Polar covalent	Linear molecule H ^{δ+} — Cl ^{δ-}	Polar
Carbon tetrafluoride	CF ₄	Polar covalent	Tetrahedral molecule $ \begin{array}{c} \text{F}^{\delta-} \\ \\ \text{F}^{\delta-} - \text{C}^{\delta+} - \text{F}^{\delta-} \\ \\ \text{F}^{\delta-} \end{array} $	Non-polar
Trifluoromethane	CHF ₃	Polar covalent	Tetrahedral molecule $ \begin{array}{c} \text{F} \\ \\ \text{H} - \text{C} - \text{F} \\ \\ \text{F} \end{array} $	Polar

Table 4.1: Polarity in molecules with different atomic bonds and molecular shapes.

It is important to be able to recognise whether the molecules in a substance are polar or non-polar because this will determine what type of intermolecular forces there are. This is important in explaining the properties of the substance.

1. Ion-dipole forces

As the name suggests, this type of intermolecular force exists between an ion and a dipole (polar) molecule. You will remember that an *ion* is a charged atom, and this will be attracted to one of the charged ends of the polar molecule. A positive ion will be attracted to the negative pole of the polar molecule, while a negative ion will be attracted to the positive pole of the polar molecule. This can be seen when sodium chloride (NaCl) dissolves in water. The positive sodium ion (Na^+) will be attracted to the slightly negative oxygen atoms in the water molecule, while the negative chloride ion (Cl^-) is attracted to the slightly positive hydrogen atoms. These intermolecular forces weaken the ionic bonds between the sodium and chloride ions so that the sodium chloride dissolves in the water (Figure 4.2).

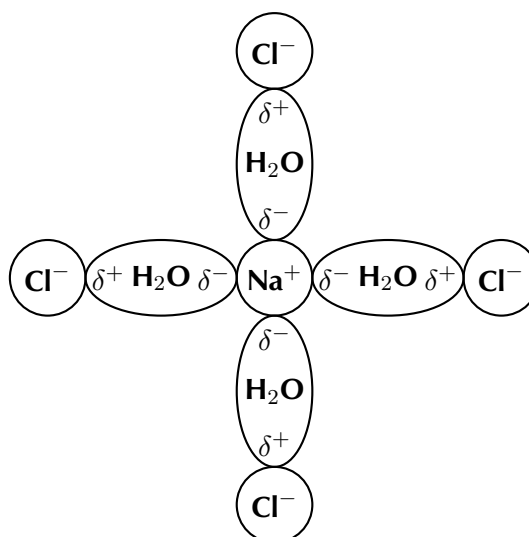
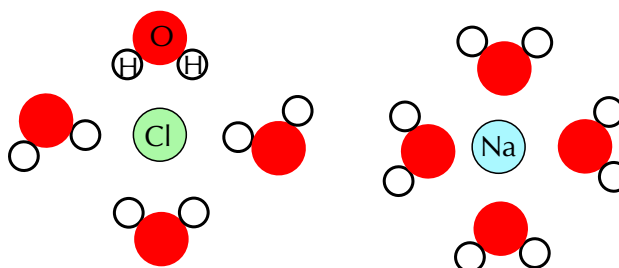


Figure 4.2: Ion-dipole forces in a sodium chloride solution.

This is a simplified diagram to highlight the regions of positive and negative charge. When sodium chloride dissolves in water it can more accurately be shown as:



2. Ion-induced-dipole forces

Similar to ion-dipole forces these forces exist between ions and non-polar molecules. The ion induces a dipole in the non-polar molecule leading to a weak, short lived force which holds the compounds together.

These forces are found in haemoglobin (the molecule that carries oxygen around your body). Haemoglobin has Fe^{2+} ions. Oxygen (O_2) is attracted to these ions by ion-induced dipole forces.

3. Dipole-dipole forces

When one dipole molecule comes into contact with another dipole molecule, the positive pole of the one molecule will be attracted to the negative pole of the other, and the molecules will be held together in this way (Figure 4.3). Examples of materials/substances that are held together by dipole-dipole forces are HCl , SO_2 and CH_3Cl .

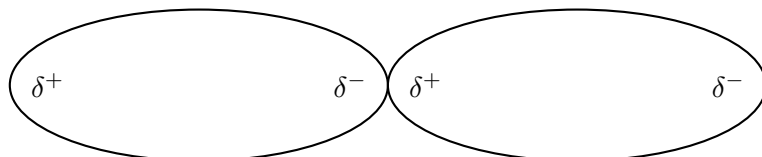


Figure 4.3: Two dipole molecules are held together by the attractive force between their oppositely charged poles.

One special case of this is hydrogen bonding.

4. Induced dipole forces

We know that while carbon dioxide is a non-polar molecule, we can still freeze it (and we can also freeze all other non-polar substances). This tells us that there must be some kind of attractive force in these kinds of molecules (molecules can only be solids or liquids if there are attractive forces pulling them together). This force is known as an induced dipole force.

In non-polar molecules the electronic charge is usually evenly distributed but it is possible that at a particular moment in time, the electrons might not be evenly distributed (remember that the electrons are always moving in their orbitals). The molecule will have a *temporary dipole*. In other words, each end of the molecules has a slight charge, either positive or negative. When this happens, molecules that are next to each other attract each other very weakly. These forces are found in the halogens (e.g. F_2 and I_2) and in other non-polar molecules such as carbon dioxide and carbon tetrachloride.

All covalent molecules have induced dipole forces. For non-polar covalent molecules these forces are the only intermolecular forces. For polar covalent molecules, dipole-dipole forces are found in addition to the induced dipole forces.

5. Dipole-induced-dipole forces

This type of force occurs when a molecule with a dipole induces a dipole in a non-polar molecule. It is similar to an ion-induced dipole force. An example of this type of force is chloroform (CHCl_3) in carbon tetrachloride (CCl_4).

The following image shows the types of intermolecular forces and the kinds of compounds that lead to those forces.

FACT

These intermolecular forces are also sometimes called “London forces” or “momentary dipole” forces or “dispersion” forces.

TIP

When the noble gases condense, the intermolecular forces that hold the liquid together are induced dipole forces.

TIP

Do not confuse hydrogen bonds with actual chemical bonds.

Hydrogen bonding is an example of a case where a scientist named something believing it to be one thing when in fact it was another. In this case the strength of the hydrogen bonds misled scientists into thinking this was actually a chemical bond, when it is really just an intermolecular force.

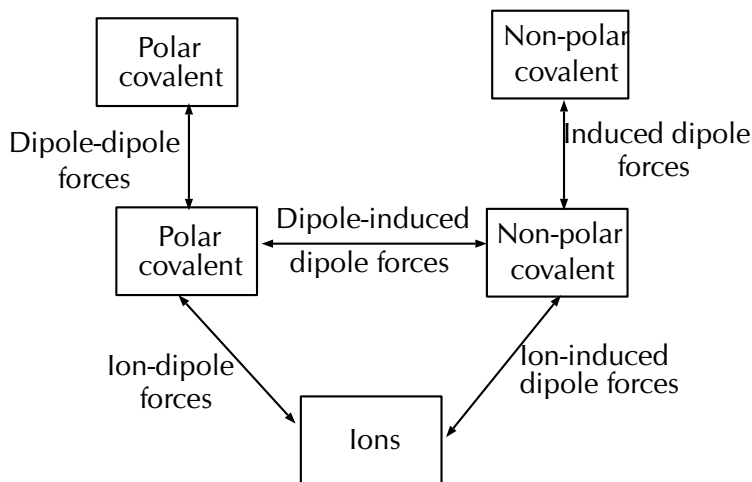


Figure 4.4: The types of intermolecular forces. The boxes represent the type of compound while the lines represent the type of force.

The last three forces (dipole-dipole forces, dipole-induced dipole forces and induced dipole forces) are sometimes collectively known as van der Waals' forces. We will now look at a special case of dipole-dipole forces in more detail.

Hydrogen bonds

As the name implies, this type of intermolecular bond involves a hydrogen atom. When a molecule contains a hydrogen atom covalently bonded to a highly electronegative atom (O, N or F) this type of intermolecular force can occur. The highly electronegative atom on one molecule attracts the hydrogen atom on a nearby molecule.

Water molecules for example, are held together by hydrogen bonds between the hydrogen atom of one molecule and the oxygen atom of another (Figure 4.5). Hydrogen bonds are a relatively strong intermolecular force and are stronger than other dipole-dipole forces. It is important to note however, that hydrogen bonds are weaker than the covalent and ionic bonds that exist between *atoms*.

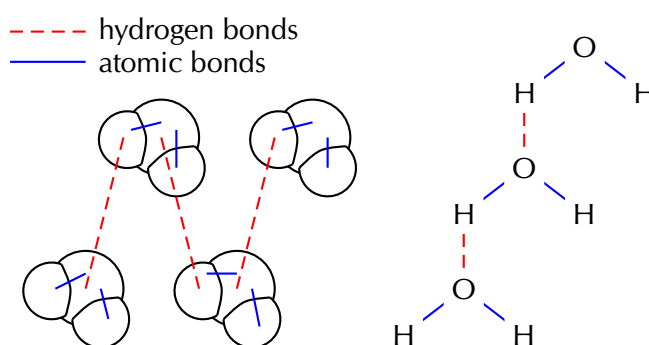


Figure 4.5: Two representations showing the hydrogen bonds between water molecules: space-filling model and structural formula.

The difference between intermolecular and interatomic forces ESBMP

It is important to realise that there is a difference between the types of interactions that occur in molecules and the types that occur between molecules. In the previous chapter we focused on the interactions between atoms. These are known as interatomic forces or chemical bonds. We also studied covalent molecules in more detail.

Remember that a covalent bond has an electronegativity difference of less than 2,1. Covalent molecules have covalent bonds between their atoms. Van der Waals' forces only occur in covalent molecules. We can show the interatomic and intermolecular forces between covalent compounds diagrammatically or in words. Intermolecular forces occur between molecules and do not involve individual atoms. Interatomic forces are the forces that hold the the atoms in molecules together. Figure 4.5 shows this.

	Interatomic forces	Intermolecular forces
Atoms or molecules	Forces between atoms	Forces between molecules
Strength of forces	Strong forces	Relatively weak forces
Distance between atoms or molecules	Very short distances	Larger distances than bonds

Table 4.2: The differences between interatomic and intermolecular forces.

Worked example 1: Intermolecular forces

QUESTION

Which intermolecular forces are found in carbon tetrachloride (CCl_4)?

SOLUTION

Step 1: Think about what you know about the molecule.

Carbon has an electronegativity of 2,5. Chlorine has an electronegativity of 3,0. The electronegativity difference between carbon and chlorine is 1,0 (recall the section on electronegativity in the previous chapter). We also know that the bond between carbon and chlorine is polar.

Also from the previous chapter we know that carbon tetrachloride is a tetrahedral molecule (recall molecular shape). Carbon tetrachloride is symmetrical and so is non-polar overall.

Step 2: Now decide which case it is

Carbon tetrachloride is non-polar and so the only kind of force that can exist is **induced dipole**.

Worked example 2: Intermolecular forces

QUESTION

Which intermolecular forces are found in the following solution: sodium chloride in water?

SOLUTION

Step 1: Think about what you know about the molecules

Sodium chloride is ionic. (the electronegativity difference is 2,1). Water has polar bonds (the electronegativity difference is 1,4). Water is a polar molecule (its molecular shape is bent or angular).

Step 2: Now decide which case it is

This is an ionic substance interacting with a polar substance. This interaction is an **ion-dipole** force.

Exercise 4 – 1:

Which intermolecular forces are found in:

1. hydrogen fluoride (HF)
2. methane (CH₄)
3. potassium chloride in ammonia (KCl in NH₃)
4. krypton (Kr)

Think you got it? Get this answer and more practice on our Intelligent Practice Service

1. [23QW](#) 2. [23QX](#) 3. [23QY](#) 4. [23QZ](#)



www.everythingscience.co.za



m.everythingscience.co.za

Understanding intermolecular forces

ESBMQ

The types of intermolecular forces that occur in a substance will affect its properties, such as its **phase**, **melting point** and **boiling point**. You should remember from the kinetic theory of matter (see grade 10), that the *phase* of a substance is determined by how strong the forces are between its particles. The weaker the forces, the more likely the substance is to exist as a gas. This is because the particles are able to move far apart since they are not held together very strongly. If the forces are very strong, the particles are held closely together in a solid structure. Remember also that the *temperature* of a material affects the energy of its particles. The more energy the particles have, the more likely they are to be able to overcome the forces that are holding them together. This can cause a change in phase.

Shown below are the three phases of water. Note that we are showing two dimensional figures when in reality these are three dimensional.

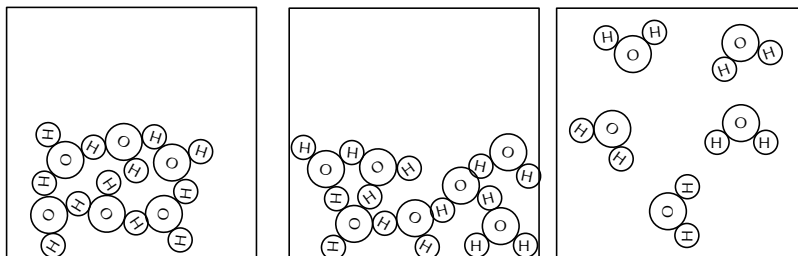


Figure 4.6: The three phases of water.



The effects of intermolecular forces

The following five experiments investigate the effect of various physical properties (evaporation, surface tension, solubility, boiling point and capillarity) of substances and determine how these properties relate to intermolecular forces. Each experiment will look at a different property.

Formal experiment: The effects of intermolecular forces: Part 1

Aim:

To investigate evaporation and to determine the relation between evaporation and intermolecular forces.

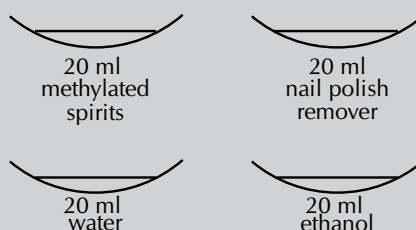
Apparatus:

You will need the following items for this experiment:

- ethanol, water, nail polish remover (acetone), methylated spirits
- evaporating dishes (or shallow basins)

Method:

1. Place 20 ml of each substance given in separate evaporating dishes.
2. Carefully move each dish to a warm (sunny) spot.
3. Mark the level of liquid in each dish using a permanent marker. Make several marks at different positions around the dish. If the permanent marker is leaving a smudge rather than a noticeable mark, carefully wipe the side of the dish and try again.
4. Observe each dish every minute and note which liquid evaporates fastest.



Results:

Record your results in the table below. You do not need to measure the level of the liquid, but rather just write how much the level had dropped (e.g. for water you might write did not notice any decrease in the level or for ethanol you might write almost all the liquid had evaporated).

Substance	1 min	2 min	3 min	4 min	5 min
Ethanol					
Water					
Nail polish remover					
Methylated spirits					

Discussion and conclusion:

You should find that water takes the longest time to evaporate. Water has strong intermolecular forces (hydrogen bonds). Ethanol ($\text{CH}_3\text{CH}_2\text{OH}$) and methylated spirits (mainly ethanol ($\text{CH}_3\text{CH}_2\text{OH}$) with some methanol (CH_3OH)) both have hydrogen bonds but these are slightly weaker than the hydrogen bonds in water. Nail polish remover (acetone (CH_3COCH_3)) has dipole-dipole forces only and so evaporates quickly.

Substances with weaker intermolecular forces evaporate faster than substances with stronger intermolecular forces.

Formal experiment: The effects of intermolecular forces: Part 2

Aim:

To investigate surface tension and to determine the relation between surface tension and intermolecular forces.

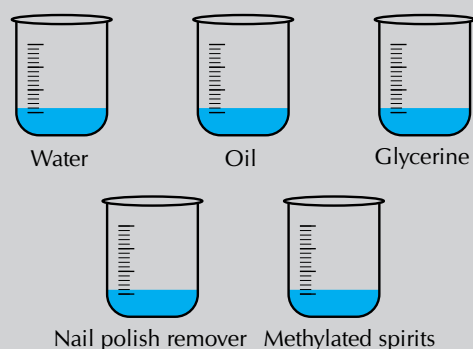
Apparatus:

You will need the following items for this experiment:

- water, cooking oil (sunflower oil), glycerin, nail polish remover (acetone), methylated spirits
- small glass beakers or glass measuring cylinders
- small piece of glass or clear plastic (about 5 cm by 5 cm.)

Method:

1. Place about 50 ml of each substance given in separate small beakers or measuring cylinders.
2. Observe the shape of the meniscus. (This is the level of the liquid). Note what happens at the edges where the liquid touches the glass. (You can place a few drops of food colouring in each substance to help you see the meniscus.)
3. Now place a drop of the substance on a small piece of glass. Observe the shape of the drop.

**Results:**

Record your results in the table below. You just need to give a qualitative result (in other words what you see in the experiment).

Substance	Shape of meniscus	Shape of droplet
Water		
Oil		
Glycerine		
Nail polish remover		
Methylated spirits		

Discussion and conclusion:

The meniscus for all these substances should be concave (i.e. higher at the edges than in the middle). This is because the forces holding the molecules in the substance together are weaker than the attraction between the substance and the glass of the tube.

You should also have noticed that water, oil and Glycerine tend to form a drop, while nail polish remover and methylated spirits do not. Strong intermolecular forces help hold the substance together, while weaker ones do not hold the molecules in the substance together as much.

Water has the strongest intermolecular forces (hydrogen bonds) of all the substances used. Glycerine and methylated spirits also have hydrogen bonds, but these intermolecular forces are slightly weaker than in water. Sunflower oil is mostly non-polar but has very long molecules which help account for the higher surface tension.

Substances with strong intermolecular forces will generally have a greater surface tension than substances with weaker intermolecular forces.

Aim:

To investigate solubility and to determine the relation between solubility and intermolecular forces.

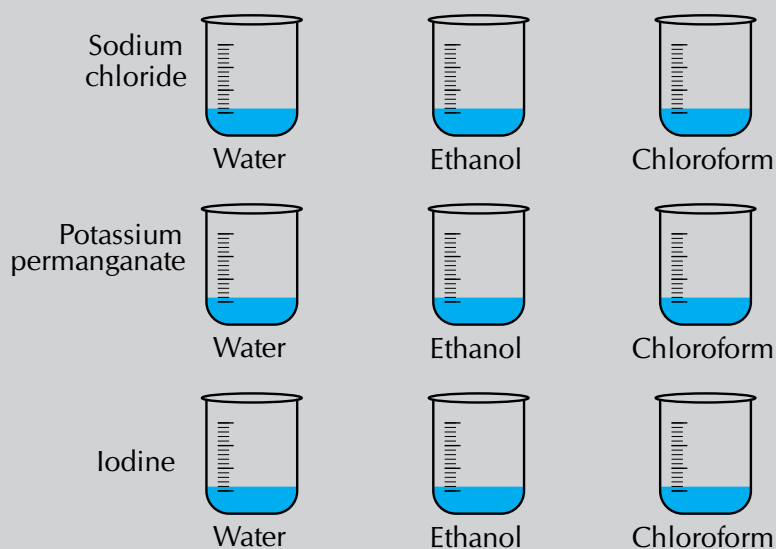
Apparatus:

You will need the following items for this experiment:

- Solids: sodium chloride (table salt), iodine, potassium permanganate
- Solvents: water, ethanol, chloroform
- 9 beakers or test-tubes
- 3 A4 sheets of paper

Method:

1. Place about 20 ml of each solvent given in separate beakers. Place this set on a piece of paper labelled "sodium chloride".
2. Repeat this step twice. The second set is for potassium permanganate (so your piece of paper will say "potassium permanganate") and the third set is for iodine (so your piece of paper will say "iodine"). You should now have nine beakers in total.
3. Into the first set, add about 2 g of sodium chloride.
4. Into the second set, add about 2 g of potassium permanganate.
5. Into the third set, add about 2 g of iodine.
6. Observe how much of each substance dissolves in the solvent.



Results:

Record your results in the table below. If you observe only a small amount of the solid dissolving then write that very little solid dissolved. If all the solid dissolves then write that all the solid dissolved.

Substance	Water	Chloroform	Ethanol
Sodium chloride			
Potassium permanganate			
Iodine			

Discussion and conclusion:

You should find that the sodium chloride and potassium permanganate dissolved (at least a bit) in all the substances. The iodine did not dissolve in any of the substances. The three solvents (water, chloroform and ethanol) are all polar and have dipole-dipole forces. Sodium chloride and potassium permanganate are both ionic substances, while iodine is non-polar.

Substances will dissolve in solvents that have similar intermolecular forces or in solvents where the ionic bonds can be disrupted by the formation of ion-dipole forces.

Formal experiment: The effects of intermolecular forces: Part 4

Aim:

To investigate boiling point and to determine the relation between boiling point and intermolecular forces.

Apparatus:

You will need the following items for this experiment:

- water, cooking oil (sunflower oil), Glycerine, nail polish remover, methylated spirits
- test-tubes and a beaker
- hot plate

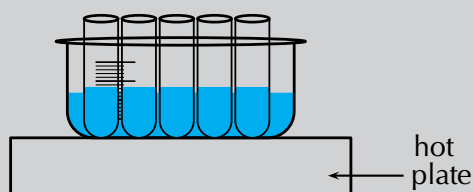
Method:

WARNING!

Methylated spirits and nail polish remover are highly flammable. They will easily catch fire if left near an open flame. For this reason they must be heated in a water bath. This experiment **MUST** be performed in a well ventilated room.

1. Place about 20 ml of each substance given in separate test-tubes.
2. Half-fill the beaker with water and place on the hot plate.
3. Place the test-tubes in the beaker.

4. Observe how long each substance takes to boil. As soon as a substance boils, remove it from the water bath.



Results:

Write down the order in which the substances boiled, starting with the substance that boiled first and ending with the substance that boiled last.

Discussion and conclusion:

You should have found that the nail polish remover and the methylated spirits boil before the water, oil and Glycerine.

Glycerine, water and methylated spirits have hydrogen bonds between the molecules. However, in water and Glycerine these intermolecular forces are very strong while in the methylated spirits they are slightly weaker. This leads to the higher boiling point for water and Glycerine. Nail polish remover has weaker dipole-dipole forces.

Although cooking oil is non-polar and has induced dipole forces the molecules are very large and so these increase the strength of the intermolecular forces.

Substances with strong intermolecular forces will have a higher boiling point than substances with weaker intermolecular forces.

Formal experiment: The effects of intermolecular forces: Part 5

Aim:

To investigate capillarity (how far up a tube a liquid rises or how far down a liquid falls) and to determine the relation between capillarity and intermolecular forces.

Apparatus:

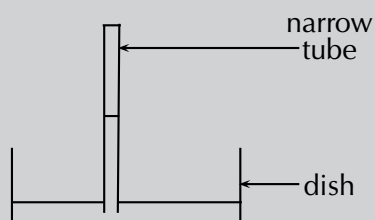
You will need the following items for this experiment:

- water, cooking oil (sunflower oil), nail polish remover, methylated spirits
- large shallow dish, narrow glass tube (with one end closed)

Method:

1. Place about 20 ml of water in the shallow dish.
2. Hold the narrow tube just above the level of the water in the dish.

3. Observe how far up the tube the water travels.
4. Repeat for the other three substances, remembering to wash and dry the dish and tube well between each one.



Results:

Record your results in the table below. You do not need to measure how far up the tube the substance travels but rather say if it only travelled a short distance or a long distance.

Substance	Distance travelled up tube
Water	
Oil	
Nail polish remover	
Methylated spirits	

Discussion and conclusion:

Water travels the greatest distance up the tube. Nail polish remover travels the least distance.

Capillarity is related to surface tension. If the attractive force between the glass walls of the tube and the substance are stronger than the intermolecular forces in the substance, then the edges of the liquid will be pulled above the surface of the liquid. This in turn helps pull the liquid up the tube.

Substances with strong intermolecular forces will travel further up a narrow tube (have a greater capillarity) than substances with weaker intermolecular forces.

From these experiments we can see how intermolecular forces (a microscopic property) affect the macroscopic behaviour of substances. If a substance has weak intermolecular forces then it will evaporate easily. Substances with weak intermolecular forces also have low surface tension and do not rise as far up in narrow tubes as substances with strong intermolecular forces. Boiling points are lower for substances with weak intermolecular forces. Substances are more likely to be soluble in liquids with similar intermolecular forces.

We will now look at some more properties (molecular size, viscosity, density, melting and boiling points, thermal expansion, thermal conductivity) in detail.

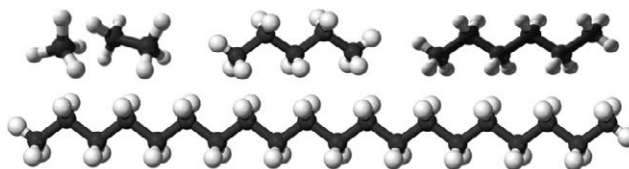
FACT

It is partly the stronger intermolecular forces that explain why petrol (mainly octane (C_8H_{18})) is a liquid, while candle wax ($\text{C}_{23}\text{H}_{48}$) is a solid. If these intermolecular forces did not increase with increasing molecular size we would not be able to put liquid fuel into our cars or use solid candles.

Molecular size

The alkanes are a group of organic compounds that contain carbon and hydrogen bonded together. The carbon atoms link together to form chains of varying lengths.

The boiling point and melting point of these molecules is determined by their molecular structure, and their surface area. The more carbon atoms there are in an alkane, the greater the surface area and therefore the higher the boiling point. The melting point also increases as the number of carbon atoms in the molecule increases. This can be seen in the table below.



Formula	CH_4	C_2H_6	C_5H_{12}	C_6H_{14}	$\text{C}_{20}\text{H}_{42}$
Name	methane	ethane	pentane	hexane	icosane
Molecular mass ($\text{g}\cdot\text{mol}^{-1}$)	16	30	72	86	282
Melting point ($^{\circ}\text{C}$)	-183	-183	-130	-95	37
Boiling point ($^{\circ}\text{C}$)	-164	-89	36	69	343
Phase at room temperature	gas	gas	liquid	liquid	solid

You will also notice that, when the molecular mass of the alkanes is low (i.e. there are few carbon atoms), the organic compounds are gases because the intermolecular forces are weak. As the number of carbon atoms and the molecular mass increases, the compounds are more likely to be liquids or solids because the intermolecular forces are stronger.

You should see that the larger a molecule is the stronger the intermolecular forces are between its molecules. This is one of the reasons why methane (CH_4) is a gas at room temperature while pentane (C_5H_{12}) is a liquid and icosane ($\text{C}_{20}\text{H}_{42}$) is a solid.

Viscosity

Viscosity is the resistance to flow of a liquid. Compare how easy it is to pour water and syrup or honey. The water flows much faster than the syrup or honey.



You can see this if you take a cylinder filled with water and a cylinder filled with glycerin. Drop a small metal ball into each cylinder and note how easy it is for the ball to fall to the bottom. In the glycerin the ball falls slowly, while in the water it falls faster.

Substances with stronger intermolecular forces are more viscous than substances with weaker intermolecular forces.

Activity: Machine and motor oils

You are given the following information about engine oils.

Oil	Use	Other info
SAE 30 monograde	Engines	Low viscosity
SAE 50 monograde	Engines	High viscosity
SAE 15W-40 multigrade	Engines	Medium viscosity
SAE 0W-40 multigrade	Engines	Medium viscosity

Multigrade oils can be used even in cold weather since they remain fluid (The first number is the rating for winter weather and the W shows that this is the rating in winter. The second number is the viscosity rating in summer). Monograde oils are given their viscosity rating at 100°C. The viscosity is an indication of how well the oil flows. The more viscous an oil the larger the molecules that are in the oil.



- Which oil has the longest molecules?
- Which oil has the shortest molecules?
- Which oil has the strongest overall intermolecular forces?
- Which oil has the weakest overall intermolecular forces?
- What can you conclude about the link between the magnitude of the intermolecular force and viscosity?

Density

DEFINITION: *Density*

Density is a measure of the mass in a unit volume.

The solid phase is often the most dense phase (water is one noteworthy exception to this). This can be explained by the strong intermolecular forces found in a solid. These forces pull the molecules together which results in more molecules in one unit volume than in the liquid or gas phases. The more molecules in a unit volume the denser that substance will be.

Melting and boiling points

Intermolecular forces affect the boiling and melting points of substances. Substances with weak intermolecular forces will have low melting and boiling points while those with strong intermolecular forces will have high melting and boiling points. In the experiment on intermolecular forces you investigated the boiling points of several substances, and should have seen that molecules with weaker intermolecular forces have

a lower boiling point than molecules with stronger intermolecular forces.

One further point to note is that covalent network structures (recall from grade 10 that these are covalent compounds that form large networks and an example is diamond) will have high melting and boiling points due to the fact that some bonds (i.e. the strong forces between atoms) have to break before the substance can melt. Covalent molecular substances (e.g. water, sugar) often have lower melting and boiling points, because of the presence of the weaker intermolecular forces holding these molecules together.

Thermal expansion

As substances are heated their molecules start moving more vigorously (their kinetic energy increases). This causes the liquid to expand on heating. You can observe this in a thermometer. As the alcohol (or mercury) is heated it expands and rises up the tube.

This is why when you tile a floor you have to leave gaps between the tiles to allow for expansion. It is also why power lines sag slightly and bridges have slight gaps for expansion.

Thermal conductivity

Different materials conduct heat differently. The following activity will highlight this.

Investigation: Thermal conductivity

Take a long thin piece of graphite and a long thin piece of copper (or other metal). Attach a bit of wax to the one end of each rod (you will need to melt the wax a bit first to make it stick). While the wax is still soft, press a toothpick into the blob of wax.

Now suspend the graphite and copper rods from a desk or chair using a piece of string and heat the other end. Observe which toothpick falls off first. Try to explain why.

Heat is transferred through a substance from the point being heated to the other end. This is why the bottom of a pot gets hot first (assuming you are heating the pot on a stove plate). In metals there are some free, delocalised electrons which can help transfer the heat energy through the metal. In covalent molecular compounds there are no free, delocalised electrons and the heat does not travel as easily through the material.

Worked example 3: Understanding intermolecular forces

QUESTION

Explain why the melting point of oxygen (O_2) is much lower than the melting point of hydrogen chloride HCl .

SOLUTION

Step 1: Write down what you know about melting points and forces

The stronger the intermolecular force, the higher the melting point. So if a substance has strong intermolecular forces, then that substance will have a high melting point.

Step 2: Write down which forces occur in the two given compounds

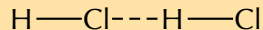
Oxygen is non-polar and has induced dipole forces. Hydrogen chloride is polar and has dipole-dipole forces.

Step 3: Combine all the facts to get the answer

We know that stronger intermolecular forces lead to higher melting points. We also know that oxygen has weaker intermolecular forces than hydrogen chloride (induced dipole versus dipole-dipole forces). Therefore oxygen will have a lower melting point than hydrogen chloride since oxygen has weaker intermolecular forces.

Exercise 4 – 2: Types of intermolecular forces

1. Given the following diagram:



- Name the molecule and circle it on the diagram
 - Label the interatomic forces (covalent bonds)
 - Label the intermolecular forces
2. Given the following molecules and solutions:
HCl, CO₂, I₂, H₂O, KI(aq), NH₃, NaCl(aq), HF, MgCl₂ in CCl₄, NO, Ar, SiO₂
Complete the table below by placing each molecule next to the correct type of intermolecular force.

Type of force	Molecules
Ion-dipole	
Ion-induced-dipole	
Dipole-dipole (no hydrogen bonding)	
Dipole-dipole (hydrogen bonding)	
Induced dipole	
Dipole-induced-dipole	

In which one of the substances listed above are the intermolecular forces:

- strongest
 - weakest
3. Use your knowledge of different types of intermolecular forces to explain the following statements:

FACT

There are about 55,5 mol of water in 1 L. This is equivalent to $3,34 \times 10^{25}$ molecules of water. That's a lot of water molecules!

- a) The boiling point of F_2 is much lower than the boiling point of NH_3
- b) Water evaporates slower than carbon tetrachloride (CCl_4).
- c) Sodium chloride is likely to dissolve in methanol (CH_3OH).

4. Tumi and Jason are helping their dad tile the bathroom floor. Their dad tells them to leave small gaps between the tiles. Why do they need to leave these small gaps?

Think you got it? Get this answer and more practice on our Intelligent Practice Service

1. [23R2](#) 2. [23R3](#) 3a. [23R4](#) 3b. [23R5](#) 3c. [23R6](#) 4. [23R7](#)



www.everythingscience.co.za



m.everythingscience.co.za

4.2 The chemistry of water

ESBMR

▶ See video: [23R8](#) at www.everythingscience.co.za

We will now look at how intermolecular forces apply to a very special liquid. This section shows how the knowledge of intermolecular forces can be applied to the case of water.

The microscopic structure of water

ESBMS

In many ways, water behaves very differently from other compounds. These properties are directly related to the microscopic structure of water, and more specifically to the *shape* of the molecule and its *polar nature*, and to the *intermolecular forces* that hold water molecules together.

In the previous chapter you learnt about molecular shape and polarity. Water has two hydrogen atoms around a central oxygen atom. The central oxygen atom also has two lone pairs of electrons. This gives water a bent (or angular) shape. It also means that water is polar since the two hydrogen atoms are not parallel to each other and so do not cancel out the bond polarity (refer back to the previous chapter on molecular shape). We can see this in the following image:

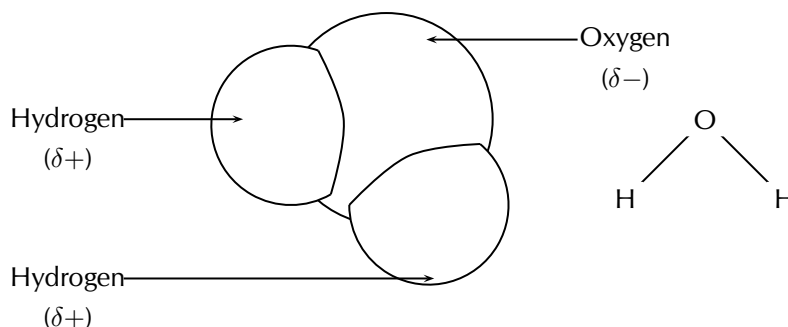


Figure 4.7: Diagrams showing the structure of a water molecule. Each molecule is made up of two hydrogen atoms that are attached to one oxygen atom.

Water molecules are held together by **hydrogen bonds**. Hydrogen bonds are a much stronger type of intermolecular force than those found in many other substances, and this affects the properties of water.

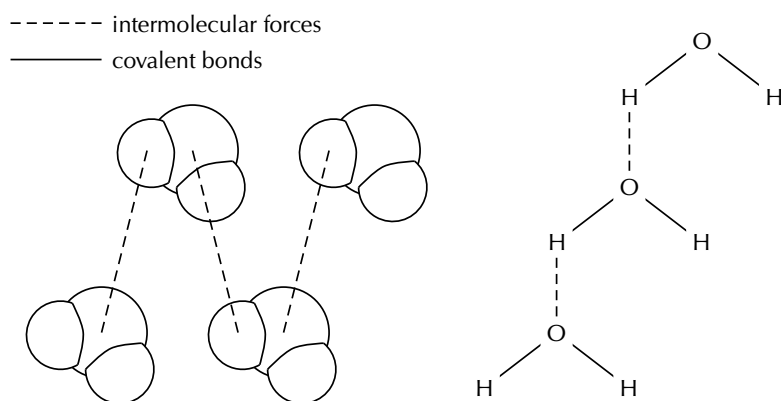


Figure 4.8: Intermolecular and covalent bonds (interatomic forces) in water. Note that the diagram on the left only shows *intermolecular* forces. The covalent bonds (interatomic forces) are between the atoms of each water molecule.

FACT

It is the high specific heat of water and its ability to absorb infra-red radiation that allows water to regulate the climate. Towns close to the sea often have less extreme temperatures than inland towns due to the oceans ability to absorb the heat.

The unique properties of water

ESBMT

We will now look at a few of the properties of water.

1. Specific heat

DEFINITION: *Specific heat*

Specific heat is the amount of heat energy that is needed to increase the temperature of a unit mass of a substance by one degree.

Water has a high specific heat, meaning that a lot of energy must be absorbed by water before its temperature changes.

You have probably observed this phenomenon if you have boiled water in a pot on the stove. The metal of the pot heats up very quickly, and can burn your fingers if you touch it, while the water may take several minutes before its temperature increases even slightly. How can we explain this in terms of hydrogen bonding? Remember that increasing the temperature of a substance means that its particles will move more quickly. However, before they can move faster, the intermolecular forces between them must be disrupted. In the case of water, these forces are strong hydrogen bonds, and so a lot of energy is needed just to break these, before the particles can start moving further apart.

2. Absorption of infra-red radiation

Water is able to absorb infra-red radiation (heat) from the sun. As a result of this, the oceans and other water bodies act as heat reservoirs, and are able to help moderate the Earth's climate.

3. Melting point and boiling point

The melting point of water is 0°C and its boiling point is 100°C (at standard pressure or 0,987 atm). This large difference between the melting and boiling point is very important because it means that water can exist as a liquid over a large range of temperatures. (This temperature range is only large in the world

FACT

When the boiling point of water is measured at sea level (e.g. towns like Cape Town and Durban), it is often very close to 100 °C since the atmospheric pressure is almost the same as the standard pressure. If you measure the boiling point of water in a town at a higher altitudes (e.g. Johannesburg or Polokwane) it will have a slightly lower boiling point.

FACT

Antarctica, the “frozen continent”, has one of the world’s largest and deepest freshwater lakes. And this lake is hidden beneath 4 km of ice! Lake Vostok is 200 km long and 50 km wide. The thick, glacial blanket of ice acts as an insulator, preventing the water from freezing.

around us, if we look at space and the universe then this is a very narrow temperature range.)

In grade 10 you studied the heating and cooling curve of water. This is given below.

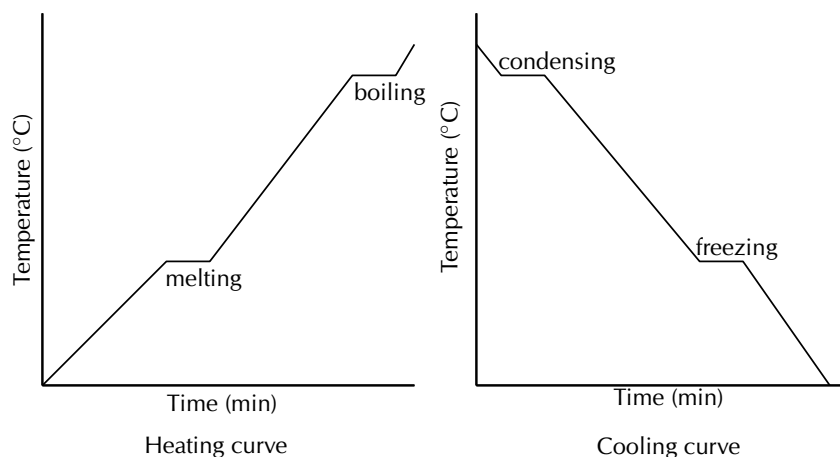


Figure 4.9: Heating and cooling curves for water.

4. High heat of vaporisation

DEFINITION: *Heat of vaporisation*

Heat of vaporisation is the energy that is needed to change a given quantity of a substance into a gas.

The strength of the hydrogen bonds between water molecules also means that it has a high heat of vaporisation. “Heat of vaporisation” is the heat energy that is needed to change water from the liquid to the gas phase. Because the forces between molecules are strong, water has to be heated to 100 °C before it changes phase. At this temperature, the molecules have enough energy to break the intermolecular forces that hold the molecules together. The heat of vaporisation for water is 40,65 kJ·mol⁻¹.

It is very important for life on earth that water does have a high heat of vaporisation. Can you imagine what a problem it would be if water’s heat of vaporisation was much lower? All the water that makes up the cells in our bodies would evaporate and most of the water on earth would no longer be able to exist as a liquid!

5. Less dense solid phase

Another unusual property of water is that its solid phase (ice) is *less dense* than its liquid phase. You can observe this if you put ice into a glass of water. The ice doesn’t sink to the bottom of the glass, but floats on top of the liquid. This phenomenon is also related to the hydrogen bonds between water molecules. While other materials contract when they solidify, water expands. The ability of ice to float as it solidifies is a very important factor in the environment. If ice sank, then eventually all ponds, lakes, and even the oceans would freeze solid as soon as temperatures dropped below freezing, making life as we know it impossible on Earth. During summer, only the upper few metres of the ocean would thaw. Instead, when a deep body of water cools, the floating ice insulates the liquid water below, preventing it from freezing and allowing life to exist under the frozen surface.

It should be clear now, that water is an amazing compound, and that without its unique properties, life on Earth would definitely not be possible.

Worked example 4: Properties of water

QUESTION

Explain why water takes a long time to heat up, but the pot that you are heating it in gets hot quickly.

SOLUTION

Step 1: Decide which property is applied here

We are asked why water takes a long time to heat up compared to the pot you are heating it in. The property that applies here is the high specific heat of water. The other properties of water do not apply here since we are comparing the pot to the water and the pot is not changing phase.

Step 2: Write the final answer

Water has a high specific heat, while the metal that the pot is made of does not. The metal pot needs less energy to heat it up and so it gets hotter faster. Water needs a lot of energy to change its temperature and so it takes longer to get hot.

Informal experiment: The properties of water

Aim:

To investigate the properties of water.

Apparatus:

- water
- ice
- beakers
- carbon tetrachloride, oil, ethanol
- various solids (e.g. sodium chloride, potassium chloride, potassium permanganate, iodine)

Method:

1. Pour about 100 ml of water into a glass beaker.
2. Place the beaker on a stand and heat it over a Bunsen burner for about a minute.
3. After this time, carefully touch the side of the beaker (Make sure you touch the glass very lightly because it will be very hot and may burn you!). Then test the temperature of the water.

4. Note what happens when you place ice into water.
5. Carefully layer water and carbon tetrachloride in a test-tube. Which substances floats on top? Try adding the water first and the carbon tetrachloride second and then the other way around. Repeat with oil and ethanol.
6. Dissolve the different solid substances in water. Observe how much of each solid (if any) dissolves.

Results:

Record all your results in the following table.

Property	Observation
Temperature	
Ice in water	
Carbon tetrachloride and water	
Oil and water	
Ethanol and water	
Solubility	

Conclusion:

You should find that the glass beaker heats up faster than the water. You should also find that water is more dense in the liquid phase than in the solid phase. Water floats on some liquids and other liquids float on water. Water is a good solvent for polar and ionic substances.

Activity: How humans have used the properties of water

Carry out some research into: water bags on cars, clay pots and carafes for water and safe or “cool” rooms to keep food cool. Find out which people groups use these things and how the properties of water help in each case.

Exercise 4 – 3: The properties of water

1. Hope returns home from school on a hot day and pours herself a glass of water. She adds ice cubes to the water and notices that they float on the water.
 - a) What property of ice cubes allows them to float in the water?
 - b) Briefly describe how this property affects the survival of aquatic life during winter.
2. Which properties of water allow it to remain in its liquid phase over a large temperature range? Explain why this is important for life on earth.

3. Which property of water allows the oceans to act as heat reservoirs? What effect does this have on the Earth's climate?

Think you got it? Get this answer and more practice on our Intelligent Practice Service

1. [23R9](#) 2. [23RB](#) 3. [23RC](#)



www.everythingscience.co.za



m.everythingscience.co.za

4.3 Chapter summary

ESBMV

▶ See presentation: [23RD](#) at www.everythingscience.co.za

- **Intermolecular forces** are the forces that act between molecules.
- The **type** of intermolecular force in a substance, will depend on the **nature of the molecules**.
- **Polar molecules** have an unequal distribution of charge, meaning that one part of the molecule is slightly positive and the other part is slightly negative. The molecule is said to be a dipole.
- **Non-polar molecules** have an equal distribution of charge.
- There are five types of intermolecular forces: ion-dipole forces, ion-induced-dipole forces, dipole-dipole forces, dipole-induced dipole forces and induced dipole forces.
- **Ion-dipole** forces exist between **ions and polar (dipole) molecules**. The ion is attracted to the part of the molecule that has an opposite charge to its own.
- **Ion-induced dipole** forces exist between **ions and non-polar molecules**. An ion induces a dipole in the non-polar molecule.
- **Dipole-dipole** forces exist between two **polar (dipole) molecules**.
- **Dipole-induced dipole** forces exist between a **polar molecule** and a **non-polar molecule**.
- **Induced dipole** forces exist between two **non-polar molecules**.
- Dipole-dipole forces, dipole-induced dipole forces and induced dipole forces are collectively called **van der Waals' forces**.
- **Hydrogen bonds** are a type of dipole-dipole force that occurs when a hydrogen atom is attached to a highly electronegative atom (oxygen, fluorine, nitrogen). A hydrogen atom on one molecule is attracted to the electronegative atom on a second molecule.
- Intermolecular forces affect the **properties** of substances.
- Substances with **larger molecules** have **stronger** intermolecular forces than substances with smaller molecules.

- **Viscosity** is the resistance to flow of a liquid. Substances that are very viscous have larger molecules and stronger intermolecular forces than substances with smaller molecules.
- **Density** is a measure of the mass in a unit volume. Solids have strong intermolecular forces and so have more molecules in one unit volume.
- Substances with **weak** intermolecular forces will have **low melting and boiling** points while those with strong intermolecular forces will have high melting and boiling points.
- **Thermal expansion** is the expansion of a liquid on heating.
- **Thermal conductivity** is a measure of how much a material conducts heat.
- Water has **strong hydrogen bonds** which hold the molecules together. It is these intermolecular forces that give water its unique properties.
- Water has the following properties: a high specific heat, absorption of infrared radiation, a large range in which it exists as a liquid, a high heat of vaporisation and has a less dense solid phase.
- **Specific heat** is the amount of heat energy that is needed to increase the temperature of a unit mass of a substance by one degree.
- **Heat of vaporisation** is the energy that is needed to change a given quantity of a substance into a gas.

Exercise 4 – 4:

1. Give one word or term for each of the following descriptions:
 - a) The attractive force that exists between molecules.
 - b) A molecule that has an unequal distribution of charge.
 - c) The amount of heat energy that is needed to increase the temperature of a unit mass of a substance by one degree.
2. Refer to the list of substances below:
 HCl, Cl₂, H₂O, NH₃, N₂, HF
 Select the true statement from the list below:
 - a) NH₃ is a non-polar molecule
 - b) The melting point of NH₃ will be higher than for Cl₂
 - c) Ion-dipole forces exist between molecules of HF
 - d) At room temperature N₂ is usually a liquid
3. The following table gives the melting points of various hydrides:

Hydride	Melting point (°C)
HI	–34
NH ₃	–33
H ₂ S	–60
CH ₄	–164

- a) In which of these hydrides does hydrogen bonding occur?
- HI only
 - NH_3 only
 - HI and NH_3 only
 - HI, NH_3 and H_2S
- b) Draw a graph to show the melting points of the hydrides.
- c) Explain the shape of the graph.

(IEB Paper 2, 2003)

4. The respective boiling points for four chemical substances are given below:

Substance	Boiling point
Hydrogen sulphide	$-60\text{ }^\circ\text{C}$
Ammonia	$-33\text{ }^\circ\text{C}$
Hydrogen fluoride	$20\text{ }^\circ\text{C}$
Water	$100\text{ }^\circ\text{C}$

- Which one of the substances exhibits the strongest forces of attraction between its molecules in the liquid state?
- Give the name of the force responsible for the relatively high boiling points of hydrogen fluoride and water and explain how this force originates.
- The shapes of the molecules of hydrogen sulfide and water are similar, yet their boiling points differ. Explain.

(IEB Paper 2, 2002)

5. Susan states that van der Waals forces include ion-dipole forces, dipole-dipole forces and induced dipole forces.

Simphiwe states that van der Waals forces include ion-dipole forces, ion-induced dipole forces and induced dipole forces.

Thembile states that van der Waals forces include dipole-induced dipole forces, dipole-dipole forces and induced dipole forces.

Who is correct and why?

6. Jason and Bongani are arguing about which molecules have which intermolecular forces. They have drawn up the following table:

Compound	Type of force
Potassium iodide in water (KI(aq))	dipole-induced dipole forces
Hydrogen sulfide (H_2S)	induced dipole forces
Helium (He)	ion-induced dipole forces
Methane (CH_4)	induced dipole forces

- Jason says that hydrogen sulfide (H_2S) is non-polar and so has induced dipole forces. Bongani says hydrogen sulfide is polar and has dipole-dipole forces. Who is correct and why?
- Bongani says that helium (He) is an ion and so has ion-induced dipole forces. Jason says helium is non-polar and has induced dipole forces. Who is correct and why?

- c) They both agree on the rest of the table. However, they have not got the correct force for potassium iodide in water (KI(aq)). What type of force actually exists in this compound?
7. Khetang is looking at power lines around him for a school project. He notices that they sag slightly between the pylons. Why do power lines need to sag slightly?
8. Briefly describe how the properties of water make it a good liquid for life on Earth.

Think you got it? Get this answer and more practice on our Intelligent Practice Service

1a. [23RF](#) 1b. [23RG](#) 1c. [23RH](#) 2. [23RJ](#) 3. [23RK](#) 4. [23RM](#)
5. [23RN](#) 6. [23RP](#) 7. [23RQ](#) 8. [23RR](#)



www.everythingscience.co.za



m.everythingscience.co.za